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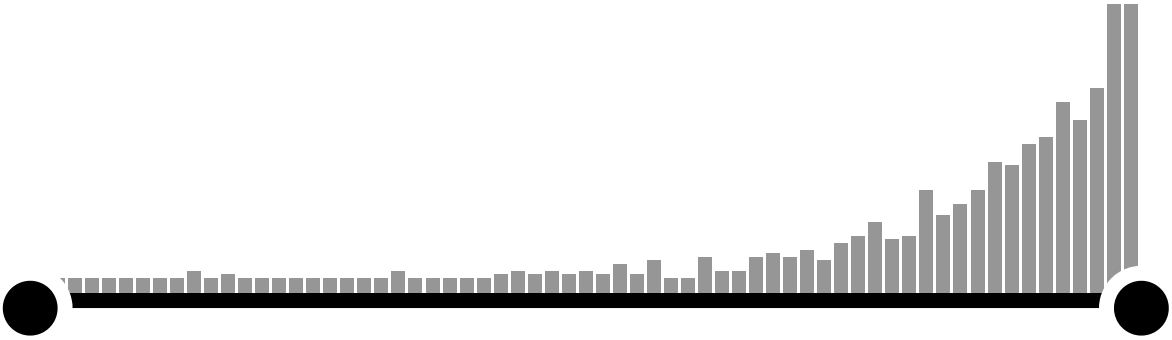
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	Document title	Authors	Source	Year	Citations
401	Conference Paper **Gender differences in environment evaluation for office building**	Bajc T.,Milanovic S.	2019 4th International Conference on Smart and Sustainable Technologies Splitech 2019, 8783130	2019	5

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Paper deals with an analysis of differences between the male and female responses regarding *thermal* comfort, *noise* and lighting level and indoor air quality in offices. The research was done in Belgrade, Serbia, for office building in a period from March to Jun 2018. Measurements for *outdoor* and indoor air parameters were done, together with the statistical survey of occupants. The results showed the significant differences in responses between male and women for the same state of the environment. It is noticed that men were more sensitive to the higher temperatures, preferring colder working environment, while women stated to feel more comfortable in slightly warmer environment. Study also showed that men were more tolerant to *noise*, while women were more sensitive to poor air quality.

402	Article **Principles of energy conversion and noise characterization in air ventilation ducts exposed to solar radiation**	Dehra H.	Applied Energy, 242	2019	9

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The aim of this paper is to present a design tool with its guiding principles for energy conversion and *noise* characterization of an exterior rectangular duct often installed on building envelopes for pre-conditioning of fresh air. The energy conversion in an exterior rectangular duct is a function of solar irradiation, air gap width, mass flow rate and pressure, wall and air temperatures. A generalized two dimensional *thermal* analysis of an *outdoor* duct is presented by placement of surface and air nodes into two adjacent stacks of control volumes representing outer and inner walls of duct. A matrix solution procedure is adopted by constituting conjugate heat exchange of conduction, convection, radiation and ventilation heat transport. The rectangular duct model is built by a metallic exterior wall exposed to a steady heat flux generation due to solar heat gain and a well-insulated back panel as its counterpart wall. The model improves the results of traditional *thermal* models for the cases where: (i) there are sections of conjugate heat exchange; and (ii) stack effect due to *thermal* buoyancy necessitates 2-D nodal analysis for distant composite nodes. The design tool is supported with some numerical and experimental results of an airflow window with a photovoltaic (PV) solar wall installed in an *outdoor* test-room. Furthermore, some examples of *noise* characterization calculations are illustrated using devised *noise* measurement equations.

Article

	Document title	Authors	Source	Year	Citations
403	**Assessing noise with provisions for ventilation and overheating in dwellings**	Harvie-Clark J.,Chilton A., Conlan N.,Trew D.	Building Services Engineering Research and Technology , 40(3)	2019	24

Hide abstract Find It @ UCL View at Publisher Related documents

In the design of residential developments, it has been common practice for façade *sound* insulation (to protect against *outdoor noise*) to be considered separately from the provisions for ventilation and for mitigating overheating. This fragmented approach has led to different designers making different, incompatible assumptions about the internal environmental quality conditions: the *acoustic* designer assumes that windows are closed to control external *noise* ingress, while the mechanical designer assumes that windows are open for ventilation or mitigating overheating. This leaves occupants with a choice between reasonable *noise* levels or *thermal* comfort, but not both. This problem is exacerbated by increased overheating risk in modern buildings and future climate scenarios. In response to this issue, the Association of *Noise* Consultants has produced the draft *Acoustics* Ventilation and Overheating Residential Design Guide – the 'AVO Guide'. The AVO guide recommends an approach to *acoustic* assessment that takes regard of the interdependence of provisions for external *noise* ingress, ventilation and overheating. This paper describes the context, background and content of the AVO Guide. There are references to passive attenuated options for ventilative cooling to help designers avoid simply specifying mechanical cooling. Practical application : This paper is entirely focused on the practical application of the guidance in the Association of *Noise* Consultants' *Acoustics*, Ventilation and Overheating Residential Design Guide. It describes the current practical problem that designers face, and the mistakes that are the common practice across the industry, of failing to integrate the design for *noise* with the ventilation strategy and provisions for overheating. It outlines how this can be achieved to enable better internal environmental quality conditions for occupants, supported by the current English Planning regime and professional good practice guidance available.

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Article

	Document title	Authors	Source	Year	Citations
404	**Valorization of sugarcane bagasse by developing completely biodegradable composites for industrial applications**	Guna V.,Ilangovan M., Hu C.,Venkatesh K., Reddy N.	Industrial Crops and Products , 131	2019	94

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Increasing constraints on the use of concrete and need to promote energy efficient green buildings are leading the research for new materials for civil engineering and architectural applications. Biobased renewable materials such as bamboo and coir are being studied for both indoor and *outdoor* applications. Utilizing agricultural residues and co-products for construction directly promotes sustainability and green building. In addition, substantial value can be added to the residues, use of synthetic polymers can be reduced and hence benefits to the environment can be accrued. In this research, we report an approach to utilize sugarcane bagasse for high value interior applications. We have used raw bagasse as reinforcement and wheat gluten as matrix to develop completely biodegradable ceiling tiles. Ratio of bagasse and gluten and density of the composite was varied to achieve desired level of performance. The biodegradable ceiling tiles were characterized for their flexural strength, *thermal*, water and *acoustic* resistance. The bagasse-gluten composite even at its lowest density had 63% higher flexural strength than the conventional gypsum based ceiling tiles. The composites exhibited good *thermal* stability until 250 °C. Maximum water absorbed by bagasse-gluten composites was 62% (w/w) as opposed to the 84% for the gypsum tiles. Furthermore, bagasse-gluten composites had similar flammability as that of the gypsum tiles and possessed moderate *acoustic* insulation and heat transmission. Bagasse based ceiling tiles meet most of the performance requirements and show high potential to replace the gypsum based tiles in current use.

	Document title	Authors	Source	Year	Citations
	Article • *Open access*				
405	**High-energy impact testing of agglomerated cork at extremely low and high temperatures**	Kaczynski P.,Ptak M., Wilhelm J., Fernandes F.A.O., de Sousa R.J.A.	International Journal of Impact Engineering , 126	2019	44

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Agglomerated cork, made from the scraps of wine stoppers, has been finding a wide set of applications due to its excellent *thermal* and *acoustic* insulation properties. The random orientation of grains makes the material nearly isotropic, while its dominant viscoelastic behaviour and nearly zero Poisson's ratio make the material also very interesting in applications where dimensional stability is highly demanded. With proven properties, agglomerated cork has been widely used for manufacturing of architectural facades, in civil construction, aerospace engineering and even home appliances production. For *outdoor* applications, the performance of cork material under different working temperatures is a vital point to be considered. This paper assesses the capability of five different types of cork agglomerates to withstand 500 J impact energy under different temperature conditions. Keeping 11.2 kg impact mass and velocity of 9.2 m/s, impact tests were performed for a wide range of temperatures starting from sub-zero temperature (–30 °C) up to 100 °C in order to cover a full span of working circumstances. Results show significant variations of amount of absorbed energy depending on testing temperature, calling the attention of designers and product developers for important aspects to be considered upon the application of this material under extreme weather conditions.

	Article				
406	**Noise-reducing vents for windows in warm, humid, tropical countries**	Satwiko P.,Prasetya A., Suhedi F.	Architectural Engineering and Design Management , 15(2)	2019	3

Hide abstract Find It @ UCL View at Publisher Related documents

Sustainable building design requires the application of natural ventilation, which has three prerequisites–a comfortable *outdoor* air temperature (20–25°C), unpolluted *outdoor* air and a low-*noise* environment (under 55 dBA). There are many locations where the first two prerequisites are met, but the environmental *noise* makes applying natural ventilation challenging. This research developed an auxiliary *noise*-reducing vent for windows (NRVW) that allows daylight and *outdoor* air to enter the room while reducing the penetration of *outdoor noise*. As an auxiliary, the NRVW is intended to be integrated into any (double) glazed windows with a *sound* transmission class (STC) above 25 dBA. The research used physical model experiments to measure the NRVW's STC and *outdoor*–indoor transmission class (OITC). Digital model simulation was used to measure the NRVW's ventilation performance. It was found that NRVW has a rating of STC 18 and OITC 16, and it generates indoor air velocity of 0.015–0.086 m/s in the occupants' zone for an *outdoor* wind speed of 1–5 m/s. Modification of the room interior increases the indoor air velocity to 0.147 m/s. Computer simulation also demonstrated that, with an occupant inside the room, an *outdoor* air temperature of 25°C and *outdoor* wind speed of 1–5 m/s maintained the indoor air temperature at 25.309–25.258°C, indicating that the resulting airflow could keep the indoor air temperature low. Although the NRVW was originally developed for warm–humid climate conditions, it is also applicable for other climates if the required *outdoor* conditions are met.

	Article • *Open access*				
407	**Thermographic Laplacian-pyramid filtering to enhance delamination detection in concrete structure**	Cheng C.,Na R.,Shen Z.	Infrared Physics and Technology , 97	2019	27

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Despite decades of efforts using *thermography* to detect delamination in concrete decks, challenges still exist in removing environmental *noise* from *thermal* images. The performance of conventional temperature-contrast approaches can be significantly limited by environment-induced non-uniform temperature distribution across imaging spaces. Time-series based methodologies were found robust to spatial temperature non-uniformity but requires extended period to collect data. A new empirical image filtering method is introduced in this paper to

	Document title	Authors	Source	Year	Citations

enhance the delamination detection using blob detection method that originated from computer vision. The proposed method employs a Laplacian of Gaussian filter to achieve multi-scale detection of abnormal *thermal* patterns by delaminated areas. Results were compared with the state-of-the-art methods and benchmarked with time-series methods in the case of handling non-uniform heat distribution issue. To further evaluate the performance of the method numerical simulations using transient heat transfer models were used to generate the 'theoretical' *noise*-free *thermal* images for comparison. Significant performance improvement was found compared to the conventional methods in both indoor and *outdoor* tests. This methodology proved to be capable to detect multi-size delamination using single *thermal* image. It is robust to non-uniform temperature distribution. The limitations were discussed to refine the applicability of the proposed procedure.

Review

☐ 408 **Smart Animal Agriculture: Application of Real-Time Sensors to Improve Animal Well-Being and Production** — Halachmi I.,Guarino M., Bewley J.,Pastell M. — Annual Review of Animal Biosciences, 7 — 2019 — 331

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Consumption of animal products such as meat, milk, and eggs in first-world countries has leveled off, but it is rising precipitously in developing countries. Agriculture will have to increase its output to meet demand, opening the door to increased automation and technological innovation; intensified, sustainable farming; and precision livestock farming (PLF) applications. Early indicators of medical problems, which use sensors to alert cattle farmers early concerning individual animals that need special care, are proliferating. Wearable technologies dominate the market. In less-value-per-animal systems like sheep, goat, pig, poultry, and fish, one sensor, like a camera or robot per herd/flock/school, rather than one sensor per animal, will become common. PLF sensors generate huge amounts of data, and many actors benefit from PLF data. No standards currently exist for sharing sensor-generated data, limiting the use of commercial sensors. Technologies providing accurate data can enhance a well-managed farm. Development of methods to turn the data into actionable solutions is critical.

Article

☐ 409 **A modified approach to estimating thermodynamic impact on buildings: A case study of poor urban setting in Lagos state** — Emetere M.E. — Journal of Engineering Design and Technology, 17(1) — 2019 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

Purpose: Managing the urban housing plan of a very fast-growing city may be difficult if the scientific input, i.e. *thermodynamic* architecture and the climate change challenges, is not factored into its initial framework. Recent building plan in some parts of a growing city located in a developing country was adopted for the purpose of this research. The purpose of this study is to investigate the impact of poor urban planning on humans. Design/methodology/approach: The reverberation time analysis was carried using the Ecotect software. In total, 15-year surface temperature data were obtained (1999-2013) from the Global Land Data Assimilation System. *Thermal* distributions were calculated using beta probability and Gaussian distribution. Also, the parametric study of the solar constant was accomplished using possible mathematical outcomes. Findings: It was discovered that irrespective of the fabrics of building, air properties and materials within a building, the total heat and *sound* absorptions are high for the life form. Necessary recommendations were made for further study. Research limitations/implications: Only the *outdoor* impact was calculated. Practical implications: There should be more proactive measures by the urban planning authorities. Social implications: There would be wide spread of diseases and very low *thermal* comfort. Originality/value: This paper illustrates on the most ignored parameter in environmental architecture.

Article

☐ 410 **Experimental investigation of wind and temperature induced scintillation effect on optical wireless communication link** — Tripathi A.,Soni G.G., Gupta S.,Mandloi A.S. — Optik, 178 — 2019 — 21

Hide abstract Find It @ UCL View at Publisher Related documents

	Document title	Authors	Source	Year	Citations

Atmospheric disturbances due to randomly distributed wind flow and temperature results in fluctuation of intensity as well as in the phase of the received light signal, impairing an optical wireless link performance. In this work, turbulence-induced scintillation on Optical Wireless Communication (OWC) is investigated experimentally on the basis of optimized signal wavelength, link length and launched power level. To mimic the *outdoor* environmental conditions, an indoor chamber equipped with fans and heating coil is made up, which gives manual control of the temperature and wind velocity within the chamber. The received signal bearing data rate of 100 Mbps is measured after traversing through the designed chamber and used to characterize the turbulence strength in terms of scintillation index (S.I.). To establish a long-range OWC link, the atmospheric channel length is increased using round-trip reflection by placing plane front surface mirrors at both ends of the chamber, also front concave mirror is used to collect distributed irradiance onto aperture area of the detector at the receiver. Experimentally, we have tried to characterize the combined effect of the wind and the temperature on the free space communication system. Various communication metrics like bit error rate (BER), signal to *noise* ratio (SNR) and Q factor have been evaluated and it is reported that the BER performance of an OWC link having wind flow of 15 km/h gets degraded by 10^9 as compared to still air.

Conference Paper

411 **Simulation of noise propagation of outdoor HVAC/R unit in surrounding space**

Poysat P.,Robinet B., Lenaerts S.

Inter Noise 2019 Madrid 48th International Congress and Exhibition on Noise Control Engineering

2019

3

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The proposed paper will describe how *acoustic* simulation technique will improve the quality of installation of *outdoor* HVAC units and its acceptance. The HVAC/R *outdoor* units are *thermodynamic* machines having 2 major sources of *noise*, a refrigerant compressor and a fan to move the air and make the *thermal* exchange. The installation of such equipment in urban area becomes more challenging due to restricting regulation and neighborhood acceptance. Furthermore, a modification of the installation after commissioning is not easy and certainly costly. A system manufacturer can now use *acoustic* solver to predict the *sound* propagation of a unit, anticipate various configuration of field installation and provide recommendations to contractor. The objective is to minimize the claims for *noise* and improve the quality of installation and acceptance of HVAC *outdoor* units.

Conference Paper • *Open access*

412 **Modeling of energy conversion and noise characterization in outdoor ducts exposed to solar radiation**

Dehra H.

Energy Procedia, 158

2019

2

Hide abstract Find It @ UCL View at Publisher Related documents

The energy conversion and *noise* characterization in an exterior rectangular duct is important, for example, in modeling PV solar wall and transpired unglazed structures. The energy conversion in an exterior rectangular duct is a function of solar irradiation, air gap width, mass flow rate and pressure, wall and air temperatures. A generalized two dimensional *thermal* analysis of an *outdoor* duct is presented by placement of surface and air nodes into two adjacent stacks of control volumes representing outer and inner walls of duct. A matrix solution procedure is adopted by constituting conjugate heat exchange of conduction, convection, radiation and ventilation heat transport. *Noise* characterization due to energy conversion in *outdoor* ducts is also presented. The energy conversion and *noise* characterization is supported with some experimental results of a photovoltaic (PV) solar wall installed in an *outdoor* test-room. Furthermore, some examples of *noise* characterization calculations are illustrated using devised *noise* measurement equations.

Conference Paper

413 **Atmospheric propagation of aircraft acoustic signature from high altitude**

Ghinet S.,Price A., Daigle G.A.,... Grewal A., Wickramasinghe V.

Inter Noise 2019 Madrid 48th International

2019

3

	Document title	Authors	Source	Year	Citations
			Congress and Exhibition on Noise Control Engineering		

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A large number of computational approaches for *acoustic* atmospheric propagation have been developed and extensively validated during the past decades. Generally, the model of choice to predict *outdoor sound* propagation is the Parabolic Equation (PE) because it can simultaneously account for all the factors that can affect the propagation. These factors include geometrical spreading, atmospheric absorption, ground effects, refraction and turbulence. However, in the case of a high-altitude aircraft flying overhead, a microphone near the ground would be located outside the solid angle of validity of the PE. In this paper an impedance plane formulation that does not account for refraction and a Fast Field Program (FFP) that does not account for turbulence were implemented to predict the *acoustic* spectra of an aircraft propagating through the atmosphere at various altitudes, distances and angles. The paper aims at presenting the two theoretical approaches, the numerical and testing validation results and discussing the influence of various factors on the aircraft *sound* propagation from high altitude. In particular it will be shown that refraction is negligible, but that meteorological measurements made only at ground level are insufficient to estimate the atmospheric absorption.

Conference Paper

☐ 414	**Measurement of the sound transmission loss of rain screen wall assemblies**	Connelly M.,Stehling E., Tamanna O.,Mahn J.	Proceedings of the 26th International Congress on Sound and Vibration Icsv 2019	2019	0

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As part of an initiative to adapt building facades for the future weather expected in British Columbia due to climate change, the *sound* insulation of rainscreen wall assemblies with superior moisture and *thermal* performance targets was investigated. A hybrid field-test method was used to screen for the critical material and design variables which affect the *sound* transmission loss of rainscreen wall assemblies. The hybrid field-test method used an *outdoor* source as per ASTM E966, the indoor *sound* levels were measured in a semi-anechoic environment as per ISO 15186. The prerequisite to evaluating the hybrid field test method was a specific investigation on the inclusion of the rainscreen drainage and ventilation cladding gaps in the test specimens, a deviation from the standard practice of testing a sealed wall assembly. A total of 57 wall assemblies were screened at the hybrid field lab and a further eleven test were completed using the ASTM E-90 test method. While the results of the hybrid field-test method did not align with ASTM E90 test results, the hybrid field-test method was shown to be viable as a quick and lower cost means of obtaining results for the initial screening of designs. The results of the testing demonstrated that the use of split-insulation in the rainscreen cavity with increased cladding mass increased transmission loss

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Conference Paper

☐ 415	**Integrated acoustic and thermo-fluid insulation modeling of an airflow window with a photovoltaic solar wall**	Dehra H.	Building Simulation Conference Proceedings , 1	2019	4

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

A simulation model is proposed for integrated *acoustic* and *thermo*-fluid insulation constituting an airflow window with a photovoltaic (PV) solar wall spandrel section. The physical model of an *outdoor* test-room comprises of a wooden framed double or cavity wall assembly with: (i) a triple glazed fenestration section with a closed roller blind; (ii) a solar wall spandrel section of double-glass PV modules and back panel of polystyrene filled plywood board; and (iii) fan pressure based manually operated inlet and exhaust dampers with ventilation through exhaust fan for transportation of heat. A generalized two-dimensional analysis of a double wall structure is illustrated by placement of surface and air nodes into two adjacent stacks of control volumes representing outer and inner walls. Integrated *noise* insulation and energy conversion model is presented. The energy conversion and *noise* insulation model is supported with some numerical results using devised *noise* measurement equations.

Conference Paper • *Open access*

☐ 416 **Users' satisfaction of indoor environmental quality conditions in ZEB+ at high latitudes**

Zaikina V.,Moscoso C., Matusiak B.,Chelli A., Balasingham I.

Iop Conference Series Earth and Environmental Science , 352(1), 012001

2019

6

Hide abstract Find It @ UCL View at Publisher Related documents

Energy efficiency policy forces architects to design buildings with increasingly well-sealed building skin. With the minimized *outdoor* connection, the indoor environment factors depend strongly on technical systems and control. In these scenarios, occupant dissatisfaction indicates a need for improvements of indoor environment and the way it is controlled. The aim of the article is to contribute to the discussion about the user perspective of indoor environmental quality in ZEB in the Nordic region. The focus on daylight as a factor for visual comfort, and on low outside temperature as an aspect of *thermal* comfort was dictated by this choice. An experimental study was conducted with 75 participants, in which the *thermal*, *acoustic* and visual conditions (controlled factors) together with a view out, humidity and CO2 level (monitored factors) were assessed by them and quantified via sensors. In most studied settings, the *thermal* comfort was the most determinant factor, followed by the *acoustic* and visual comfort. Other significant factors were mean illuminance in the room, mean temperature at the participants' desks and a mode value of the *noise* level. The daylight levels much lower than recommended in regulations were accepted by participants as comfortable if they were sitting by the window. Also, participants preferred a higher indoor temperature than the recommended in Norway.

Conference Paper • *Open access*

☐ 417 **Experimental investigations on tire/road friction dependence from thermal conditions carried out with real tread compounds in sliding contact with asphalt specimens**

Arricale V.M.,Carputo F., Farroni F.,Sakhnevych A., Timpone F.

Key Engineering Materials , 813 KEM

2019

3

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

The understanding of tire's adherence with a rough surface is a common goal for several fields in the automotive sector. In fact, grip is synonym of safety and performance, playing a decisive role for braking distance and vehicle stability, fuel consumption, wear rate [1], *noise* generation and for the vehicle dynamics control system (e.g. ABS, TCS, AYC and other) [2]. This paper deals with tire tread grip experimental investigation and evaluation under different conditions that influence it during the sliding contact [3]. In this regard the test campaign involves the use of different tire compounds (in terms of viscoelastic characteristics), tested in several conditions: different contact pressure, sliding speed, temperature, sliding contact length and road surface. The test bench employed by the UniNa Vehicle Dynamic Research Group is an upgrade of the British Pendulum, an instrument for *outdoor* tribological tests on road sections. The principal sensors installed on the test bench are an encoder, for the evaluation of the sliding speed of the tire specimen, and a load cell, for the measurement of the force arising at tire/road interface in the longitudinal and vertical directions [4]. In fact, the grip shall be determined as the ratio of the longitudinal force and corresponding load on the tire. The paper's aim is the description of the experimental campaign after an accurate introduction on the test setup and an illustration of the equipment. Finally, the preliminary results and the methodologies used to process the acquired data are described.

Conference Paper

418 **Enhancing indoor health comfort in adaptively reused heritage building** — Susan,Wardhani D.K. — Proceedings of the Annual International Conference on Architecture and Civil Engineering — 2019 — 0

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A heritage building that being adaptively reused has some concerns regarding indoor health and comfort. They are prone to cause discomfort to the occupants. The assessment based on GBCI rating tools of Indoor Health and Comfort gives the existing building (Perpustakaan Bank Indonesia) 10 credits. 1 credit for *outdoor* air introduction, 6 credits for chemical pollutant, 1 credit for outside view, and 1 credit for *thermal* comfort. This research aims to propose proper strategies, to make the building has better condition of indoor health and comfort, which represent by higher credits from GBCI rating tools. Strategies are observed, simulated, and manually calculated to enhance indoor health and comfort, as well as giving the building more credits. No-smoking campaign should be applied as prequisite criteria of indoor health and comfort. CO2 sensors are placed in computer room and kids reading area, which gives the building additional 1 credit. Maintenance and treatment for ducting system are proposed to give the building 1 more credit. Other 2 credits come from the planning of artificial lighting and individual switching system. 78% transmittance glass is proposed to enhance illuminance level, which leads the building to get another 1 credit. Standing AC unit gives higher control to the occupants in every different rooms, and this strategy gives the building another 1 credit. Treatment for *noise* level using green wall fence gives the building 1 credit. Interior plants (2 credits) and pest management (1 credit) are the last two strategies proposed to enhance indoor health and comfort. Those strategies give the building another 10 credits, which lead them to achieve 19 credits from 28 total credits available in the rating tools.

Conference Paper

419 **Visible light propagation characteristics under turbulent atmosphere and its impact on communication performance of traffic system** — Guo L.-D.,Cheng M.-J.,Guo L.-X. — Proceedings of SPIE the International Society for Optical Engineering, 11170, 1117047 — 2019 — 5

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In this paper, mathematical modeling and statistical methods are used. establishing a simple bit error rate evaluation model of intelligent transportation optical communication system under ideal atmospheric conditions. The lognormal turbulence distribution model is introduced into the geometric model of visible optical communication system to further calculate the parameters of the whole communication system. The effects of signal-to-*noise* ratio, aperture size, visibility, wavelength and other factors on the bit error rate and channel capacity of visible optical

	Document title	Authors	Source	Year	Citations

communication are calculated and analyzed. After calculation, the bit error rate increases with the increase of signal-to-*noise* ratio (SNR), decreases with the increase of visible wavelength and increase with distance. besides, the stronger the turbulence is, the more violent the fluctuation of BER is. The simulation results based on this model are basically consistent with the actual situation. It can simulate the visible light communication of intelligent transportation system better and provide reference for the selection of communication parameters in different environments.

Conference Paper

420 **Simulative study and potential analysis of inflatable facade elements to balance environmental disturbances**

Guenther J.,Oei M., Harder N.,Sawodny O.

9th Eccomas Thematic Conference on Smart Structures and Materials Smart 2019

2019

2

Hide abstract Find It @ UCL View at Publisher Related documents

The high *thermal* mass of conventional buildings is an important factor for maintaining comfortable indoor conditions. The building itself serves as a heat storage and damps the effects of changing *outdoor* temperatures or varying solar loads. Moving away from heavyweight to ultra-light-weight buildings allows to rethink the conventional design of facades as sole *thermal* or *noise* insulation. Intelligent facade structures with integrated capability for *thermal* actuation can be used to control indoor conditions in order to reduce the energy demand. Inflatable membrane facades allow for a variation of heat capacity by introducing air and thus increasing the *thermal* mass of a facade element. This can be used to control the heat exchange between the building interior and the environment depending on the outside conditions. Furthermore, introducing connections between adjacent wall elements enables the transport of *thermal* energy from one side of the building to another. This allows for intelligent distribution of solar loads by transferring heated air from the south- to the north-facing side of the building. In this work, a *thermal* network model of a building using inflatable membrane wall elements is developed. The numerous couplings of element-specific mass and temperature dynamics introduce various effects on the *thermal* conditions and the related energy demand. The theoretical potential of this type of actuation is determined through a detailed simulative study. Recordings of the temperate climate in Germany are combined with a geometrical solar irradiation model to determine the environmental disturbances. Subsequently, an analysis of different operational conditions is performed to identify the energy saving potential of rewarding use cases. While the energy demand rather depends on system design parameters (e.g. solar absorption), the reduction of radiant temperature asymmetry and demand-oriented heat transfer from unoccupied to occupied zones are promising applications.

Conference Paper

421 **Occupants' behaviour in office building: Stochastic models for window opening.**

Fabi V.,Maggiora V., Corgnati S.P.,Andersen R.

Proceedings Windsor Conference 2014 Counting the Cost of Comfort in A Changing World

2019

2

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The interactions between building occupants and control systems have a high influence on energy consumption and on internal environmental quality. In the perspective of a future of “nearly-zero” energy buildings, it is crucial to analyse the energy-related interactions deeply to predict a realistic energy use during the design stage. Since the reaction to *thermal*, *acoustic* or visual stimulus is not the same for every human being, monitoring the behaviour inside buildings is an essential step to assert differences in energy consumption related to different interactions. Based on measurements in eight offices in Prague, seven models concerning occupants' window opening behaviour in office buildings were proposed. The dataset was gathered through a monitoring campaign lasting 11 months, from February 2012 until January 2013. Indoor and *outdoor* environmental conditions were collected in eight south-west exposed rooms, mechanically ventilated and heated through radiators. Two rooms were used by two people while the other six were single offices. Seven models for opening and closing windows were inferred using multiple logistic regression. The model's outputs were probabilities of a window opening and window closing event within the next five minutes. Implementing all the models into dynamic energy simulation software would lead to different simulated

	Document title	Authors	Source	Year	Citations

behaviour patterns and consequently different demands and uses of energy, representing a more reliable variability of energy consumption due to building occupants'.

Article • *Open access*

422 **Development of an Indoor Environmental Quality assessment tool for the rating of offices in real working conditions** — Devitofrancesco A., Belussi L.,Meroni I., Scamoni F. — Sustainability Switzerland , 11(6), 1645 — 2019 — 21

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The Indoor Environmental Quality (IEQ) refers to the quality of indoor spaces in relation to the health and well-being of users. IEQ is a holistic concept considering various components of the overall indoor comfort: *thermo*-hygrometric, lighting, air quality and *acoustics*. Each component is described through specific performance indicators and benchmarks. The quality of the built environment is assessed at different stages from design to operational phase. The scientific literature reports several case studies related to the assessment of the individual components of the IEQ Tools aimed at the evaluation of the overall IEQ. The paper proposes an assessment tool based on the SB Method (Sustainable Building Method) and the Multi Criteria Analysis for the evaluation of IEQ during the operational phase of a building. Each component of IEQ is analysed through objective indicators and calculation methods. The tool provides two main outcomes: a global score expressing the overall performance of the building from the IEQ perspective; quantitative evaluations of all indoor comfort components through monitoring and measurement of the environmental variables. The above contributes to select intervention areas to optimize indoor design and to identify technologies aimed at ensuring the best IEQ levels for users at the operational stage. The system was applied to an open-plan working space of an office building. Monitoring activities and measurements are carried out to detect the indoor and *outdoor* variables affecting the IEQ. All aspects of IEQ were analysed and quantified so to evaluate the overall performance of the building and provide data to improve the working conditions.

Conference Paper

423 **Greenship assessment of indoor health comfort in adaptive reused building** — Wardhani D.K.,Susan — Proceedings of the Annual International Conference on Architecture and Civil Engineering — 2019 — 1

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When a heritage building can no longer function, a new function must be proposed to preserve that building. Of all several colonial buildings that have been changed, the space adjustments made to the existing conditions of those buildings have not paid attention to the tenant' condition and comfort. The application of adaptive reuse in Surabaya is mainly carried out in colonial buildings, one of which is the Library of Bank Indonesia. Indoor Health and Comfort (IHC) as one of the Greenship Interior Space criteria is used to assess the condition of the building of Library of Bank Indonesia since it is related to the health and comfort's health of tenant. The use of Greenship Interior Space as criteria for assessing building condition is especially useful for design sustainability in reducing energy and saving resources. The criteria of Indoor Health and Comfort (IHC) of Greenship Interior Space which are used as observation for the Library of Bank Indonesia consist of No Smoking Campaign (IHC P), *Outdoor* Air Introduction (IHC 1), CO2 Monitoring (IHC 2), Chemical Pollutant (IHC 3), Indoor Pollutant Source Control (IHC 4), Biological Pollutant (IHC 5), Visual Comfort (IHC 6), Outside View and Daylight (IHC 7), *Thermal* Comfort (IHC 8), *Acoustic* Level (IHC 9), Interior Plants (IHC 10), Pest Management (IHC 11), and Room Occupant Survey (IHC 12). Of all 31 points as the total points of IHC values,, the Library of Bank Indonesia has got 9 points. Adaptive reused buildings is one way of sustainable development on an urban scale, especially in extending the life of buildings, reducing waste from building destruction, saving energy and providing socio-economic benefits to the community.

Conference Review

424 **Proceedings of the 2018 12th Spanish Conference on Electron Devices, CDE 2018** — [No Authors Found] — Proceedings of the 2018 12th Spanish Conference on — 2018 — 0

Document title	Authors	Source	Year	Citations
		Electron Devices Cde 2018		

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The proceedings contain 35 papers. The topics discussed include: lab-on-PCB: low cost 3D microelectrode array device for extracellular recordings; fabrication process of non-linear planar diodes based on GaN; Electron injection model for linear and parabolic 2D materials: *noise* as a parabolic or linear band detector; *thermophotovoltaic* efficiency measurement: design and analysis of a novel experimental method; electrical impedance spectroscopy microflow cytometer for cell viability tests; constrained many-objective evolutionary extraction procedure for an OTFT compact model including contact effects; and analyzing *outdoor* degradation of flexible PEDOT-free P3HT:PCBM organic solar cells using impedance spectroscopy.

Conference Paper • *Open access*

425 **Self-Interference Channel Measurements for In-Band Full-Duplex Street-Level Backhaul Relays at 70 GHz** | Haneda K.,Jarvelainen J.,Karttunen A.,Putkonen J. | IEEE International Symposium on Personal Indoor and Mobile Radio Communications PIMRC, 2018-September, 8580687 | 2018 | 16

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This paper reports measurements of self-interference (SI) channels in an in-band full-duplex (IBFD) relay scenario for millimeter-wave street-level backhauling. The street-level backhauling suffers from clutters surrounding the relay devices and hence SI originating from the environment may be dominant. An ultrawideband channel sounder operating at 70 GHz carrier frequency is used to measure the SI channels in a chamber, indoor window-side, indoor and *outdoor* corner scenarios. The transmit and receive antennas are waveguide-fed lens antennas. The measurements showed that cross-polarized feeding of waveguides to the lenses provided higher isolation than co-polarized feeding. In the corner scenario, observed minimum isolation level is 85 dB in the frequency domain, and 100 dB in the delay domain. The results confirmed that isolation is not limited by the direct coupling because of the high directivity of the lens antenna, but by the environmental coupling. Further link budget analysis for *outdoor*-to-indoor backhauling showed that the IBFD relay with practical transmit power is capable of enhancing the signal-to-*noise* ratio and hence extending the coverage, as SI at the relay receiver can be almost the same level as *thermal noise*.

Review • *Open access*

426 **A review of performance specifications and studies of trickle vents** | Biler A.,Tavil A.U.,Su Y.,Khan N. | Buildings, 8(11), 152 | 2018 | 25

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The air quality of indoor spaces is the most significant parameter in providing a healthy living environment. Low indoor air quality (IAQ) leads to Sick Building Syndrome-one of the major reasons for labor loss in office buildings. The fundamental measure to ensure indoor air quality is ventilation, which includes two basic types: mechanical ventilation and natural ventilation. Natural ventilation is an exchange of stale indoor air with fresh *outdoor* air by means of a pressure difference due to wind and/or stack effect. Trickle Vents, known also as background ventilators, are natural ventilation devices which can be integrated into façades or window systems as an alternative to operable vents, specifically in high-rise buildings. The major design criteria of Trickle Vents are ventilation capacity, controllability, actuation, *thermal* insulation, air permeability, water tightness, climatic adaptation, security, and *acoustic* attenuation. Other important parameters in Trickle Vents design are positioning, equivalent area, and control strategy. This paper aims to review all these aspects, particularly with reference to building regulations and commercial products. Furthermore, simulation, experimental, monitoring, and survey studies of Trickle Vents are also discussed. This literature review is presented from the perspective of performance parameters, control strategies, positioning, etc., with an aim to provide a comprehensive overview of such technology.

Article

	Document title	Authors	Source	Year	Citations
427	**Identification of urban park activity intensity at different thermal environments and visible sky by using sound levels**	Tsai K.-T.,Lin Y.-H.	International Journal of Biometeorology, 62(11)	2018	20

Hide abstract Find It @ UCL View at Publisher Related documents

An efficient urban environmental design considers appropriate *thermal* comfort condition, shaded space, and activity intensity. Physiologically equivalent temperature (PET) and visible sky, i.e., the sky view factor (SVF), are usually used as indicators to determine the *outdoor thermal* comfort and amount of shaded space, respectively. The activity intensity in urban parks, which is dependent on culture and micrometeorological conditions, was represented inappropriately by attendance density in Taiwan. To optimize the park design and improve the park utilization rate in Taiwan, several environmental factors such as *sound* pressure levels and numbers of park visitors were measured, and PET values and SVF values were calculated from primary micrometeorological data such as wind speed and globe temperature. This study proposed equivalent continuous *sound* pressure level (Leq) as a novel indicator to represent park activity intensity and investigated the correlation between Leq and SVF at different PET values. Leq was more appropriate than was attendance density in representing the park activity intensity in Taiwan. In addition, Leq was highly negatively correlated with SVF when visitors felt that the *outdoor thermal* comfort condition was hot or very hot. In other words, a lower degree of shading in the park resulted in lower activity intensity. Park visitors tended to engage in activities in the shaded regions because of more favorable *thermal* comfort conditions (i.e., neutral PET). The established quantitative relationships among Leq, PET, and SVF can serve as a reference for park planning.

Article

428	**Open-plan office post-occupancy evaluation: inquiring Algerian gender specificity**	Benammar A.,Anouche K., Lesgaa H.,Hamza Cherif Y.	Facilities, 36(11-12)	2018	7

Hide abstract Find It @ UCL View at Publisher Related documents

Purpose: This paper aims to examine the impact of an open-plan office (OPO) space organisation on a user's attitude in the Algerian context; more specifically, it investigates gender differences in the occupants' perception of such working environment. It, principally, aims to explore the employees' reaction towards OPO and sees how much such local office type complies with indoor environment quality (IEQ) and psychological comfort.
Design/methodology/approach: The theoretical framework of the study is mainly related to environmental psychology referring to the interaction between users and their environment. Post-occupancy evaluation was carried out using exploratory study and questionnaires, followed by statistical analyses. It was performed on a large-scale sample of employees (296 employees) working in recently built OPO situated in Oran (Algeria). Findings: Fundamentally, women appear to show more concern regarding comfort. They do not show much reluctance to be mixed with men in a large office space as opposed to more conservative reaction towards mixing up in *outdoor* public space environment. As for environmental factors (IEQ), indicators have shown the inadequacy of most buildings in terms of *thermal*, light or *noise* comfort. The study has also revealed that a majority of users recognise the professional advantages of the OPO, although it is suggested that their preferred type would be the individual office. Originality/value: The paper provides a concise starting point for future research interested in developing Algerian context OPO design in terms of both indoor environmental and psychological comfort.

Article • *Open access*

429	**Ventilation behavior in residential buildings with mechanical ventilation systems across different climate zones in China**	Lai D.,Qi Y.,Liu J.,... Zhao L., Wei S.	Building and Environment, 143	2018	72

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A mechanical ventilation system in a residential building provides a more reliable, controllable, and comfortable means of ventilation than natural ventilation through an open window. However, the operation of a mechanical ventilation system consumes energy and may generate *noise* and secondary air pollutants. Thus, it is worthwhile to study residents' actual ventilation patterns. This study investigated the usage of natural and mechanical ventilation in 46 apartments across five different climate zones in China by on-site monitoring and a questionnaire survey. On average, the daily natural and mechanical ventilation durations were 11 h and 7.2 h, respectively. Large differences existed among climate regions and seasons. From north to south, as the climate became warmer, the usage of

Document title	Authors	Source	Year	Citations

natural ventilation increased. From a seasonal perspective, the natural ventilation duration was the longest in summer and the shortest in winter. The trend of mechanical ventilation usage was opposite to that of natural ventilation. Generally, as the *outdoor* air temperature increased, the duration of natural ventilation increased and the duration of mechanical ventilation decreased. This study found that occupants prioritized their *thermal* comfort needs over healthy indoor air quality (IAQ). However, occupants were willing to spend money on energy when the health requirement could be met by mechanical ventilation. This study provides further understanding of occupants' ventilation behavior, and the results can serve as more accurate boundary conditions for analyses of IAQ, energy consumption, and *thermal* comfort in Chinese residential buildings.

Article • *Open access*

430 **Open windows for natural airflow and environmental noise reduction** — Mediastika C.E., Kristanto L.,Anggono J., Suhedi F.,Purwaningsih H. — Architectural Science Review , 61(5) — 2018 — 12

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For buildings in tropical climates, the use of open windows for natural ventilation can not only provide low cost and low energy comfort but also provide *thermal* delight for occupants. However open windows let in environmental *noise*. The size and location of windows in walls are key but this study set out to determine whether there are any window forms that can effectively reduce the level of *sound* ingress into a building. A top-hung window was chosen for this study looking at the dimensions of the window opening and its orientation in relation to the environmental *noise* source. The top-hung form was selected for its potential to balance the functions of allowing airflow while potentially blocking and reducing *noise* levels with its window pane angle. The window pane was tested in a laboratory at three opening angles: 0° (closed), 5°, and 10° to let the *outdoor* air in. The angles were also tested in three different orientations in relation to the *noise* source position: perpendicular, sideways 60°, and sideways 90°. The test was conducted at 1/3 octave band frequency as specified by ASTM E90-09 to obtain the transmission loss, then ASTM E1332-90 was referred to calculate the *outdoor*-indoor transmission class (OITC) of the specimens. The study revealed that window orientation and extent of the openings and window pane angle have little effect on *noise* reduction. The paper concludes with a discussion of how higher levels of natural ventilation can be achieved, particularly in noisy urban areas. The top-hung window, once open, barely blocks environmental *noise*. However, when the window was closed, the perpendicular orientation offered more *noise* reduction when compared windows placed sideways to the *noise* source. The adjustable pane-angle of a top-hung window placed perpendicular to the airflow, and thus the *noise* source, seemed to have the most potential to balance the functions of allowing airflow when opened and reducing significant *noise* when closed. Nonetheless, an open window that through its design alone can significantly reduce the ingress of ambient *noise* into a building is still an issue.

Article

431 **Zonal network solution of temperature profiles in a ventilated wall module** — Rakotomahefa T.M.J., Wang F.,Zhang T.T., Wang S. — Journal of Building Performance Simulation , 11(5) — 2018 — 1

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A ventilated wall module is an *outdoor* air-intake device that can clean the drawn air, reduce heat loss or gain, and insulate *outdoor noises*. It contains a solar board, air cavities, and an air filter. This investigation proposes a zonal network model to efficiently predict temperature profiles in a ventilated wall module. The *thermal* network is established by applying an energy balance to each network node. The *outdoor* temperature, indoor temperature, and solar irradiation intensity are specified as boundary parameters. The proposed model is validated by the measured temperatures in a prototype model. The results show that the proposed model provides reasonably accurate temperatures with a maximum gap of 3°C between the prediction and the measurement. The temperature profiles approach straight lines when the mass flow rate significantly increases. A similar trend is observed when the solar irradiation intensity, the solar board absorptivity, or the glass transmitivity dramatically decreases.

Article

	Document title	Authors	Source	Year	Citations
432	**Acoustic inversion for Monin-Obukhov similarity parameters from wind noise in a convective boundary layer**	Hart C.R.,Nykaza E.T., White M.J.	Journal of the Acoustical Society of America , 144(3)	2018	3

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The prediction accuracy of *outdoor sound* is in large part limited by uncertainties in the state of the atmosphere. These uncertainties can potentially be reduced by inferring scaling parameters of the atmospheric surface layer from wind *noise*. Screened microphones sense wind *noise* as a result of mean atmospheric flow, turbulent eddy interaction with the windscreen, and pressure fluctuations within the turbulent flow. Under conditions of terrain homogeneity and atmospheric quasi-steadiness, the Monin-Obukhov similarity theory (MOST) states that only a handful of parameters governs the dynamics of the atmospheric surface layer. This study explores the relationships of atmospheric similarity parameters to the *acoustic* spectrum of wind *noise* in a convective boundary layer. Ambient *noise* data collected in a high desert during a 2007 long-range *sound* propagation experiment are analyzed for the purposes of establishing a nondimensional empirical relationship between the *acoustic* power spectrum and MOST parameters. Furthermore, this paper examines the consequences of inferring surface-layer scaling parameters with different parameter priors. This study shows that, for minimizing the variance in the inversion, the most important parameter to constrain is the Obukhov length.

	Document title	Authors	Source	Year	Citations
	Conference Paper				
433	**Outdoor People Detection in Low Resolution Thermal Images**	Cerutti G.,Milosevic B., Farella E.	2018 3rd International Conference on Smart and Sustainable Technologies Splitech 2018, 8448322	2018	9

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Presence detection is a main functionality to make our living spaces smarter and is implemented through several kinds of sensors and smart devices. Recent advancements in embedded systems market and technology enable the design of sophisticated solutions in a low-cost and scalable fashion. However, applications of presence detection, such as surveillance or occupancy detection, home automation or smart lighting are built for indoor scenarios. Therefore, many systems weaken their performance when applied *outdoor*, where ambient conditions have higher variability. In this work, we describe our exploratory study on people detection in *outdoor* scenarios by use of an 8x8 pixels resolution *thermal* sensor. We tested different techniques to extract the presence of a person crossing the detection area. We observed that signal to *noise* ratio depends on the difference between background and human body temperature. To address this, we collected a dataset spanning a wide range of background conditions and different user clothing and we used it to tune and evaluate the proposed detection techniques. As a possible solution, we propose to adapt the threshold with temperature, providing a regression curve to select it and demonstrate benefits against the use of a fixed threshold with all explored techniques.

	Document title	Authors	Source	Year	Citations
	Article • *Open access*				
434	**Measuring and mitigating overheating risk in solid wall dwellings retrofitted with internal wall insulation**	Tink V.,Porritt S., Allinson D.,Loveday D.	Building and Environment , 141	2018	79

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Upgrading the *thermal* insulation of UK houses to improve wintertime energy efficiency raises concerns about potential summertime overheating risk. To address these concerns, experiments were conducted in a pair of *thermally* matched, solid walled houses, located in the UK. One of the pair was retrofitted with internal wall insulation, while the other remained uninsulated; both houses were monitored for four weeks during the summer of 2015. Operative temperatures in the living room and main bedroom were observed to be higher in the internally insulated house in comparison to the uninsulated house. The houses were again monitored for a further three weeks with a simple overheating mitigation strategy applied consisting of night ventilation and shading using internal blinds. The data were normalised for variations in external weather conditions using a linear regression model, with the exponentially-

	Document title	Authors	Source	Year	Citations

weighted *outdoor* running mean air temperature as the predictor variable of indoor operative temperature. The results showed that the mitigation strategy was effective at reducing the internal temperature in the internally insulated house to a level similar to that observed in the uninsulated house. The marginal increase in overheating risk should not be considered a barrier to the uptake of IWI in this type of house and location, at this time. Shading devices and secure *noise* attenuating vents for existing dwellings may be needed as part of a package of refurbishment in the future. It could become a requirement within the Building Regulations [1] to reduce overheating risk when retrofitting existing homes.

Article

435 **Part I: Thermal analysis of naturally ventilated BIPV system: Experimental investigation and convective heat transfer coefficients estimation** — Agathokleous R.A., Kalogirou S.A. — Solar Energy, 169 — 2018 — 64

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The purpose of this two-part study is to present the experimental analysis carried out on a naturally ventilated Building Integrated Photovoltaic (BIPV) system and the new correlations developed for the estimation of the convective heat transfer coefficients (CHTC) in the air gap, and use the developed correlations to construct a simulation model which is validated with the experimental data. In BIPV systems the air gap is responsible to cool the PVs and remove excess heat to avoid building overheating. The ventilation of the air gap can be natural or mechanical. However, naturally ventilated systems are less studied although they have important advantages over the mechanically ventilated ones, such as the avoidance of extra energy of the fans, maintenance and *noise*. The present Part I of this study presents an experimental based *thermal* analysis of a naturally ventilated vertical BIPV system. A series of experiments on a custom made BIPV system were carried in real *outdoor* conditions as well as indoors with the use of a large scale solar simulator to measure the *thermal* characteristics of the system and its *thermal* behaviour. Indoor experiments were performed to avoid external disturbances from wind that may occur outside. The results show that an open-ended air gap of 0.1 m can create adequate air flow on naturally ventilated systems and can ensure low PV temperatures to avoid PV efficiency decrease. The experimental data are then used to estimate the convective heat transfer coefficients to fit the real conditions of the BIPV systems. Then two correlations are proposed for the estimation of the Nusselt number that fits best the *thermal* characteristics of a naturally ventilated BIPV system.

Conference Paper

436 **A multi-parametric PV solar wall device** — Dehra H. — IEEE International Conference on Power Control Signals and Instrumentation Engineering Icpcsi 2017 — 2018 — 10

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A multi-parametric evaluation of a photovoltaic (PV) solar wall device was performed. The PV solar wall device was built with a pair of PV modules, a ventilated air cavity and an insulating back panel of plywood board filled with polystyrene installed in an *outdoor* test-room. The full scale *outdoor* test-room experimental facility for a PV solar wall device was set up at Concordia University, Montréal, Québec, Canada. The experiments were conducted for obtaining currents, voltages, temperatures, air velocities, sensible heat capacity and *thermal* storage capacity of a PV solar wall device with active ventilation through an *outdoor* room. The effect of heat conduction and radiation exchange was considered in evaluating a PV solar wall device through an energy analysis model. Photovoltaic amplification was attained with power output from a potentiometer through rotation of its circular knob. A PV solar wall device was further analyzed for its electrical parameters as Resistance-Capacitance (RC) electrical analog circuit. The effect of inductive and capacitive heating losses was considered in evaluating electrical characteristics of a PV solar wall device exposed to solar radiation. A definition, measurement expression and unit for an electrical *noise* were also coined.

Article

	Document title	Authors	Source	Year	Citations
437	**Effects of thermophysiological and non-thermal factors on outdoor thermal perceptions: The Tomebamba Riverbanks case**	Galindo T.,Hermida M.A.	Building and Environment , 138	2018	61

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Outdoor thermal comfort (OTC) is an indicator of urban-environmental sustainability. A better understanding of OTC requires exploring the effects of contextual, non-*thermal* and human-related factors on *thermal* sensation votes (TSV) in addition to the long-studied *thermophysiological* and microclimatic factors. In this research, the Socio-Ecological Model (SEM) structure was used for grouping *thermophysiological* and non-*thermal* factors in three environments: corporeal (level of physical activity, level of clothing, gender, age and skin tone), mental (perceived urban agreeability, perceived urban insecurity and perceived urban *noise*) and social (company, occupation and cultural background). Field surveys were performed in three representative weeks of the annual climate-type in Cuenca, Ecuador during the hottest month (January), a cooler month (July) and an intermediate month (April), surveying 2321 users of two urban sites located at representative areas of the Tomebamba riverbanks. Statistical descriptive analysis and inferential methods were employed for exploring the effect of these SEM factors on the TSV using the Physiological Equivalent Temperature (PET) calculated with RayMan1.2 from on-site measured data. The analytical results evidenced a low influence of all three SEM environments on the TSV. Additionally, differences in male and female OTC requirements were identified, and two local PET-TSV scales were calculated from (1) urban global radiation data and (2) mean radiant temperature, calculated according to ISO 7726 with Ø 150 mm black-globe-*thermometer* measurements. The main results show Acceptable Temperature Ranges (rTa) in Cuenca from 29 °C to 34 °C PET (1) and from 26 °C to 37 °C PET (2), for 85% acceptability.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
438	**Predicting integrated thermal and acoustic performance in naturally ventilated high-rise buildings using CFD and FEM simulation**	Yu X.,Zhang Q.,Kang J., Cui F.	Building Simulation, 11(3)	2018	21

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The study of ventilation windows for both natural ventilation and *noise* mitigation has drawn significant attention recently. This paper presents the numerical approaches to analyse the integrated *thermal* and *acoustical* performance of ventilation windows, for a residential building in tropical climate which employs double-layer *noise* mitigation window for natural ventilation. Given a set of *outdoor* wind conditions, the distributions of indoor flow and temperature fields are simulated using Computational Fluid Dynamics (CFD) model. The *thermal* comfort is evaluated using statistical Predicted Mean Vote (PMV) method. For the *acoustic* performance, *noise* radiation from road traffic is assumed as the *noise* source, and the *sound* insulation of building façade is simulated using Finite Element Method (FEM). From the simulation results, it is found that the *thermal* satisfaction response is closely related to the inlet wind temperature and speed, and the window opening size greatly affects the ventilation performance. From the case study in Singapore, during certain season, day/night time and with sufficient wind flow, the ventilation window can provide enough fresh air, maintain adequate *thermal* comfort and quiet *acoustic* environment for the occupants. The numerical approaches presented in this paper are applicable to general window design studies, and the simulation findings can be incorporated into green building planning. The advantages of using simulation approaches are highlighted and their limitations are discussed.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
439	**Outdoor comfort conditions in urban areas: On citizens' perspective about microclimate mitigation of urban transit areas**	Piselli C.,Castaldo V.L., Pigliautile I.,Pisello A.L., Cotana F.	Sustainable Cities and Society , 39	2018	120

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Understanding citizen environmental perception is a crucial issue to improve *outdoor* environmental and landscape quality. This paper is aimed at investigating the perspective of travelling citizens about local microclimate conditions in a transportation open-air hub of an urban district in central Italy, to propose effective mitigation strategies. Therefore, a survey was submitted to pedestrians while crossing the area to understand their actual perception of visual-*thermal*-*acoustic* conditions characterizing the *outdoor* environment, with varying weather and personal characteristics. Simultaneously, the continuous in-situ monitoring of the main environmental parameters was

	Document title	Authors	Source	Year	Citations

performed. Finally, the benefits generated by selected microclimate mitigation and landscape improvement strategies were quantitatively assessed by means of validated microclimate district models. Results of the field survey highlighted the minor tolerance of the local environment by local citizens compared to tourists, especially those coming from denser and more polluted cities. Moreover, the simulations confirmed the capability of selected microclimate mitigation strategies against anthropogenic heat, to improve pedestrians' *outdoor thermal* comfort conditions in summer, without winter penalties. In particular, the vegetation increase, according to pedestrians' request for additional green areas, combined to other solutions for sustainable landscape change, showed the most significant impact in summer overheating mitigation and urban resilience to anthropogenic climate change.

Article

440 **Thermal behavior of a translucent superinsulated latent heat energy storage wall in summertime** — Souayfane F.,Biwole P.H.,Fardoun F. — Applied Energy, 217 — 2018 — 66

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This paper investigates the *thermal* performance of a translucent solar wall providing, concurrently, storage and restitution of heat, super *thermal-acoustic* insulation and daylighting to the interior environment. The wall is composed of glazing, silica aerogel used as a transparent insulation material (TIM) and glass bricks filled with fatty acid, an eutectic phase change material (PCM). To assess the TIM–PCM wall *thermal* behavior, experimentations were conducted in-situ in a full-sized test cell located in Sophia Antipolis, southern France. Experimental data shows that the tested wall is more effective in winter and might cause overheating during the summer mainly due to solar gains and un-cycling behavior of PCM which remains in liquid state. To enhance the energy performance of the wall in summertime, a numerical model describing the heat transfer mechanisms occurring in the PCM layer in combination with the other transparent wall layers is developed. Then, the model of the wall is linked to TRNSYS software to assess the *thermal* performance of the whole building. The numerical model is validated experimentally and a good agreement is shown comparing the simulated values with the measured data for seven consecutive days in summer and winter. The importance of considering the natural convection effect in the liquid PCM is also demonstrated. Moreover, it was shown that shading devices can effectively reduce overheating while natural night ventilation decreases the indoor temperature without affecting the PCM performance since the *outdoor* temperature is always higher than the phase change temperature. The use of a glass with selective solar reflection properties depending on the season instead of the ordinary glazing is shown also to be very effective way to overcome the overheating problem. Finally, the TIM-PCM wall is tested under different climate conditions and passive solutions are given to ensure *thermal* comfort in summer season.

Book Chapter

441 **Novel Building Materials** — Fabiani C.,Pisello A.L.,Paksoy H. — Comprehensive Energy Systems, 2-5 — 2018 — 4

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The proper choice of building envelope material may optimize building energy efficiency and indoor-*outdoor* comfort conditions in the built environment, together with its environmental sustainability. Therefore, a deep and multidisciplinary understanding of building envelopes and the new potentialities associated to building materials is needed in order to design green and smart constructions. In this perspective, the present chapter concerns, firstly, basics of building physics about hygro-*thermal*, *acoustics*, and lighting performance of the built environment. Then the main building material properties and the key traditional and innovative typologies are assessed with a specific focus on energy efficiency and environmental sustainability targets.

Article • *Open access*

442 **An acoustic thermometer for air refractive index estimation in long distance interferometric measurements** — Pisani M.,Astrua M.,Zucco M. — Metrologia, 55(1) — 2018 — 14

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We present a method to measure the temperature along the path of an optical interferometer based on the propagation of *acoustic* waves. It exploits the high sensitivity of the speed of *sound* to air temperature. In particular, it

	Document title	Authors	Source	Year	Citations

takes advantage of a technique where the generation of *acoustic* waves is synchronous with the amplitude modulation of a laser source. A photodetector converts the laser light into an electronic signal used as a reference, while the incoming *acoustic* waves are focused on a microphone and generate the measuring signal. Under this condition, the phase difference between the two signals substantially depends on the temperature of the air volume interposed between the sources and the receivers. A comparison with traditional temperature sensors highlighted the limit of the latter in the case of fast temperature variations and the advantage of a measurement integrated along the optical path instead of a sampling measurement. The capability of the *acoustic* method to compensate for the interferometric distance measurements due to air temperature variations has been demonstrated to the level of 0.1 °C corresponding to 10^{-7} on the refractive index of air. We applied the method indoor for distances up to 27 m, *outdoor* at 78 m and finally tested the *acoustic thermometer* over a distance of 182 m.

Article

443 **Indoor environment and comfort of Huizhou traditional dwellings | 徽州传统民居室内环境及舒适度** Huang Z.,Yu M.,Zheng L., Gong C. Tumu Jianzhu Yu Huanjing Gongcheng Journal of Civil Architectural and Environmental Engineering, 40(1) 2018 5

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This study was conducted based on one year field measurement of indoor and *outdoor* environment parameters in a typical traditional dwelling of Huizhou in China. The indoor comfort was evaluated based on the measured multi-environment parameters. According to the analysis, the characteristic of cold winter and cool summer in Huizhou traditional dwellings is demonstrated by the measured data. It shows that the dwelling has a good indoor environment and *thermal* comfort in spring and autumn. The indoor *thermal* comfort is also good during summer due to good natural ventilation and sun shading, etc. While the indoor *thermal* comfort is not fully satisfactory during winter due to the poor air tightness. In addition, *acoustic* environment is complying with relevant standards for all the rooms. However, the visual environment can't reach the standards of modern architecture design lighting in the most of time.

Article • *Open access*

444 **Designing an air-to-air heat exchanger dedicated to single room ventilation with heat recovery** Gendebien S.,Martens J., Prieels L.,Lemort V. Building Simulation, 11(1) 2018 7

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The present paper focuses on the development steps of heat exchangers dedicated to single room ventilation unit with heat recovery (SRVHR) by proposing a numerical approach. A methodology is suggested in order to determine the best trade-off between hydraulic and *thermal* performance given a specific geometry. The methodology consists in a mapping of the coefficient of performance (COP) of the unit. The latter is defined as the ratio between recovered heat and the fan energy use, given a specific indoor/*outdoor* temperature difference. However, the energy performance should not be the only criterion to be taken into account in the frame of the design steps of a heat recovery exchanger: technical, economic and *acoustic* aspects should also be considered. This numerical methodology is illustrated by means of a real example of a newly developed heat exchanger dedicated to a SRVHR. The optimization is first performed while using a semi-empirical model (based on the use of correlations and on a spatial division of the studied heat exchanger). The semi-empirical model allows for the creation of a COP map in order to identify the most effective geometry parameters for the heat exchanger. The decision concerning the final geometry is made accounting for the so-called technical, economic and *acoustic* considerations. A discussion on some parameters needed for the COP establishment is also proposed.

Conference Paper

445 **Acoustic performance test assessment on sound generated by rainfall on multilayer-etfe-cushion systems** Chivante M.R.P.,Toso S., Moritz K. 8th International Conference on Textile Composites and 2018 0

	Document title	Authors	Source	Year	Citations
			Inflatable Structures Structural Membranes 2017		

Hide abstract Find It @ UCL View at Publisher Related documents

Ethylene-tetra-fluor-ethylene (ETFE) membrane structure presents itself as an innovative construction element, taking advantage of its flexibility, lightweight, transparency and *thermal* insulation properties. However, for large enclosed areas, concerns exist related to the *sound* insulation characteristic of the system, especially related to the internal *noise* levels generated during rainfall conditions. Therefore, aiming to both assess the effect of rain *noise* on these structural elements, as well as generating a clear picture to this condition to designers and clients, Taiyo Europe promoted a measurement campaign on a predefined set of pneumatic ETFE roof elements. The set of pneumatic cushions investigated was produced from the same 250-micron thick ETFE foil. The number of layers, number of air chambers and the level of the pressure inside the cushion was variable, following a carefully predefined test plan. The elements were tested with and without an additional rain suppressor fabric mesh on top of the specimen. The tested elements were installed on a special *outdoor* structure designed and built for this purpose. This work will describe the tests performed, as well as an assessment of the results achieved and their impact on the design of enclosed buildings cladded with ETFE pneumatic structures.

Conference Paper

446 **Using infrared HOG-based pedestrian detection for outdoor autonomous searching UAV with embedded system**

Shao Y.,Mei Y.,Chu H., ... He Y.,Zhan H.

Proceedings of SPIE the International Society for Optical Engineering , 10615, 106153Z

2018

5

Hide abstract Find It @ UCL View at Publisher Related documents

Pedestrian detection (PD) is an important application domain in computer vision and pattern recognition. Unmanned Aerial Vehicles (UAVs) have become a major field of research in recent years. In this paper, an algorithm for a robust pedestrian detection method based on the combination of the infrared HOG (IR-HOG) feature and SVM is proposed for highly complex *outdoor* scenarios on the basis of airborne IR image sequences from UAV. The basic flow of our application operation is as follows. Firstly, the *thermal* infrared imager (TAU2-336), which was installed on our *Outdoor* Autonomous Searching (OAS) UAV, is used for taking pictures of the designated *outdoor* area. Secondly, image sequences collecting and processing were accomplished by using high-performance embedded system with Samsung ODROID-XU4 and Ubuntu as the core and operating system respectively, and IR-HOG features were extracted. Finally, the SVM is used to train the pedestrian classifier. Experiment show that, our method shows promising results under complex conditions including strong *noise* corruption, partial occlusion etc.

Conference Paper • *Open access*

447 **Comparative analysis of thermal and acoustic performance of composites made from wood fibres, recycled rubber and ABS**

Georgescu S.V., Cosereanu C.

Iop Conference Series Materials Science and Engineering, 399(1), 012019

2018

2

Hide abstract Find It @ UCL View at Publisher Related documents

The paper investigates the *thermal* and *acoustic* properties of composites made from wood fibres (WF) and recycled rubber (R) crumbs and acrylonitrile butadiene styrene (ABS) shavings resulted in the particleboards' edge banding. Panels with a target density of 300 kg/m^3 were manufactured for testing the *thermal* conductivity coefficient and *sound* absorption coefficient. Mixed panels WF:ABS, R:ABS with participation rates (in %) of 10:90, 20:80, 30:70; 40:60; 50:50 and WF:R:ABS with participation rates (in %) of 5:5:90, 10:10:80, 15:15:70; 20:20:60; 25:25:50 were investigated in this paper. The experiment simulated the indoor and *outdoor* temperature conditions for the winter and summer seasons, namely 20 °C for indoor and -10 °C to 35 °C for the *outdoor* temperatures. The results show that, with the increase of ABS share, the *thermal* performance of the panel increases. The presence of WF in the composition has a good influence on the *thermal* performance of the panels, whilst the presence of rubber brings a better *acoustic* performance of the composites. The analysis of the results show that the *thermal* conductivity coefficient (λ)

	Document title	Authors	Source	Year	Citations

experimentally determined depends on the *outdoor* temperature and recorded the best value for the composite WF-ABS (0.0434 $Wm^{-1}K^{-1}$) followed by WF-R-ABS (0.0460 $Wm^{-1}K^{-1}$) and R-ABS (0.0477 $Wm^{-1}K^{-1}$). Maximum values were recorded for R-ABS structure at a temperature of 35 °C (0.0573 $Wm^{-1}K^{-1}$), but this structure recorded the best *sound* absorption coefficient (0.87).

Article • *Open access*

448 **Contemporary Study of Radio Noise Characteristics in Diverse Environments** — Palaios A.,Miteva V.M., Mahonen P. — IEEE Access, 6 — 2018 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

We present results from the first phase of our measurement campaign designed to provide information on the current radio *noise* levels. Our focus is to understand if the increasing number of users and devices with inbuilt low-cost wireless transceivers has increased the *noise* levels considerably. We also study the stochastic properties of contemporary radio *noise*. In the literature, it is almost universally assumed that radio *noise* is well modeled by a white Gaussian stochastic process, and we study how often deviations from this baseline are found. Our measurement approach is able to capture frequency- and time-domain data with very high accuracy in diverse indoor and *outdoor* locations, providing a broad overview of the influence the environment has on *noise* and man-made interference. The results cast doubts on the widely used assumptions that consider man-made interference as an extra AWGN component over *thermal noise* floor. We also discuss the issue of modeling the measured radio *noise* and we explain the major research challenges for future work.

Conference Paper

449 **Outdoor comfort simulation of complex architectural designs a review of simulation tools from the designer perspective** — Naboni E.,Silvia C., Meloni M.,Scartezzini J.-L. — ASHRAE and Ibpsa Usa Building Simulation Conference — 2018 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

The expanding design interest in *outdoor* comfort design is today supported by a few software tools. Given their different user-friendliness, modelling environments and simulation engines, a rational inquiry is how they are integrated into the digital design flow of architects and urban planners, which may include complex forms developments. Preliminary work is conducted to select architect-friendly tools that support the analysis of urban microclimate. A minimum set of criteria (amongst them the use of 3D models, the capability of calculating Mean Radiant Temperature and visual graphical outputs) led to the pre-selection of CitySim Pro, ENVI-met, Autodesk *Thermal* CFD, Grasshopper plug-ins Honeybee / Ladybug. The complex modelling experiment is conducted by simulating the *outdoor thermal* comfort of the space underneath the Rolex Learning Centre in Lausanne with each of the tools above. The paper describes and compares the principles, the procedures and the resources needed to prepare *sound* models that reduce the time of computation without compromising the quality of results. The potential applicability of tools in design is finally discussed from a user's point of view. The tools' capability of creating a 3D complex geometric model, or of importing one from a typical architectural tool, such as Rhino, is studied. Furthermore, it is debated how models can be used in the broader digital environment.

Article • *Open access*

450 **Design and development of a low-cost, portable monitoring device for indoor environment quality** — Tiele A.,Esfahani S., Covington J. — Journal of Sensors, 2018, 5353816 — 2018 — 96

Hide abstract Find It @ UCL View at Publisher Related documents

This article describes the design and development of a low-cost, portable monitoring system for indoor environment quality (IEQ). IEQ is a holistic concept that encompasses elements of indoor air quality (IAQ), indoor lighting quality (ILQ), *acoustic* comfort, and *thermal* comfort (temperature and relative humidity). The unit is intended for the monitoring of temperature, humidity, PM2.5, PM10, total VOCs (x3), CO_2, CO, illuminance, and *sound* levels. Experiments were conducted in various environments, including a typical indoor working environment and *outdoor* pollution, to evaluate the unit's potential to monitor IEQ parameters. The developed system was successfully able to monitor parameter variations, based on specific events. A custom IEQ index was devised to rate the parameter

	Document title	Authors	Source	Year	Citations

readings with a simple scoring system to calculate an overall IEQ percentage. The advantages of the proposed system, with respect to commercial units, is associated with better customisation and flexibility to implement a variety of low-cost sensors. Moreover, low-cost sensor modules reduce the overall cost to provide a comprehensive, portable, and real-time monitoring solution. This development facilities researchers and interested enthusiasts to become engaged and proactive in participating in the study, management, and improvement of IEQ.

Book Chapter

451 **Assessment of Outdoor Comfort Conditions Based on the Application of a Participative Model in Open Urban Spaces** — Monteiro L.M., Okamoto Junior J., Peixoto E.V.,... Suzuki K.M., Prata-Shimomura A.R. — World Sustainability Series — 2018 — 1

Hide abstract Find It @ UCL View at Publisher Related documents

This paper presents the development of a mobile device application for the implementation of a participative model for evaluation of open spaces. Participants would be users of the Cidade UniversitÃ¡ria Armando Salles Oliveira (CUASO) Campus. The objective is to verify their perception of open spaces with regards to *thermal* environmental conditions; *acoustics*; daylight and ergonomic, and enable quick acquisition of these opinions. Thus, the name of the application: Opine. The participative model will provide quantitative subsidies as to the number of users of open spaces and their perception of these spaces and point towards a possible calibration which represents the process of climate adaptation. Studies in the area of environmental comfort and climate in open spaces presuppose the acquisition of data pertaining to environmental conditions, the physical characterization of spaces and usersâ€™ opinions, gathered by subjective answers. The last years have shown that, despite the technology used in field researches, part of this information could be acquired in a more dynamic manner. The development of the application led us to think of it as merely a questionnaire, as it would be more direct and easy to understand. Furthermore, an application with a questionnaire could be interesting, as campus users can voice their opinion and make their criticisms (or compliments) heard. The mobile application was developed, initially, for the investigation of user comfort in open spaces of the Campus Cidade UniversitÃ¡ria Armando Salles Oliveira, in SÃ£o Paulo, Brazil, but can also be used for the analysis of other open urban spaces. The application was developed for Android systems, with the following characteristics regarded as relevant: easy to understand; clear; organized; easy to read. It should also be visually instinctive to the user, making the data acquisition process more agile and also enabling the rapid and dynamic treatment of the data, providing fast and easy further calculations based on the index of Temperature of Equivalent Perception (TEP) for the assessment of *outdoor* comfort.

Conference Paper

452 **Effects of environmental perception on thermal sensation in sub-tropical and high-density cities: A case study of Hong Kong** — Chung M.S.C.,Lau K.K.L. — Proceedings of 10th Windsor Conference Rethinking Comfort — 2018 — 2

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Outdoor thermal comfort is crucial in sub-tropical regions where summer heat stress impede *outdoor* space usage. Research on the effects of environmental perception on *thermal* comfort has increased in recent years as a result of intensified urban development and it has been found that urban geometry design can modify the relationship between climate and *thermal* comfort. Despite this, there is a lack of holistic studies focusing on the effects of qualities of urban space on *thermal* adaptation and comfort. This study investigates the relationship between environmental perception and *outdoor thermal* comfort under typical summer conditions in streets and parks in Hong Kong by conducting questionnaire survey on *thermal* sensation and environmental perception in terms of convenience, visual and *acoustic* comfort, air quality and safety. Simultaneous micrometeorological measurements were also conducted to obtained objective conditions of *thermal* comfort in designated urban spaces. A total of 1921 responses were collected between June 2017 and September 2017 in 12 locations. Overall, we found that environmental perceptions, particularly the perceived air quality, *acoustic* comfort and visual comfort, significantly affect *thermal* sensation and comfort. Improved perception of each investigated environmental parameter could lead to a substantial increase in the percentage of people feeling *thermally* comfortable.

Conference Paper

	Document title	Authors	Source	Year	Citations
453	**Shading effect and heat reflection of the green façade: Measurements of an external corridor building in Munich, Germany**	Lin H.,Musso F.,Xiao Y.	PLEA 2018 Smart and Healthy within the Two Degree Limit Proceedings of the 34th International Conference on Passive and Low Energy Architecture, 3	2018	1

Hide abstract Find It @ UCL View at Publisher Related documents

Risks of urban heat island (UHI) on the *outdoor* comfort and human health are rising in the trends of high-density urbanism. Green façade (GF) is one of the technologies which interested by researchers and architects because of the effects of the diminution on UHI, urban *acoustic*, and air pollution. This paper focuses on the shading effect of the GFs on the semi-*outdoor thermal* environment. A student dormitory with GFs is invited to investigate on a summer day in Munich, Germany. Through the measurements comparing the shaded and exposed area, results show that the Ave. and the Max. air temperatures (Temps) are decreased by 0.7℃ (W) and 2.1 ℃ (W), the Ave. surface Temp of the glazing is reduced by 1.2℃ (E), and which of the aluminium is reduced by 3.5℃ (W)in the shaded area. Results confirm the shading effect of the GFs and reveal the potential of the decreasing heat reflection on building façade to the surrounding environment.

	Book Chapter				
454	**Novel Building Materials**	Fabiani C.,Pisello A.L., Paksoy H.	Comprehensive Energy Systems Volumes 1 5, 2	2018	0

Hide abstract Find It @ UCL View at Publisher Related documents

The proper choice of building envelope material may optimize building energy efficiency and indoor–*outdoor* comfort conditions in the built environment, together with its environmental sustainability. Therefore, a deep and multidisciplinary understanding of building envelopes and the new potentialities associated to building materials is needed in order to design green and smart constructions. In this perspective, the present chapter concerns, firstly, basics of building physics about hygro-*thermal*, *acoustics*, and lighting performance of the built environment. Then the main building material properties and the key traditional and innovative typologies are assessed with a specific focus on energy efficiency and environmental sustainability targets.

	Conference Review				
455	**Proceedings of SPIE - The International Society for Optical Engineering**	[No Authors Found]	Proceedings of SPIE the International Society for Optical Engineering , 10766	2018	0

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The proceedings contain 22 papers. The topics discussed include: high operating temperature MCT discrete detector data and analysis; tunable mid-infrared optical resonator on nanopatterned chalcogenide glasses; review of the development of 2k2 IR FPAs for astronomy and space in Europe; nanostructured antireflection coatings for infrared sensors and applications; high throughput, *outdoor* characterization of silicon photovoltaic modules by moving electroluminescence measurements; study on mid-IR spectroscopy on whole blood samples for human glucose quantification applications; anomalous *thermal* expansion of mercurous halides; polarimetric hyperspectral imaging with *acousto*-optic tunable filter in the visible to shortwave infrared range; and design of a Fabry-Perot interferometer based on silicon wafer for dielectric gas sensing applications.

Book Chapter • *Open access*

	Document title	Authors	Source	Year	Citations
456	**Towards a Generic Solution for Inspection of Industrial Sites**	Hutter M.,Diethelm R., Bachmann S.,... Meyer K., Hoepflinger M.	Springer Proceedings in Advanced Robotics , 5	2018	27

Hide abstract Find It @ UCL View at Publisher Related documents

Autonomous robotic inspection of industrial sites offers a huge potential with respect to increasing human safety and operational efficiency. The present paper provides an insight into the approach taken by team LIO during the ARGOS Challenge. In this international competition, the legged robot ANYmal was equipped with a sensor head to perform visual, *acoustic*, and *thermal* inspection on an oil and gas site. The robot was able to autonomously navigate on the *outdoor* industrial facilty using rotating line-LIDAR sensors for localization and terrain mapping. Thanks to the superior mobility of legged robots, ANYmal can omni-directionally move with statically and dynamically stable gaits while overcoming large obstacles and stairs. Moreover, the versatile machine can adapt its posture for inspection. The paper additionally provides insight into the methods applied for visual inspection of pressure gauges and concludes with some insight into the general learnings from the ARGOS Challenge.

Conference Paper

457	**Outage performance analysis for outdoor vehicular visible light communications**	Lin S.-H.,Wang J.-Y.,Bao X., Li Y.	2017 9th International Conference on Wireless Communications and Signal Processing Wcsp 2017 Proceedings, 2017-January	2017	10

Hide abstract Find It @ UCL View at Publisher Related documents

Outdoor vehicular visible light communication (V2LC) has been proposed recently as a new solution to increase road safety and realize autonomous driving. This paper investigates the outage probability performance for the *outdoor* V2LC system. In the system, the main distortion is caused by the ambient-induced shot *noise* and the *thermal noise*. By considering the impacts of path loss and atmosphere turbulence, a statistical channel model is established for *outdoor* V2LC. Based on the established channel model, the probability density function of the composite channel gain is derived. Moreover, the closed-form expression of the outage probability is derived. Numerical results show that the derived theoretical expression of the outage probability is very accurate and can be used to evaluate the system performance without time-intensive simulations.

Article

458	**The evolution of human sleep: Technological and cultural innovation associated with sleep-wake regulation among Hadza hunter-gatherers**	Samson D.R., Crittenden A.N., Mabulla I.A., Mabulla A.Z.P.	Journal of Human Evolution , 113	2017	36

Hide abstract Find It @ UCL View at Publisher Related documents

Sleep is necessary for the survival of all mammalian life. In humans, recent investigations have generated critical data on the relationship between sleep and ecology in small-scale societies. Here, we report the technological and social strategies used to alter sleep environments and influence sleep duration and quality among a population of hunter-gatherers, the Hadza of Tanzania. Specifically, we investigated the effects that grass huts, *sound* levels, and fire had on sleep. We quantitatively compared *thermal* stress in *outdoor* environments to that found inside grass hut domiciles to test whether the huts function as *thermoregulated* microhabitats during the rainy season. Using physiological equivalent temperature (PET), we found that the grass huts provide sleep sites with less overall variation in *thermal* stress relative to outside baseline environments. We also investigated ambient *acoustic* measures of nighttime environments and found that *sound* significantly covaried with sleep-wake activity, with greater *sound* levels associating with less sleep. Finally, after controlling for ecological variables previously shown to influence sleep in this population, fire was shown to neither facilitate nor discourage sleep expression. Insofar as data among contemporary sub-tropical foragers can inform our understanding of past lifeways, we interpret our findings as

	Document title	Authors	Source	Year	Citations

suggesting that after the transition to full time terrestriality, it is likely that early Homo would have had novel opportunities to manipulate its environments in ways that could have significantly improved sleep quality. We further conclude that control over sleep environment would have been essential for migration to higher latitudes away from equatorial Africa.

Article • *Open access*

459 **Uncover the Degradation Science of Silicone under the Combined Temperature and Humidity Conditions** — Singh P.,Tan C.M. — IEEE Access, 6 — 2017 — 16

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In this paper, we report the mechanistic insights into *thermal* and humidity induced degradation of silicone employed in high power LEDs. High power blue and white light emitting diodes (LEDs) are used for experimentation. The silicone encapsulant of both the blue and white LEDs are degraded due to hydrolysis, likewise for the molding part of the blue LED. However, the molding part of the white LED is degraded via *thermal* oxidation. We find that lumen degradation is rapid for white LEDs, whereas material degradation is unexpectedly rapid for blue LEDs. The reasons for such differences in the degradation of the packaging materials are explained. We also found that the degradation of LEDs under high temperature alone is different from that under high temperature and humid condition, such as those used in the *outdoor* applications.

Article • *Open access*

460 **Wind noise spectra in small Reynolds number turbulent flows** — Zhao S.,Cheng E.,Qiu X., Burnett I.,Liu J.C.-C. — Journal of the Acoustical Society of America, 142(5) — 2017 — 8

Hide abstract Find It @ UCL View at Publisher Related documents

Wind *noise* spectra caused by wind from fans in indoor environments have been found to be different from those measured in *outdoor* atmospheric conditions. Although many models have been developed to predict *outdoor* wind *noise* spectra under the assumption of large Reynolds number [Zhao, Cheng, Qiu, Burnett, and Liu (2016). J. *Acoust.* Soc. Am. 140, 4178-4182, and the references therein], they cannot be applied directly to the indoor situations because the Reynolds number of wind from fans in indoor environments is usually much smaller than that experienced in atmospheric turbulence. This paper proposes a pressure structure function model that combines the energy-containing and dissipation ranges so that the pressure spectrum for small Reynolds number turbulent flows can be calculated. The proposed pressure structure function model is validated with the experimental results in the literature, and then the obtained pressure spectrum is verified with the numerical simulation and experiment results. It is demonstrated that the pressure spectrum obtained from the proposed pressure structure function model can be utilized to estimate wind *noise* spectra caused by turbulent flows with small Reynolds numbers.

Review • *Open access*

461 **Association between environmental factors and occupant health in temporary housing built after the great east Japan Earthquake of 2011: Questionnaire survey in Sendai Area** — Hasegawa K.,Yoshino H., Yanagi U.,... Osawa H., Kagi N. — Journal of Environmental Engineering Japan, 82(741) — 2017 — 8

Hide abstract Find It @ UCL View at Publisher Related documents

The Great East Japan Earthquake occurred on March 11, 2011, causing severe damage to cities in the Tohoku district along the coast of the Pacific Ocean. Many people in the Tohoku area lost their homes, and as a result, more than 50,000 temporary housing units were built in three affected prefectures. Most temporary housing units were terrace houses built closely together with subpar *thermal* insulation and energy conservation, so potential problems concerning indoor *thermal* and *acoustic* environments, as well as the air quality, are a serious concern. Therefore, the objective of the present study was to clarify the relationship between the indoor and *outdoor* environment of temporary housing and the overall living situation of temporary housing residents. We also sought to propose an optimal method for the environmental design of temporary housing and to suggest ways to improve environmental policies. We conducted a questionnaire survey regarding the environmental factors and self-reported health

	Document title	Authors	Source	Year	Citations

conditions of residents living in temporary housing in and around Sendai city in the summer of 2011 and the winter of 2012. Questionnaires were distributed by mail to 99 temporary housing units in the summer and 152 in the winter, with response rates of 77.8% and 57.9%, respectively. In addition, we performed multivariable logistic regression analysis to investigate the association between adverse health effects and influencing factors related to living in temporary housing. Based on the results, 78.7% of the respondents answered that there was visible condensation on the surfaces of doors and windows, and 46.6% answered that they had observed mold growth in the winter. About 40% of the respondents reported that they used exhaust fans in the lavatory or bathroom continuously for ventilation. On the other hand, about half of the respondents only used exhaust fans when using the toilet or taking a bath. These results suggest that air circulation and ventilation are inadequate in temporary housing. The questionnaires also included items regarding the following health-related symptoms: ocular symptoms, nasal symptoms, respiratory symptoms, dermal symptoms and mental symptoms. The results showed that the prevalence of each symptom was about 40%, and that symptom onset was within 3 months after moving into the temporary housing. Adjusted odds ratios for each health symptom in the summer and winter were calculated using a multivariable logistic regression model adjusted for age, gender, and smoking status. In the summer, the risk of ocular, nasal, respiratory and dermal symptoms increased significantly with closer proximity to industrial plants (p<0.05> In the winter, the nsk of nasal, respiratory, and dermal symptoms increased significantly with increased frequency of hanging the laundry indoors (p<0.05). These results suggest that among residents of temporary housing, the *outdoor* air environment is associated with health symptoms during the summer and the indoor air environment is associated with adverse health effects during the winter.

Conference Paper

☐ 462 **Indoor environmental quality (IEQ) of school classrooms: Case study in Malaysia** Samad M.H.A.,Aziz Z.A., Isa M.H.M. Aip Conference Proceedings , 1892, 180001 2017 10

Hide abstract Find It @ UCL View at Publisher Related documents

Thermal Comfort is one of the key criteria for occupants' comfort and productivity in a building. In schools, it is vital for a conduciveness for teaching and learning environment. *Thermal* comfort is dependent on air temperature, humidity, radiation, internal lighting, air movement, activities, clothing and climatic change and is part of the Indoor Environmental Quality (IEQ) components which have has significant effects on occupants. The main concern over energy and running cost has meant that most public schools in Malaysia are designed for natural ventilation and not air-conditioning. The building envelope plays a significant role in reducing the radiant heat and keeps the interior cooler than the *outdoor* temperature for acceptable *thermal* comfort level. The requirement of Industrial Building System (IBS) as the envelope system for school building in Malaysia could affect the role of envelope as a climate moderator. This paper is based on a research conducted on two schools in Malaysia of varied construction materials as the building envelopes to ascertain the *thermal* comfort level of the classrooms. Elements of IEQ such as air temperature, air movement, daylighting and *noise* level were taken of various classrooms to fulfill the required objectives of determining the level of quality. The data collected and analysed from the study shows that in terms of air temperature which range from 28°C to 34.5°C, the schools do not achieve the recommended comfort level for tropical climate. As for daylighting element, results also show that some classrooms suffered with too much glare whilst others had insufficient daylighting. The findings also show the unsatisfactory level of air movement in the classrooms as well as an unacceptable *noise* level exceeding the allowable threshold. This research also concluded that the use of materials and orientation in the school design are the major determinant factors towards good IEQ levels in school buildings.

Article

☐ 463 **Outage and bit error rate analysis for vehicular visible light communications** Lin S.-H.,Wang J.-Y.,Bao X., Li Y. Optical Engineering, 56(8), 086114 2017 6

Hide abstract Find It @ UCL View at Publisher Related documents

Recently, vehicular visible light communication (V2LC) is proposed as a solution to increase road safety and realize autonomous driving. The system performance for the *outdoor* V2LC system is investigated. In the system, the main distortion is caused by ambient-induced shot *noise* and *thermal noise*. With path loss and atmosphere turbulence, a statistical channel model is established for *outdoor* V2LC. Based on the channel model, the probability density

	Document title	Authors	Source	Year	Citations

function of the composite channel gain is derived. Moreover, the theoretical expressions of the outage probability and the bit error rate are derived, respectively. Numerical results show that the derived theoretical expressions are very accurate and can evaluate the system performance without time-intensive simulations.

Article

464 **Frame effects on outdoor–indoor transmission class of fixed and open glass windows** — Mediastika C.,Kristanto L., Anggono J.,Suhedi F., Purwaningsih H. — Advanced Science Letters, 23(7) — 2017 — 5

Hide abstract Find It @ UCL View at Publisher Related documents

The effect of a frame around glass fixed windows on *outdoor*-indoor transmission class (OITC) was studied here. Three frame materials were selected, i.e., timber, Aluminum and unplasticized Poly Vinyl Chloride (uPVC). The use of real open window (to differentiate it from partially open double layered window) for natural ventilation was also studied. The top hung style was selected due to common usage and the possibility of *noise* blockage by the shutter. Laboratory test complies with ASTM E-90 was employed. The study showed that weight and density, which play significant roles in *sound* insulation quality of a material was not borne out for the window frame. Here, timber as the heaviest material gave lowest OITC and transmission loss (TL) contour. This was due to the absence of sealant and rubber strips which are unusual for timber caused by large *thermal* expansion coefficient. Top hung openings of 5° and 10° showed similar transmission loss contours, but the OITC of the 5° open window was slightly higher than the 10°.

Article

465 **High-accuracy localization platform using asynchronous time difference of arrival technology** — He S.,Dong X. — IEEE Transactions on Instrumentation and Measurement , 66(7), 7869397 — 2017 — 73

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Despite extensive research efforts on ranging and localization modeling and simulation, research on practical implementations is limited. For the first time, a complete prototype based on asynchronous time difference of arrival (A-TDOA) technique is implemented in hardware. The A-TDOA technique requires neither clock synchronization between a target and anchor nodes nor wired infrastructure among anchor nodes, both of which are necessary for time of arrival and TDOA systems, respectively. All subsystems, including transmitter, receiver, antenna, and baseband processing unit, are developed from scratch and undergone significant updates for improved reliability. The implemented system has been extensively tested in an *outdoor* and indoor line of sight radio environments, and the accuracies obtained are 20.7 and 15.2 cm in 8 m × 8 m and 6 m × 6 m areas, respectively. In nonline of sight indoor environment, the achieved accuracy is 21.3 cm in 5 m × 5 m area. The comparison with the literature published to date proves the excellent quality of these results. To better understand the localization accuracy, the error sources due to *thermal noise*, hardware limitation, and radio propagation channel are identified and investigated. Mitigation methods are proposed to reduce errors. The implemented prototype supports many unique applications including cargo tracking, tourist guiding, emergency evacuation, and so on.

Article • *Open access*

466 **A novel vertical greenery module system for building envelopes: The results and outcomes of a multidisciplinary research project** — Serra V.,Bianco L., Candelari E.,... Larcher F., Schiavi A. — Energy and Buildings , 146 — 2017 — 79

Hide abstract Find It @ UCL View at Publisher Related documents

Vegetation in architecture can be considered a proper design strategy that is aimed at improving not only the performances of buildings, but also the *outdoor* climate. Different technological solutions have been proposed over the years to cover buildings with vegetation, i.e. green roofs, green walls and green balconies. A particular typology of green wall, which has recently been gaining high consensus among designers, is the vertical greenery modular system (VGMS). The positive impact of this type of technology on the performance of buildings is related to several factors, such as the façade orientation, the use of the building, climatic conditions, the type of plants, the substrates and wall assemblies, as well as mechanical and technological issues. A multidisciplinary approach is therefore needed, and different skills have to be joined together right from the early design phase in order to optimize and

	Document title	Authors	Source	Year	Citations

balance all the aspects that are involved. In this framework, a research project has been carried out in Turin (North West Italy), with the aim of developing a novel VGMS, constituted by a modular box covered with vegetation, made up of recycled/natural and highly performing materials from the energy/environmental point of view. After the design phase, the actual performance of the VGMS was assessed, through laboratory and long-term in field monitoring, and at the same time, the technological issues, biometric parameters, and the *acoustic*, *thermal* and mechanical aspects were investigated.

Article • *Open access*

467 **Meteorological effects on the noise reducing performance of a low parallel wall structure** — Van Renterghem T., Taherzadeh S., Hornikx M., Attenborough K. — Applied Acoustics, 121 — 2017 — 5

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Numerical calculations, scale model experiments and real-life implementations have shown that the insertion of a closely spaced array of low parallel walls of finite dimension beside a road is potentially useful for road traffic *noise* abatement. However, previous studies did not consider atmospheric effects. In this work, numerical techniques have been used to predict the *sound* reduction provided by a low parallel wall structure, subject to wind and temperature related atmospheric effects. Three full-wave prediction schemes show very good agreement when looking at the insertion loss of a low 6 m wide parallel wall structure, consisting of 24 regularly spaced 0.2-m high rigid walls. Meteorological effects are predicted not to deteriorate the insertion loss (relative to rigid flat ground) of the parallel wall array in the low frequency range. However, at high *sound* frequencies the insertion loss is strongly reduced by downward refraction at a distance of 50 m in case of strong wind. Consequently, overall A-weighted road traffic *noise* insertion loss will be significantly lower during wind episodes. Although weak turbulence does not alter the energy time-averaged insertion losses, strong turbulence reduces the *noise* shielding in the high frequency range also. As with conventional *noise* walls, when considering use of low parallel wall structures for *noise* reduction outdoors, even at short distances, atmospheric effects should be considered.

Article

468 **Bio Based Building Materials** — [No Authors Found] — Construction and Building Materials, 139 — 2017 — 0

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The 2017 Special Issue of Construction and Building Materials discusses 12 selected papers from the First International Conference on Biobased Building Materials, which was held in Clermont-Ferrand, France (ICBBM2015). Six papers concerning 3BM design, multi-physical properties, mechanical performances of bio-composite fibers, *acoustical* and *thermal*, durability, fire resistance and *outdoor* climate behavior of coating with 3BM. Two papers highlight the performance of 3BM made with wool and mussel shells. Three papers discussed the innovations of bio-admixtures (such as polysaccharide, humins, and starches) and interactions between polysaccharide and superplasticiser. Finally, one paper addressed the biomineralizing bacteria used in self-healing concrete.

Article

469 **Performance Analysis of Visible Light Communications-Based Hotspots in Indoor and Outdoor Environments** — Chowdhury H., Pouttu A., Katz M. — Wireless Personal Communications, 93(3) — 2017 — 4

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In this work, we study the performance of optical-based wireless hotspots in indoor and *outdoor* environments. The performance of optical-based wireless networks depends on many factors such as in point-to-point link alignment between transmitter and receiver, field of view of the receiver, semi-angle at half power of light emitting diode transmitter, background *noise* such as ambient and *thermal noise*, velocity of the user and so on. In this paper, we use the throughput-distance relationship model which has been derived analytically by taking into account above mentioned factors for evaluating the performance of VLC hotspot networks in indoor and *outdoor* environments at both, day and night conditions. Simulations are also performed to evaluate the performance of such hotspot network in the context of data download on move scenario. Simulation results reveal that there is a large impact of

	Document title	Authors	Source	Year	Citations

background *noise* on the performance of VLC hotspots. As expected, in both indoor and *outdoor* environments VLC hotspot performs better at night than at day time. Performance of VLC hotspot networks is also quantified in terms of received file size at different bit error rate requirements and velocities of mobile user. For example, in a typical indoor environment at night time a mobile user at velocity 1 m/s can download four megabyte file while passing through a VLC hotspot with a BER requirement of $10^{(-3)}$.

Article

470 **Note: A temperature-stable low-noise transimpedance amplifier for microcurrent measurement** — Xie K.,Shi X.,Zhao K., Guo L.,Zhang H. — Review of Scientific Instruments, 88(2), 026101 — 2017 — 5

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Temperature stability and *noise* characteristics often run contradictory in microcurrent (e.g., pA-scale) measurement instruments because low-*noise* performance requires high-value resistors with relatively poor temperature coefficients. A low-*noise* transimpedance amplifier with high-temperature stability, which involves an active compensation mechanism to overcome the temperature drift mainly caused by high-value resistors, is presented. The implementation uses a specially designed R-2R compensating network to provide programmable current gain with extra-fine trimming resolution. The temperature drifts of all components (e.g., feedback resistors, operational amplifiers, and the R-2R network itself) are compensated simultaneously. Therefore, both low-temperature drift and ultra-low-*noise* performance can be achieved. With a current gain of 10^{11} V/A, the internal current *noise* density was about 0.4 fA/√Hz, and the average temperature coefficient was 4.3 ppm/K at 0-50 °C. The amplifier module maintains accuracy across a wide temperature range without additional *thermal* stabilization, and its compact size makes it especially suitable for high-precision, low-current measurement in *outdoor* environments for applications such as electrochemical emission supervision, air pollution particles analysis, radiation monitoring, and bioelectricity.

Article

471 **Impact of outdoor PM2.5 on natural ventilation usability in California's nondomestic buildings** — Martins N.R., Carrilho da Graca G. — Applied Energy, 189 — 2017 — 71

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In California, the majority of office and other nondomestic buildings use mechanical cooling and ventilation even when an optimized natural ventilation (NV) system could meet cooling and fresh air requirements. Unfortunately, in most large California cities, the *outdoor* environment is contaminated with *noise*, fine particles, heat, toxic gases or, in most cases, a combination of all four. This contaminated environment has a detrimental impact on naturally ventilated buildings due to their lack of filtration and *outdoor noise* attenuation systems. This paper presents a study on the impact of airborne particle pollution on the potential for NV cooling of office buildings in California. The study uses a multi-year database of measured hourly weather data and PM2.5 data for the five largest metropolitan areas in California, representative of 90% of the state's population. The analysis is performed in two stages with increasing complexity. The first stage is a statistical analysis that identifies coincidence between high PM2.5 and *outdoor* air temperatures that are suitable for NV. In addition, this phase includes multivariable correlation to identify particular weather events or time periods that affect PM2.5 levels. The second level of analysis is more complex, using building *thermal* simulation (EnergyPlus) to perform a detailed assessment of NV potential in the five urban locations, calculating NV flow rates, resulting indoor exposure to PM2.5 and supplemental HVAC system energy consumption. The results show that using NV in moments when the outside weather is favorable can result in HVAC energy savings of 25–80%. However, limiting NV use to moments with *outdoor* particle levels below 12 $\mu g/m^3$decreases this energy saving potential to 20–60%. In addition, in the majority of the cities analyzed in this study, the use of NV leads to an increase in indoor exposure to PM2.5 of *outdoor* origin of 400–500%.

Conference Paper • *Open access*

472 **Numerical analysis of indoor thermal comfort in a cross-ventilated space with top-hung windows** — Deng X.,Cooper P.,Ma Z., Kokogiannakis G. — Energy Procedia, 121 — 2017 — 11

Hide abstract Find It @ UCL View at Publisher Related documents

	Document title	Authors	Source	Year	Citations

Natural ventilation can be used in residential buildings in several Australian climatic zones and well-designed windows have the potential to facilitate indoor *thermal* comfort by allowing occupants to control volumetric *outdoor* airflow and indoor air velocities. Top-hung window is one of the most popular window types in Australia. This paper investigates the effect of the attributes of top-hung windows (i.e. window length, aspect ratio, height above the ground, window opening angle and the fly screen porosity) and *outdoor* air conditions (i.e. *outdoor* air temperature, wind speed and direction) on indoor *thermal* comfort during cross ventilation using CFD simulations. The Taguchi method was used to design the simulation scenarios and analysis of variance was used to determine the most significant factors influencing *thermal* comfort optimisation so as to reduce the number of CFD simulation cases. For the range of parameters considered and a particular case study building, results show that *outdoor* air temperature, window height and window opening angle are the most important factors influencing indoor *thermal* comfort in this room. The optimal window configurations for indoor *thermal* comfort of the case study building are also identified using a signal-to-*noise* ratio analysis.

Conference Paper

473 **Circadian House as a vision for healthy and human-centric building design** — Roy N., Foldbjerg P., Andersen P.A., Christoffersen J. — Proceedings of 33rd PLEA International Conference Design to Thrive PLEA 2017, 1 — 2017 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

Much attention on sustainable buildings has been put on energy aspects. However, health is the most precious resource we have, and energy is only one aspect of sustainability. A primary goal for sustainability should be to sustain good human health and provide healthy living environments. This was the starting point for a series of workshops with international experts. The result is a vision to realize healthy homes that support the different biological needs of their occupants with a focus on the entrainment of circadian rhythms, sleep-wake cycles, light exposure, *thermal* comfort, air quality, and contact to *outdoor* environments. Three key principles were identified: Live in balance with nature, Adaptability and Sensibility. Furthermore, those principles are supported by ten key factors including: Variation, Stimulation/absence of stimulation, *Outdoor*/indoor relation, Light/darkness, Electrical lighting, Cool/warm, Silence/*sounds*, Rest/activity, Control and Flexibility related to seasons. The principles and guidelines presented here can be used to guide and improve the design of residential buildings of all types, including apartment buildings, and are applicable to both new and existing dwellings.

Conference Paper

474 **Object parameters optimization on pure and mixed pixels in thermal hyperspectral imagery** — Miao X., Zhang Y., Zhang J., Zhong S. — Proceedings of SPIE the International Society for Optical Engineering, 10403, 104030O — 2017 — 1

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The hyperspectral image in *thermal* infrared domains provide information, such as temperature and emissivity, about different kinds of materials. These information can be used for a wide number of applications such as mineral mapping, bathymetry, indoor and *outdoor* detection of chemicals. But because of the limitation of spatial resolution and the characteristics of *thermal* infrared sensor, there are many mixed pixels in the data, whose temperature emissivity and abundance of different components can be hard to estimate. In this paper, a new method to estimate the parameters in pure and mixed pixels is proposed based on linear and nonlinear optimization. Firstly, the standard temperature and emissivity separation (TES) algorithm is applied on pure pixels of different materials selected by supervise or unsupervised methods to get the initial temperature. Secondly, the emissivity in different bands can be retrieved by minimizing the reconstruction error, which the more accurate temperature is optimized with. The emissivity in one band is trained by the samples in the same band but in different pixels, while the temperature is trained by different bands in one pixel. Lastly, the abundance and temperature of components in mixed pixels are estimated based on a linear mixture model of the bottom of atmosphere radiance as full constraint linear optimization problem and nonlinear optimization problem. The method is also analyzed with respect to sensitivity to the *noise* and different parameters' influences on estimation errors.

	Document title	Authors	Source	Year	Citations
	Article • *Open access*				
475	**Thermal damping effect due to a green barrier which includes Arundo donax as bioclimatic element in buildings | Efecto de amortiguamiento térmico de una barrera verde de Arundo donax como elemento de bioclimatización en edificios**	Rodriguez-Salinas P., Ruiz Morales M.,Franco A., Perez-Fernandez A.R., Lobato-Calleros O.	Informes De La Construccion , 69(547), e216	2017	1

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Among the main environmental impacts of the operation of residential buildings are those due to greenhouse gases generation as a result of electric consumption of air conditioning systems. The use of vegetation systems in residential buildings represents an alternative to reduce this energy consumption. Green vegetation systems barriers are often used as protection against winds, but recently they are also being used as *acoustic* dampers. This work explores their use as *thermal* insulation systems for buildings. Specifically, we report the behavior of an Arundo donax green barrier as a bioclimatic element. The results are analyzed based on indoor and *outdoor* temperature measurement in prototype buildings, in function of the green barrier presence. Additionally Arundo donax transpiration under extreme environmental conditions was determined.

	Article				
476	**Thermal performance of double skin envelope in full scale testing module in Mexico City**	Sanchez R.B.,Ahmad S.S., Zeevaert Alcantara L.B., Flores A.V.	Pertanika Journal of Social Sciences and Humanities , 25(S)	2017	0

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The study aims to understand the *thermal* performance of a double skin envelope system in a test cell by configuring a module that has *thermal* stability with little indoor temperature variations. A rooftop observation deck and laboratory was installed at the "J" Building of the Postgraduate Unit, National Autonomous University of Mexico campus, UNAM (19°18'33.59" N, 99°11'5.73" W). This platform was designed to test materials and construction systems under the Mexico City microclimate, which consists of two full-scale testing modules. Each cubic module measures 3m x 3m x 3m. All of its facades have several layers of insulation. Indoor and *outdoor* parameters such as *thermal*, lighting, *acoustic* and air quality were measured during the experiment. The study aimed to achieve indoor *thermal* stability, benchmarked to be close to zero during the 24-hour *thermal* gradient. Findings showed that during warmer months, between May 19 and June 21, 2016, the envelope played an important role in decreasing the indoor air temperature by 7°C. The diurnal *thermal* gradient was 3.87°C.

	Book Chapter				
477	**Applications of artificial neural networks to energy and buildings**	Buratti C.,Palladino D., Orestano F.C.	Artificial Neural Networks New Research	2017	0

Hide abstract Find It @ UCL View at Publisher Related documents

Energy consumption in buildings and indoor *thermal* comfort nowadays issues in engineering applications. A deep analysis of these problems generally requires many resources. Many studies were carried out in order to improve the methodology available for the evaluation of the energy consumption or indoor *thermal* conditions; interesting solutions with a very good feedback found in the Literature are the Artificial Neural Networks (ANNs). The peculiarity of ANNs is the opportunity of simulating and resolving complex problems thanks to their architecture, which allows to identify the combination of the involved parameters even when they are in a large amount. The Artificial Neural Networks (ANNs) are very common in engineering applications for simulating the energy performance of buildings, for predicting a particular parameter, or for evaluating the indoor *thermal* conditions in specific environments. However, many different Artificial Neural Networks are available and each of them should be applied in a specific field. This chapter examines and describes the ANNs generally used in the engineering field. Studies of ANNs applied in topics such as energy consumption in buildings, gas emissions evaluation, indoor and *outdoor thermal* conditions calculation, renewable energy sources investigation, and lighting and *acoustics* applications are reported. After a

	Document title	Authors	Source	Year	Citations

brief description of the main characteristics of ANNs, which allows to focus on the main peculiarity and characteristics of this kind of algorithms, some applications shown in the Literature and applied to engineering issues are described. In the first part of the chapter an analysis of the main parameters which influence the ANN implementation in the examined papers was carried out, then some applications of ANN in energy and buildings field found in the Literature are described. In particular, the main studies were described considering five different clusters: in the first group the ANN applications to buildings and traditional energy plants are showed, in the second one the ANN implementation for the *thermal* and energy performance evaluation of renewable energy sources are reported. In the third and forth clusters the applications found in the Literature for the indoor *thermal* parameters investigation and *outdoor thermal* conditions calculation are described, while in the last one other topics investigated using ANN models such as lighting and *acoustics* issues are considered.

Conference Paper • *Open access*

478 **One-year Field Study on Indoor Environment of Huizhou Traditional Vernacular Dwellings in China** — Huang Z.,Yu M.,Zheng L., Gong C.,Wu Z. — Procedia Engineering, 205 — 2017 — 28

Hide abstract Find It @ UCL View at Publisher Related documents

To create a comfortable indoor environment is the main development direction of the current architecture based on energy conservation, which has aroused a great interest in passive design strategies of traditional vernacular dwellings. This study conducted a field measurement in a typical traditional vernacular dwelling in Huizhou to investigate the building performance. On the basis of the monitored indoor and *outdoor* environment parameters by auto-recorders, one-year evaluation is obtained in terms of four aspects, i.e. *thermal* environment, visual environment, *acoustic* environment and *thermal* comfort. Results show that the dwelling has a good indoor environment and *thermal* comfort in spring and autumn. The indoor *thermal* comfort is also good during summer due to good natural ventilation and sun shading. However, the indoor *thermal* comfort is not fully satisfactory during winter due to the poor air tightness, and the indoor relative humidity is relative high all the year round. The *acoustic* environment is found to comply with relevant standards for all the rooms. The daylight of living room and bedroom is mainly depending on the patio, which leads to insufficient visual environment as required for some time.

Article • *Open access*

479 **Theoretical and experimental evaluation of the temperature distribution in a dry type air core smoothing reactor of HVDC Station** — Yu W.,Chen X.,Pan Z., ... Bin C.,Chen T. — Energies, 10(5), 623 — 2017 — 20

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The *outdoor* ultra-high voltage (UHV) dry-type air-core smoothing reactors (DASR) of High Voltage Direct Current systems are equipped with a rain cover and an *acoustic* enclosure. To study the convective heat transfer between the DASR and the surrounding air, this paper presents a coupled model of the temperature and fluid field based on the structural features and cooling manner. The resistive losses of encapsulations calculated by finite element method (FEM) were used as heat sources in the *thermal* analysis. The steady fluid and *thermal* field of the 3-D reactor model were solved by the finite volume method (FVM), and the temperature distribution characteristics of the reactor were obtained. Subsequently, the axial and radial temperature distributions of encapsulation were investigated separately. Finally, an optical fiber temperature measurement scheme was used for an UHV DASR under natural convection conditions. Comparative analysis showed that the simulation results are in good agreement with the experimental data, which verifies the rationality and accuracy of the numerical calculation. These results can serve as a reference for the optimal design and maintenance of UHV DASRs.

Conference Paper • *Open access*

480 **Energy Efficiency Renovation in Historical Building. Case Study: Maha Chulalongkorn Building and Maha Vajiravudh Building, Chulalongkorn University, Thailand** — Rungruengsri C. — Energy Procedia, 138 — 2017 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

Document title	Authors	Source	Year	Citations

Humanities Information Center (HIC) Faculty of Arts, Chulalongkorn University, Thailand located in Maha Vajiravudh building. It is a National preserved building built since Chulalongkorn University established. This antique building made of brick and mortar in Thai contemporary architectural style. With external corridor around building, it had been used as lecture rooms with good ventilation at the upper doors and windows. In the old days, the micro climate was mostly in comfort zone during morning and evening. Hundred years passed by, university micro climate was changed. The ambient temperature has been risen up about 10 degrees Celsius during peak time of the day. Therefore, air temperature is out of comfort zone. Furthermore, building use was changed from lecture room to library. Heat sources from computers, copy machine, especially halogen lamp had been installed during renovation last few years. Air condition systems, split type system, were installed in some reading areas. The condensing units make the *outdoor* air temperature raises and blows into the building. Thailand locates in hot humid climate which is too hot and too humid exceed human comfort zone most of the time. Therefore, library users feel extremely uncomforted for *thermal*, visual, and *acoustics*. The University policy indicates that the library hall should be natural ventilation. Then, the *acoustics* problems would be excluded. The research conducted with finding all *thermal* and visual factors. It was found that appliances are the internal heat sources, and poor illumination on working plane. The inside air temperature ranged 30 to 32 degree Celsius. The illuminate on working plane was 200 to 256 lux. Appliances were relocated with exhaust heat to the openings. All halogen lamps were replaced by LED. The illuminance was improved to 400 lux on working plane. The air temperature was reduced to 2 to 3 degree Celsius. After, internal heat sources were reduced, then seeking for cool sources. There are air conditioning rooms nearby with exhaust fans which still has low air temperature. Then, all exhaust cool air was used to bypass to the library hall. The inside air temperature, now, reach to comfort zone with low mean radiant temperature (MRT). This study showed the way to make historical building to reach comfort condition.

Book Chapter • *Open access*

481 **Achieving net-zero status while emphasizing the heritage architecture landmark of the organ pipe cactus national monument in Arizona, USA**

Chalfoun N.

Wit Transactions on the Built Environment , 171

2017

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The House Energy Doctor program at the University of Arizona is aimed at advanced understanding of the theory and principles related to sustainability in green building design, energy conservation, and passive solar high-performance buildings. Faculty and students of the program are currently engaged in a multi-year effort towards accomplishing a vision and a goal that would preserve the heritage of the Organ Pipe Cactus National Monument buildings while transforming its status into a world-class first net-zero park in the nation. The project is a collaboration with a team of experts in heritage architecture from the park. During the first fiscal year of 2015, three major tasks have been accomplished that include the development of seven net-zero design schemes for the visitor center using energy performance simulation and analysis and the integration of advanced environmental systems. To achieve these goals, 3 courses, 7 faculty members, and over 32 students collaborated on the work through first, an advanced integrative environmental design and research studio that focuses on regional sustainable energy efficient high performance buildings and emphasizes on latest environmental technologies for indoor and *outdoor* spaces. The studio investigated the heritage conservation aspects of the visitor center pertinent to Mission 66 standards. Second, a House Energy Doctor course that provides service while fostering hands-on inquiry based learning. Buildings' energy conservation and passive solar design were investigated through up-to-date computer energy simulation techniques and Level III energy audits and field investigation of the buildings. And, third, a luminous, *thermal*, and *acoustic* laboratory-based course where theoretical learning is verified by empirical research and experimentation. Many of the technologies that were prescribed to buildings on the Organ Pipe Cactus National Monument campus were tested in the laboratory. This paper demonstrates the methods used during the first year and illustrates the findings and results of the research.

Review • *Open access*

482 **The role of flooring materials in health, comfort and performance of children in classrooms**

Bluyssen P.M.

Cogent Psychology, 3(1), 1268774

2016

7

	Document title	Authors	Source	Year	Citations

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From previous and recent studies, it is clear that an indoor environment of a school may affect the health, comfort and performance of schoolchildren. Besides the occupants and their activities, the windows and the *outdoor* environment and the type of ventilation system, the interior (furnishing incl. flooring materials) is an important contributor to that indoor environment quality. It seems that flooring materials (having the largest surface area) are able to effect indoor air, *acoustical*, lighting and *thermal* quality. This paper focuses on the role of flooring materials in health, comfort and performance of children in classrooms, what do we know and which directions of research are needed.

Article

483 **Investigation of the thermal adaptability for a mobile cold atom gravimeter** — Wang Q.-Y., Wang Z.-Y., Fu Z.-J., Lin Q. — Chinese Physics B, 25(12), 123701 — 2016 — 10

Hide abstract Find It @ UCL View at Publisher Related documents

The cold atom gravimeter offers the prospect of a new generation of inertial sensors for field applications. We accomplish a mobile atom gravimeter. With the compact and stable system, a sensitivity of 1.4×10^{-7} g•Hz$^{-1/2}$ is achieved. Moreover, a continuous gravity monitoring of 80 h is carried out. However, the harsh *outdoor* environment is a big challenge for the atom gravimeter when it is for field applications. In this paper, we present the preliminary investigation of the *thermal* adaptability for our mobile cold atom gravimeter. Here, we focus on the influence of the air temperature on the performance of the atom gravimeter. The responses to different factors (such as laser power, fiber coupling efficiency, etc.) are evaluated when there is a great temperature shift of 10 °C. The result is that the performances of all the factors deteriorate to different extent, nevertheless, they can easily recover as the temperature comes back. Finally, we conclude that the variation of air temperature induces the increase of *noise* and the system error of the atom gravimeter as well, while the process is reversible with the recovery of the temperature.

Article

484 **Theoretical and experimental analysis of a solar thermoelectric power generation device based on gravity-assisted heat pipes and solar irradiation** — Zhang Z., Li W., Kan J., Xu D. — Energy Conversion and Management, 127 — 2016 — 39

Hide abstract Find It @ UCL View at Publisher Related documents

Solar *thermoelectric* power generation has been widely used to solve the power supply limitation issue for low-power wireless sensors because of its light weight, high reliability, low cost, lack of *noise*, and environmental friendliness. A solar *thermoelectric* power generation system based on gravity-assisted heat pipes and solar radiation is devised in this paper, and its behavior is continuously measured in realistic *outdoor* circumstances. The effects of key parameters, including solar luminous flux, load resistance, a proportional coefficient, and a relative Seebeck coefficient, are analyzed. Related experimental results show that the device can output a voltage of 1057 mV and an electrical current of 343 mA, resulting in an output power of 362.56 mW. With a stable external energy conversion module under aluminous flux of 7.81×10^4 lx, the voltage converted from the nature solar radiation is boosted from 1057 mV to 4.40 V, which meets the rated operating voltage of low power consumption components, such as low-power wireless sensors and ZigBee modules. An economic analysis of the system shows that the solar *thermoelectric* power generation device is both economically and technically competitive when it is applied in a low-voltage wireless sensor network.

Article

485 **Using the ICF's environmental factors framework to develop an item bank measuring built and natural environmental features affecting persons with disabilities** — Heinemann A.W., Lai J.-S., Wong A., ... Miskovic A., Gray D.B. — Quality of Life Research, 25(11) — 2016 — 23

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Objectives: To develop a measure of natural environment and human-made change features (Chapter 2 of the international classification of functioning, disability, and health) and evaluate the influence of perceived barriers on

	Document title	Authors	Source	Year	Citations

health-related quality of life. Methods: A sample of 570 adults with stroke, spinal cord injury, and traumatic brain injury residing in community settings reported their functioning in home, *outdoor*, and community settings (mean age = 47.0 years, SD = 16.1). They rated 18 items with a 5-point rating scale to describe the influence of barriers to moving around, seeing objects, hearing *sounds*, hearing conversations, feeling safe, and regulating temperature and indicated whether any difficulties were due to environmental features. We used Rasch analysis to identify misfitting items and evaluate differential item functioning (DIF) across impairment groups. We computed correlations between barriers and patient-reported outcomes measurement information system (PROMIS) social domain measures and community participation indicators (CPI) measures. Results: The 18 items demonstrated person reliability of.70, discriminating nearly three levels of barriers. All items fit the Rasch model; impairment-related DIF was negligible. Ceiling effects were negligible, but 25 % of the respondents were at the floor, indicating that they did not experience barriers that they attributed to the built and natural environment. As anticipated, barriers correlated moderately with PROMIS and CPI variables, suggesting that although this new item bank measures a construct that is related to participation and health-related quality of life, it also captures something unique. Known-groups validity was supported by wheelchair users reporting a higher level of barriers than did ambulatory respondents. Conclusions: Preliminary evidence supports the reliability and validity of this new measure of barriers to the built and natural environment. This measure allows investigators and clinicians to measure perceptions of the natural environment and human-made changes, providing information that can guide interventions to reduce barriers. Moderate relationships between barriers and PROMIS and CPI variables provide support for the measurement and theory of environmental influences on social health and participation.

Article • *Open access*

☐ 486 **A global modelling approach of natural ventilation with acoustic and daylighting constraints** — Cui S.,Perret-Gentil M., Gourdon E.,... Stabat P., Marchio D. — International Journal of Ventilation, 15(3-4) — 2016 — 8

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Natural ventilation helps to cool buildings in summer without energy consumption, but its application may be restricted by *acoustic* and glare constraints. Here, a modelling approach integrating different physical aspects was developed and connected to TRNSYS. Different control strategies of glazed openings were defined and simulations were performed with an office room and a detached family house. We find the natural ventilation potential is largely reduced due to *acoustic* and visual comfort constraints. An *acoustic* shutter, which facilitates more frequent window opening during occupancy by decreasing *outdoor noise* transmission, was equally studied in detail with the help of this modelling approach. The *acoustic* shutter improves the *acoustic* and *thermal* comforts for intelligent control strategy and is an attractive solution to enhance the natural ventilation potential under a moderate noisy environment.

Conference Paper • *Open access*

☐ 487 **Experimental study on HVAC sound parameters** — Bujoreanu C.,Benchea M. — Iop Conference Series Materials Science and Engineering, 147(1), 012051 — 2016 — 10

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HVAC system represent major source of buildings internal *noise* and therefore they are designed to provide a human *acoustic* comfort besides the *thermal* and air quality requirements. The paper experimentally investigates three types of commercial air handler units (AHU) with different ducts cross-section sizes and inlet-outlet configuration. The measurements are performed in an anechoic room. The measurements are carried out at different fan's speeds, ranging the power-charge from 30-100% while the duct air flow is slowly adjusted from full open to full closed, between 0-500 Pa. The *sound* pressure levels of the radiant units are rated using NR curves. Also, the supply and the *outdoor* ducts *sound* levels are compared in order to point the frequencies where the *noise* must be reduced. Third-octave band analysis of random *noise* of an air handling unit from a HVAC system is realized, using measurement procedures that agrees the requirements of the ISO 3744:2011 and ISO 5136:2010 standards. The comparatively results

	Document title	Authors	Source	Year	Citations

highlight the effects of the geometry, air flow pressure and power-charging dependencies upon the *sound* level. This is the start for a *noise* reduction strategy.

Conference Paper

488 **Higher sound classes possible also in areas with traffic noise, combining high façade sound insulation with fresh air supply and efficient thermal control**

Simmons C.

Proceedings of the Inter Noise 2016 45th International Congress and Exposition on Noise Control Engineering Towards A Quieter Future

2016

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Sweden has recently introduced a new governmental regulation that allows dwellings close to noisy roads, railways and industries. However, the buildings must then provide at least one so called "quiet side", which is defined as a façade with $L_{pAeq,24h,freefield}$ < 55 dBA, corresponding to a L_{DEN} of about 58 dBA road traffic *noise*. However, looking close to the results of some questionnaire surveys, it becomes apparent that their results may be interpreted in a new way. The role of the façade *sound* insulation seems to have been systematically underestimated, as well as the effects of poor indoor air quality and over-temperature. If dwellings would provide high *sound* insulation in the façades as well as clean air supply and comfortable temperature at all times, occupants would not be forced to sleep with their windows opened and therefore not need a quiet side to the same extent. Thus, such buildings could provide satisfying indoor *noise* conditions also in areas with heavy traffic in all directions, where no quiet side can be realized. However, there should still be *outdoor* areas designed for social activities and rest that are easily accessed, e.g. on top of the roof or in a nearby space. These conclusions will be suggested to the working group ISO TC 43/SC 2/WG 29, working on a new international standard on *sound* classification of dwellings (ISO/NP 19488).

Conference Paper

489 **Analysis of indoor environmental quality influence toward occupants' work performance in Kompleks Eureka, USM**

Zainon M.R.,Baharum F.,Seng L.Y.

Aip Conference Proceedings , 1761, 020110

2016

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The indoor environment much more important for people health and comfort than the *outdoor* environment. This scenario would make the performance of occupants at their work more important than energy costs in the building. So, this task is to upgrade indoor environmental quality conditions for comfort and work performance of occupants in Kompleks Eureka, USM while conserving energy of the building.. Recent studies have shown an important impact of the indoor *thermal* environment on occupants' work performance. Also studies on occupants medical leave show a very high loss of work time and working performance, which have important economical consequences for companies. The paper will mainly dealing with the indoor environmental qualities, such as *thermal* comfort level, air quality, lighting, and *acoustic* quality. The studies before showing that comfortable room temperatures, increased air ventilation above normal recommendation, comfortable *acoustic* surrounding will increases the work performance of occupants in Kompleks Eureka, USM.

Article

490 **Detection method of porcelain insulator contamination grade based on infrared-thermal-image**

Fang J.,Wang J.,Yang L., ... Han J.,Guo S.

Nongye Gongcheng Xuebao Transactions of the Chinese Society of Agricultural Engineering , 32(15)

2016

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Document title	Authors	Source	Year	Citations

With the popularization of high or extra-high voltage power transmission and the continuous improvement of grade of wire and voltage, some power grids frequently have pollution flashover accidents. Therefore, it is very important to effectively and accurately obtain the insulator contamination grade for the safe operation of power grid. Currently, major detecting methods of insulator contamination grades include equivalent salt deposit density method, leakage current method, frequency spectrum method, and so on. Although above mentioned methods are feasible, basically, all of them mainly focus on the treatment of single insulator without paying attention to the entire group of insulators' heating features and appearance characteristics of the heating zone. Therefore, according to the heating features of polluted insulators, a new detecting method of porcelain insulator contamination grade, which was based on color image treatment, was proposed under the purpose of simplifying the procedures as well as reducing the misdetection ratio of the present detecting method. It meant that the assessment of samples' contamination grades would be performed under a color-image-based treatment on the obtained insulator infrared *thermogram*. Firstly, the infrared *thermograms*, which were taken at the scene, were denoised by two-step method (Gaussian *noise* was denoised by the wavelet adaptive diffusion after the impulse *noise* was denoised by the median filtering method) to obtain the filtered insulator *thermogram*. Secondly, they were segmented based on the differences of the component R, G, and B respectively shown by the normal insulators, the broken insulators, and the corresponding infrared *thermograms*. Although the traditional OTSU segmentation algorithm is simple and effective, it is not suitable to apply on the image segmentation without clear bimodal histogram. Therefore, the presented paper completed the image threshold segmentation by combining the characteristics of wavelet transform and OTSU. The modified OTSU was adaptive to choose the optimal threshold value, and improved the obtained image quality after the segmentation. However, the binary images obtained from the segmentation still had lots of *noise* points and cavities. Therefore, the mathematical morphology of the obtained binary image was modified twice in order to obtain a polluted area of the insulator. Thirdly, 5 feature components, including ratio of area to perimeter, average value, extreme value, standard deviation as well as major-minor axis ratio of the smallest outer ellipse, were extracted from the R channel of the polluted area by taking advantage of a statistical method, and the back propagation (BP) neural network was trained. During the training, the transfer function of the hidden layers was the logarithmic transfer function, logsig. The neural transfer function of the output layers was the linear activation function, purelin, which regarded image characteristic values as the input of the network, and the salt deposit density as the output that would be transformed to the corresponding contamination grade. Consequently, a detecting model of contamination grades was constructed on the basis of the color image. Finally, the running conditions of the insulators in the Qitaihe Power Supply Company were tested, among which the relative humidity, one of the shooting conditions, was at 78%-91% in the experiment. However, the temperature of the experiment was adjusted as 10-30℃ because the infrared *thermogram* was hardly affected by the shooting temperature. The contamination grade of each insulator was recorded according to the regulations of GB/T 5582-4993 when each infrared *thermogram* was shot. A total of 500 groups of infrared *thermograms* were obtained from XP-70 porcelain insulator with 0, I, II, III and IV polluted grade respectively, which were selected as experimental samples. This system was tested repeatedly, and the overall detecting precision reached above 91%. The experimental result shows that the insulator contamination grades are more accurate when the appearance characteristics of polluted insulators and the features of infrared *thermogram* are combined during the detection. It can also be noted that such a method can be served as the foundation of the detection of insulator contamination grades in the complicated *outdoor* environment.

Conference Paper

☐ 491 **Path loss channel models for 5G cellular communications in Riyadh city at 60 GHz**

Sulyman A.I.,Alwarafy A., Seleem H.E.,Humadi K., Alsanie A.

2016 IEEE International Conference on Communications Icc 2016, 7510953

2016

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This paper presents propagation path loss channel models developed from real-field measurement campaigns that were conducted in indoor and *outdoor* Line-of-Site (LOS) propagation environments in Riyadh city, Saudi Arabia, using highly directional antennas at 60 GHz. The setup used in these measurements emulates the future fifth-generation (5G) cellular systems for both access and backhaul services, as well as for Machine-to-Machine (M2M) communications. We used our measurement data to develop the corresponding large-scale propagation path loss models at 60 GHz, using the log-distance and the floating intercept modeling approaches. It is shown that cellular

	Document title	Authors	Source	Year	Citations

radio links can be established in short-range distances up to 134 m indoors, and up to 77 m outdoors, when employing highly directional antennas at both the transmitter and receiver sides. It is also shown that path loss at 60 GHz in hot and sunny weather during the day, is higher than those obtained in cool and clear weather at night. This is partly due to solar radio *noise* effect arising from the intense solar radiation that characterizes summer afternoon in Riyadh city, which can cause a decrease in carrier-to-*noise* ratio at the input of receiving antennas. It is also partly due to the increase in *thermal noise* when electronics components in the measurement device become hot. The results presented here are thus very useful in 5G cellular design and infrastructure planning in the gulf region, where daytime temperature could reach 43° C or more.

Article • *Open access*

492 **Degradation Physics of High Power LEDs in Outdoor Environment and the Role of Phosphor in the degradation process** — Singh P.,Tan C.M. — Scientific Reports, 6, 24052 — 2016 — 76

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A moisture-electrical-temperature (MET) test is proposed to evaluate the *outdoor* reliability of high power blue LEDs, with and without phosphor, and to understand the degradation physics of LEDs under the environment of combined humidity, temperature and electrical stresses. The blue LEDs with phosphor will be the high power white LEDs. Scanning *acoustic* microscopy is used to examine the resulted delamination during this test for the LEDs. The degradation mechanisms of blue LEDs (LEDs without phosphor) and white LEDs (LEDs with phosphor) are found to be different, under both the power on (i.e. with 350 mA through each LED) and power off (i.e. without current supply) conditions. Difference in the coefficient of *thermal* expansion between the molding part and the lens material as well as the heat generated by the phosphor layer are found to account for the major differences in the degradation mechanisms observed. The findings indicate that the proposed MET test is necessary for the LED industry in evaluating the reliability of LEDs under practical *outdoor* usage environment.

Article

493 **Green roofs for energy conservation and sustainable development: A review** — Vinod Kumar V., Mahalle A.M. — International Journal of Applied Engineering Research, 11(4) — 2016 — 5

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Global warming has become a threat of our time to the existence of habitat on this earth which has made a clear impact on the level of energy and water consumption. No doubt, increase in the ambient temperature increases indoor and *outdoor* temperature level of the buildings which catalyzes the use of energy and cost intensive mechanical air-conditioning systems. Depletion of fossils also points to alternate energy sources which are to be essentially sought for. Hence thirst for alternate energy is a passion of time for the conducive existence of living creatures in this globe. Research in this area has attained a faster pace where much time and money have been spent to ensure the coexistence of man and living things. Green roof tops belong to such an idea for cutting down the energy consumption and enhancing the comfort level of commercial and residential buildings. A number of collaborative research programmes are underway to reduce the heat in leak to the building and to improve the indoor conditions. Basically green roof technology is a passive cooling system that resists the heat penetrated due to solar radiation. In green roofs, solar radiation is balanced by sensible and latent heat fluxes from the plants and the soil surfaces and with conduction through the soil substrate. Efficiency of green roof depends on the climatic conditions particularly temperature. Green roofs also play an important role in reducing CO2 in atmosphere, used a medium for storm water runoff, vegetation and reducing the *noise* level in the building. This article critically reviews the research undertaken on green roof buildings *thermal* potential. It is an effort to shed light into the potential application of green roofs with an emphasis on energy conservation. The cooling potential and *thermal* performance of intensive and extensive green roofs, green roofs with solar shading and *thermal* insulation for commercial and residential buildings are also analyzed. This article critically reviews the various research works on energy efficiency and benefits of green roofs around the world.

Article

	Document title	Authors	Source	Year	Citations
494	**BOSSA: A multidimensional post-occupancy evaluation tool**	Candido C.,Kim J., De Dear R.,Thomas L.	Building Research and Information , 44(2)	2016	133

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Research findings point to three methodological shortcomings of current post-occupancy evaluation (POE) tools: (1) contextualizing results, (2) adding instrumental data side by side to survey results and (3) producing meaningful feedback to its key stakeholders. This paper introduces the holistic BOSSA (Building Occupants Survey System Australia) and tools developed under this project's scope in close collaboration with industry. It aims to present and discuss the statistical analysis used in the BOSSA tool, distilling the survey results down to nine indoor environmental quality (IEQ) dimensions and their association with four overall indices. Principal component analysis (PCA) extracted nine IEQ dimensions that were uncorrelated with each other: spatial comfort, indoor air quality, personal control, *noise* distraction and privacy, connection to the *outdoor* environment, building image and maintenance, individual space, *thermal* comfort, and visual comfort. Four separate multiple regression analyses were conducted, one for each global evaluation item as an independent variable: work area comfort, building satisfaction, productivity and health. This statistical analysis provided the rational basis of BOSSA's scoring system, designed to simplify how occupant survey results are communicated to key stakeholders from the property industry and researchers.

Article

495	**Research on ground-based LWIR hyperspectral imaging remote gas detection**	Zheng W.-J.,Lei Z.-G., Yu C.-C.,... Liao N.-F., Su J.-H.	Guang Pu Xue Yu Guang Pu Fen Xi Spectroscopy and Spectral Analysis , 36(2)	2016	5

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The new progress of ground-based long-wave infrared remote sensing is presented, which describes the windowing spatial and temporal modulation Fourier spectroscopy imaging in details. The prototype forms the interference fringes based on the corner-cube of spatial modulation of Michelson interferometer, using cooled long-wave infrared photovoltaic staring FPA (focal plane array) detector. The LWIR hyperspectral imaging is achieved by the process of collection, reorganization, correction, apodization, FFT etc. from data cube. *Noise* equivalent spectral radiance (NESR), which is the sensitivity index of CHIPED-1 LWIR hyperspectral imaging prototype, can reach 5.6×10^{-8} W·(cm^{-1}·sr·cm^{2})$^{-1}$ at single sampling. The data is the same as commercial temporal modulation hyperspectral imaging spectrometer. It can prove the advantage of this technique. This technique still has space to be improved. For instance, spectral response range of CHIPED-1 LWIR hyperspectral imaging prototype can reach 11.5 μm by testing the transmission curve of polypropylene film. In this article, choosing the results of *outdoor* high-rise and diethyl ether gas experiment as an example, the authors research on the detecting method of 2D distribution chemical gas VOC by infrared hyperspectral imaging. There is no observed diethyl ether gas from the infrared spectral slice of the same wave number in complicated background and low concentration. By doing the difference spectrum, the authors can see the space distribution of diethyl ether gas clearly. Hyperspectral imaging is used in the field of organic gas VOC infrared detection. Relative to wide band infrared imaging, it has some advantages. Such as, it has high sensitivity, the strong anti-interference ability, identify the variety, and so on.

Conference Paper • *Open access*

496	**A Maternity Hospital in Shenyang Indoor Environment Performance Influence Factor Analysis**	Yu L.,Chen S.,Jia J.,Ding R., Wang S.	Procedia Engineering, 146	2016	2

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In recent years, the city air pollution level of $PM_{2.5}$, ozone, nitrogen oxides and sulfur dioxides in our country occupies first place. Indoor environmental problems caused by *outdoor* air pollution are increasingly significant. Indoor environment of hospital which is a special public building has special requirement. The feature of maternity hospital is crowded, heavy energy consumption, weakness of maternal and neonate, generate the sick and the patients need the better indoor environment quality. Therefore, this paper takes hospital building as research object by using questionnaire method, which analyzes the indoor comprehensive environmental performance of hospital, such as the

	Document title	Authors	Source	Year	Citations

air quality, the *thermal* environment, the light environment, and the *acoustic* environment. Results show that most of the patients is satisfied with the indoor comprehensive environmental performance of hospital, and only a small part is dissatisfied with the *sound* environment and the *outdoor* vision. This paper provides a research base for establishing a complete evaluation system and method of comprehensive indoor environmental performance of maternity hospital, which will help in building environmental monitoring systems and improving the level of public participation in the construction of hospital management.

Conference Paper

497 **A UML/MARTE extension for designing energy harvesting in wireless sensor networks** — Saida R.,Kacem Y.H., Ben Saleh M.S.,Abid M. — Smart Innovation Systems and Technologies , 55 — 2016 — 1

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Power supply is the major concern in the wireless sensor networks (WSNs) applications. Currently, the node lifetime is limited by a battery supply which is a short lifetime, unmanageable and uneconomical. Energy Harvesting was proposed as a promising alternative to power sensor nodes in many application fields. Several energy harvesting concepts are considered in WSNs systems such as solar, vibration, *thermal*, kinetic, *acoustic noise*, radio frequency (RF), biochemical and hybrid energy sources. The existing modeling design for the power supply section of sensor nodes is limited to the design of solar energy harvesting method which is mostly employed in *outdoor* applications with sufficient sun light. However, other energy harvesting concepts are potential ambient sources of energy which offer an enough amount of power for sensor nodes. In this paper, we propose a high level methodology based on UML/MARTE standard to model specifications of outlined energy harvesting devices in the WSNs. We define new packages extending the “HW_Harvesting” package which is extending the “HW_PowerSupply” package. A case study of a WSNs system regarding leak detection in water pipeline monitoring is used to evaluate the practical use of our proposal.

Conference Paper

498 **Influence of atmospheric turbulence on optical measurement: A case report and review of literature** — Yao L.,Shang Y.,Fu D. — Proceedings of SPIE the International Society for Optical Engineering , 9796, 979604 — 2016 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

When videogammetry (optical measurement) was carried *outdoor* or under cruel indoor circumstance, the results would be inevitably affected by the atmosphere turbulence. As a result, the precision of surveying was destroyed. The field of air turbulence's impact on optical measurement was neglected by scholars for a long time, the achievements massed about laser optics and optical communications. The mostly adapted method was *noise* filtration when the pixel wandering could not be rejected in engineering application, which got little improvement on usual conditions. The principle of influence under atmospheric turbulence on optical measurement is presented in this paper. And experiments data and applications are carried out to announce the impact of atmospheric turbulence. Combining with relevant researches, some essential issues and expectations of the atmospheric turbulence research are proposed.

Conference Paper • *Open access*

499 **Self-adaptive Multi-purpose Modular Origami Structure** — Andreozzi S.,Bessone G.I., Poala M.B.,... Papetti V., Mariani S. — Procedia Engineering, 161 — 2016 — 6

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A research line in architecture and interior design has been focused for years on the selection of materials with properties specifically tailored for light, *thermal* and *acoustic* comforts. An adaptive origami-based structure is here proposed in order to overcome the limited capability of a single material to adjust its response to environmental changes. Such structure is highly flexible, with applications ranging from indoor to *outdoor* environments. We focus

	Document title	Authors	Source	Year	Citations

on building facades, to show some results relevant to a small-scale prototype aimed to provide shading to the sunlight.

Conference Paper

500 **Multi-waveband simulation-based signature analysis of camouflaged human dismounts in cluttered environments with TAIThermIR and MuSES**

Packard C.D.,Klein M.D.,Viola T.S.,Hepokoski M.A.

Proceedings of SPIE the International Society for Optical Engineering, 9997, 99970I

2016

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The ability to predict electro-optical (EO) signatures of diverse targets against cluttered backgrounds is paramount for signature evaluation and/or management. Knowledge of target and background signatures is essential for a variety of defense-related applications. While there is no substitute for measured target and background signatures to determine contrast and detection probability, the capability to simulate any mission scenario with desired environmental conditions is a tremendous asset for defense agencies. In this paper, a systematic process for the *thermal* and visible-through-infrared simulation of camouflaged human dismounts in cluttered *outdoor* environments is presented. This process, utilizing the *thermal* and EO/IR radiance simulation tool TAIThermIR (and MuSES), provides a repeatable and accurate approach for analyzing contrast, signature and detectability of humans in multiple wavebands. The engineering workflow required to combine natural weather boundary conditions and the human *thermoregulatory* module developed by *ThermoAnalytics* is summarized. The procedure includes human geometry creation, human segmental physiology description and transient physical temperature prediction using environmental boundary conditions and active *thermoregulation*. Radiance renderings, which use Sandford-Robertson BRDF optical surface property descriptions and are coupled with MODTRAN for the calculation of atmospheric effects, are demonstrated. Sensor effects such as optical blurring and photon *noise* can be optionally included, increasing the accuracy of detection probability outputs that accompany each rendering. This virtual evaluation procedure has been extensively validated and provides a flexible evaluation process that minimizes the difficulties inherent in human-subject field testing. Defense applications such as detection probability assessment, camouflage pattern evaluation, conspicuity tests and automatic target recognition are discussed.

Article • *Open access*

501 **A survey on annual change of window opening in detached houses a method to monitor open-width of sliding window and the annual characteristics of open-width**

Hayashi M.,Sugawara M.,Honma Y.,Kurabuchi T.

Journal of Environmental Engineering Japan, 81(719)

2016

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The aim of this study is to estimate the influence of indoor air quality upon the dweller's health in detached houses. The indoor air quality influences the dwellers health continuously for log times. Especially the long-term exposure of chemical compounds increase the risk of multiple chemical sensitivity. Therefore it is very important to know the annual characteristics of indoor air quality. The annual change of indoor quality depends on the annual change of emission and that of ventilation. The ventilation rate depends on not only the ventilation system but also the dweller's habit of operating windows. The window operation is influenced by not only indoor conditions but also the dwellers living style, the environment of circumstance and the other factors. The sliding windows were used in common detached houses. The dwellers can adjust the open wide of sliding window considering any factors, for example indoor *thermal* condition, air quality *sound*, and these *outdoor* conditions. Therefore the state of the open width is thought to be varied and the prediction is not easy. In order to know the annual characteristics of indoor air quality and to predict the long term exposure of pollutant, the following studies are carried out. An instrument to measure the open-width of sliding window was developed and the open-widths were measured in six detached houses through a year. The results are the followings. 1. An instrument using mechanical system can measure open width of sliding window continuously through a year. 2. The open-width depends on not only indoor temperature but also the daily dweller's schedule ant the other factors. 3. The annual reaction of open-width to the indoor temperature is behind when the room is not cooled normally. 4. When the indoor temperature is high in summer, the room is cooled normally and the open-width is large when the room is not cooled. 5. When the indoor temperature is not high in

	Document title	Authors	Source	Year	Citations

summer, the room is not cooled normally and the windows open for long time and open-width is operated from middle to large when the room is not cooled. In this study the basic annual characteristics of operating windows became clear. The survey on the window opening in more houses, the modeling of window opening and the prediction of annual indoor air quality are necessary in future.

Article

502 **Analysis of indoor environmental quality influence toward occupants' work performance in Kompleks Eureka, USM** — Baharum F.,Zainon M.R.,Seng L.Y.,Nawi M.N.M. — International Journal of Supply Chain Management, 5(4) — 2016 — 2

Hide abstract Find It @ UCL View at Publisher Related documents

Indoor environment much more important for people health and comfort than the *outdoor* environment for most people who spend most of their living times in industrialized countries lives in an artificial indoor environment. Along these lines, this research is to upgrade indoor environmental quality conditions for comfort and work performance of occupants in Kompleks Eureka, USM while conserving energy of the building. Detailed field investigations of the indoor environment in large office buildings in many parts of the world have documented that the indoor environmental quality is typically rather mediocre, with many people dissatisfied and many suffering from sick-building syndrome symptoms. Recent studies have shown an important impact of the indoor *thermal* environment on occupants' work performance. Also studies on occupants medical leave show a very high loss of work time and working performance, which have important economical consequences for companies. The paper will mainly dealing with the indoor environmental qualities, such as *thermal* comfort level, air quality, lighting, and *acoustic* quality. The studies before showing that comfortable room temperatures, increased air ventilation above normal recommendation, comfortable *acoustic* surrounding will increases the work performance of occupants in Kompleks Eureka, USM.

Article

503 **Integrated improvement of occupants' comfort in urban areas during outdoor events** — Rossi F.,Anderini E.,Castellani B.,Nicolini A.,Morini E. — Building and Environment, 93(P2) — 2015 — 63

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Outdoor public spaces contribute to improve the quality of life in urban areas available for *outdoor* activities, leisure and social interactions. The vitality and the functionality of *outdoor* urban spaces depend significantly on comfort conditions of people in these places. For this reason, the analysis of comfort conditions in urban public environments is important also in relation to the intended use of the considered spaces.Almost all the studies about *outdoor* comfort mainly focus on the *thermal* comfort; other surveys consider aspects, such as *acoustic* and visual quality. However, generally *thermal*, *acoustic* and visual comfort conditions are treated separately. There are cases in which it is appropriate to deal with the three aspects simultaneously (e.g. spaces for temporary urban events), in order to guarantee global comfort of the occupants. To enhance simultaneously *acoustic*, lighting and *thermal* comfort conditions of an *outdoor* public environment, an integrated approach is provided and a simple, innovative and low cost solution is studied. The suggested solution is a system of suspended modular and temporary white *acoustic* sails appropriately positioned during *outdoor* entertainment events on summer. The particular case study of Matteotti Square in Città di Castello, Italy is analyzed. The results showed that an improvement on global comfort conditions of the square may be obtained through the use of proper architectural solutions and materials, which additionally result easily applicable to similar (for issues, geometry and destination of use) *outdoor* environments.

Article

504 **Occupant perceptions and a health outcome in retail stores** — Zhao M.,Kim Y.-S.,Srebric J. — Building and Environment, 93(P2) — 2015 — 19

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Indoor Environmental Quality (IEQ) in commercial buildings, such as retail stores, can affect employee satisfaction, productivity, and health. This study administered an IEQ survey to retail employees and found correlations between measured IEQ parameters and the survey responses. The survey included 611 employees in 14 retail stores located in

	Document title	Authors	Source	Year	Citations

Pennsylvania (climate zone 5A) and Texas (climate zone 2A). The survey questionnaire featured ratings of different aspects of IEQ, including *thermal* comfort, lighting and *noise* level, indoor smells, overall cleanness, and environmental quality. Simultaneously with the survey, on-site physical measurements were taken to collect data of relative humidity levels, air exchange rates, dry bulb temperatures, and contaminant concentrations. This data was analyzed using multinomial logit regression with independent variables being the measured IEQ parameters, employees' gender, and age. This study found that employee perception of stuffy smells is related to formaldehyde and PM_{10} concentrations. Furthermore, the survey also asked the employees to report an annual frequency of common colds as a health indicator. The regression analysis showed that the cold frequency statistically correlates with the measured air exchange rates, *outdoor* temperatures, and indoor $PM_{2.5}$ concentrations. Overall, the air exchange rate is the most influential parameter on the employee perception of the overall environmental quality and self-reported health outcome.

Conference Review

505 **E3S Web of Conferences** | [No Authors Found] | E3s Web of Conferences, 5 | 2015 | 0

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The proceedings contain 390 papers. The topics discussed include: evaluation of integrated time-temperature effect in pyrolysis process of historically contaminated soils with cadmium (Cd) and lead (Pb); testing single and combinations of amendments for stabilization of metals in contrasting extremely contaminated soils; risk-based approach for *thermal* treatment of soils contaminated with heavy metals; effects of chemical applications to metal polluted soils on cadmium uptake by rice plant; characterization of natural adsorbent material for heavy metal removal in a petrochemical site contamination; economics and science in higher education: the need for a common language; future scenarios of urban mobility and their sustainability; VELUXlab: a sustainable prototype of active house for innovation; nature, economics, property and commons. subversive notes inspired by Elinor Ostrom; towards identifying contribution of wake turbulence to inflow *noise* from wind turbines; hybrid turbulence models for atmospheric flow: a proper comparison with RANS models; satisfaction perception of indoor environment of low-cost housing: a case study of flat Taman Desa Sentosa; criteria for affordable housing performance measurement: a review; *outdoor* environment of low-cost housing: a case study of flat Taman Desa Sentosa; simulations of the muon flux sensitivity to rock perturbation associated to hydrogeological processes; and muon telescope based on micromegas detectors: from design to data acquisition.

Conference Paper • *Open access*

506 **Low complexity FPGA based background subtraction technique for thermal imagery** | Imran M.,O'Nils M., Munir H.,Thornberg B. | ACM International Conference Proceeding Series, 08-11-Sep-2015 | 2015 | 1

Hide abstract | Find It @ UCL | View at Publisher | Related documents

Embedded smart camera systems are gaining popularity for a number of real world surveillance applications. However, there are still challenges, i.e. variation in illumination, shadows, occlusion, and weather conditions while employing the vision algorithms in *outdoor* environments. For safety-critical surveillance applications, the visual sensors can be complemented with beyond-visual-range sensors. This in turn requires analysis, development and modification of existing imaging techniques. In this work, a low complexity background modelling and subtraction technique has been proposed for *thermal* imagery. The proposed technique has been implemented on Field Programmable Gate Arrays (FPGAs) after in-depth analysis of different sets of images, characterizing poor signal-to-*noise* ratio challenges, e.g. motion of high frequency background objects, temperature variation and camera jitter etc. The proposed technique dynamically updates the background on pixel level and requires a single frame storage as opposed to existing techniques. The comparison of this approach with two other approaches show that this approach performs better in different environmental conditions. The proposed technique has been modelled in Register Transfer Logic (RTL) and implementation on the latest FPGAs shows that the design requires less than 1 percent logics, 47 percent block RAMs, and consumes 91 mW power consumption on Artix-7 100T FPGA.

	Document title	Authors	Source	Year	Citations
	Review				
507	**Mechanical faults analysis of high voltage disconnectors and review of diagnosis technology**	Qiu Z.,Ruan J.,Huang D.,Zhang Y.,Zhang E.	Gaoya Dianqi High Voltage Apparatus, 51(8)	2015	46

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Frequent faults of high voltage (HV) disconnectors seriously threaten the safety and stability of power grid. Mechanical faults are the most common failure type of *outdoor* disconnectors. In this study, the causes of the typical mechanical faults of a disconnector, including rupture of porcelain post insulator, malfunction and corrosion, were analyzed. And the present researches on mechanical fault diagnostic methods of HV disconnectors, such as ultrasonic flaw detection, *acoustic* vibration technique, infrared *thermography*, stress-strain measurement, and driving motor current signal detection, were reviewed. Furthermore, sticking faults of transmission or rotating mechanism of a disconnector were detected by resistance strain gauge and motor current signal analysis, and the results were preliminarily analyzed. On this basis, several suggestions about comprehensive mechanical fault diagnosis of HV disconnector were offered to provide reference for installation, debugging, operation and maintenance of HV disconnectors in *outdoor* substations.

	Conference Paper				
508	**Big brother for bees (3B) - Energy neutral platform for remote monitoring of beehive imagery and sound**	Edwards-Murphy F., Magno M.,O'Leary L., ... Whelan P.,Popovici E.M.	Proceedings 2015 6th IEEE International Workshop on Advances in Sensors and Interfaces Iwasi 2015, 7184943	2015	33

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Honey bees have played a major role in the history and development of humankind, in particular for nutrition and agriculture. The most important role of the western honey bee (Apis mellifera) is that of pollination. A large amount of crops consumed throughout the world today are pollinated by the activity of the honey bee. It is estimated that the total value of these crops stands at 155 billion euro annually. The goal of the work outlined in this paper was to use wireless sensor network technology to monitor a colony within the beehive with the aim of collecting image and audio data. These data allows the beekeeper to obtain a much more comprehensive view of the in-hive conditions, an indication of flight direction, as well as monitoring the hive outside of the traditional beekeeping times, i.e. during the night, poor weather, and winter months. This paper outlines the design of a fully autonomous beehive monitoring system which provided image and *sound* monitoring of the internal chambers of the hive, as well as a warning system for emergency events such as possible piping, dramatically increased hive activity, or physical damage to the hive. The final design included three wireless nodes: a digital infrared camera with processing capabilities for collecting imagery of the hive interior; an external *thermal* imaging camera node for monitoring the colony status and activity, and an accelerometer and a microphone connected to an off the shelf microcontroller node for processing. The system allows complex analysis and sensor fusion. Some scenarios based on *sound* processing, image collection, and accelerometers are presented. Power management was implemented which allowed the system to achieve energy neutrality in an *outdoor* deployment with a 525 x 345 mm solar panel.

Article

	Document title	Authors	Source	Year	Citations
509	**Ventilation and condensation water failure in coupled windows - Experimental laboratory- and outdoor measurement about the ventilation and the condensation water failure in den window interspace of a coupled window | Hinterlüftung und Tauwasserausfall bei Verbundfenstern: Experimentelle Labor- und Freilanduntersuchungen zu Hinterlüftung und Tauwasserausfall im Fensterzwischenraum eines Verbundfensters Hinterlüftung und Tauwasserausfall bei Verbundfenstern**	Bauer M.,Baumler A., Bauer M.	Bauphysik, 37(4)	2015	0

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Coupled windows are always used when increased physical construction requirements are placed on the *sound* and heat protection in winter. This is achieved through the double skin window structure, with the additional benefit that a sunblind can be placed in the window space between the outer and inner window sash [1]. In practical use, condensation water occurs on different glazing types. This condensation water occurs especially on the inside of the outer glazing (Fig. 1, pos. 2) where it is least wanted. In the course of detailed studies on a coupled window of the WICONA brand, this transient condensation water occurrence is scientifically evaluated based on laboratory and field measurements. The monitoring program for the experimental determination of the condensation water occurrence is elaborated on the basis of a construction methodological model. For this purpose, the coupled window is divided into its elementary components and the material and energy flows accounted for. Through the combined approach of building physics and *thermodynamics* probes can be attached preferably to the surfaces where there is potential risk for condensation water occurrence. The metrological investigations show that the ventilation and the risk of condensation water on pos. 2 can be influenced mainly by placing opening sections at the base and head of the coupled window. Trough small opening cross sections, very little condensation water shall form on pos. 2. An increased air exchange between the window interspace and the environment can be detected with larger aperture cross-sections, which also increases the occurrence of condensation water.

Article

510	**Effects of atmospheric turbulence on CO2 detection accuracy in coherent lidar**	Dong X.,Hu Y.-H., Zhao N.-X.,Wang Y.-Y., Xu S.-L.	Guangdianzi Jiguang Journal of Optoelectronics Laser , 26(7)	2015	5

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Atmospheric turbulence can induce intensity scintillation, beam drift, wavefront distortion and polarization degradation of returned signal, which affects the spatial light coupling ratio of the detection system as well as the heterodyne efficiency, and as a result, the detection accuracy will be reduced. The signal to *noise* ratio (SNR) of the returned atmospheric backward scattering signal is calculated at different heights;also the atmospheric scintillation modulation factor is analyzed based on the Hufnagel-Valley model, and the near-surface intensity scintillation variance is acquired. The method of pattern matching numerical integration is used to simulate the wavefront aberration, and the space light-signal mode fiber coupling efficiencies at different heights are obtained. Based on the complex amplitude of the light field and the airy spot model on the detector surface, the effects of the space phase fluctuation on heterodyne efficiency are analyzed. Combined with CO_2 differential absorption detection experiment data using a 20 m-optical path gas cell, the CO_2 concentration detection accuracy of this experiment setup is determined;and the effects of atmospheric turbulence on *outdoor* experiment are studied. The results show that the atmospheric turbulence will reduce system detection resolution by 1.21 times, and through the 20 m optical path, the system resolution is changed from 424.83×10^{-6} to 514.04×10^{-6}. This analysis is of importance to the *outdoor* experiment of coherent differential absorption lidar (DIAL).

Article • *Open access*

	Document title	Authors	Source	Year	Citations
511	**Selection diversity-aided subcarrier intensity modulation/spatial modulation for free-space optical communication**	Lee J.-H.,Hwang S.-H.	Iet Optoelectronics, 9(2)	2015	6

Hide abstract Find It @ UCL View at Publisher Related documents

In this study, the authors investigate the performance of selection diversity-aided subcarrier intensity modulation (SIM)/spatial modulation (SM) for free-space optical communications. Selection diversity is a receiver diversity technique which can enhance reception robustness in atmospheric turbulence-induced fading. Moreover, it is most attractive in terms of the complexity when compared with the other diversity schemes. Through computer simulations, the performance of the proposed selection diversity-aided SIM/SM is compared in terms of bit error rate with that of the conventional SIM in an additive white Gaussian *noise* and in an *outdoor* environment with log-normal atmospheric channel. Furthermore, the error performance are mathematically analysed in the atmospheric turbulence channel at the spectral efficiency of 2 bits/s/Hz. Numerical results show that the performance of the proposed SIM/SM with two receivers is better than that of a conventional SIM with four receivers at the spectral efficiency of 4 bits/s/Hz. It means that the SIM/SM can release the complexity with less diversity photodetectors over the atmospheric turbulence channel. Additionally, it is shown that the performance of the proposed SIM/SM outperforms that of a conventional SIM regardless of the irradiance variance value. Therefore, the proposed selection diversity-aided SIM/SM can be a good solution to achieve reliable communication with low complexity under high spectral efficiency, even under the impaired conditions of a turbulence fading channel.

Article

512	**Algorithms for crowd surveillance using passive acoustic sensors over a multimodal sensor network**	Agarwal R.,Kumar S., Hegde R.M.	IEEE Sensors Journal, 15(3), 6953090	2015	28

Hide abstract Find It @ UCL View at Publisher Related documents

Crowd detection and monitoring is an active area of research because of its significance in many areas, especially law enforcement. Various sensor modalities, such as infrared imaging, video feed, received signal strength indicator, Radio-frequency identification, GPS signals, and audio tones through mobiles have been used in earlier work. In this paper, a method that uses passive *acoustic* sensors in a multimodal sensor network for crowd monitoring is described. This multimodal system uses three modalities, namely, carbon dioxide level, *sound* intensity level, and received signal strength for crowd detection and monitoring. The first two modalities are sensed using passive sensors, whereas the last one is an active sensor. This combination makes the proposed algorithms energy efficient and computationally less complex. The proposed multimodal crowd monitoring algorithm requires an effective clustering method. Hence, three clustering algorithms that utilize temporal, spatial, and spatio-temporal information are also proposed. Subsequently, an algorithm that fuses the information in different modalities is also proposed for efficient crowd monitoring. Additional contributions of this paper are the development of attenuation, reverberation, and additivity models, using real sensor deployments. Both simulation and real field experiments are conducted to evaluate the performance of the proposed algorithms in indoor and *outdoor* spaces. The results of crowd detection and monitoring obtained from these methods are motivating enough to use the proposed method in real small-scale deployment scenarios.

Article • *Open access*

513	**Sound propagation over soft ground without and with crops and potential for surface transport noise attenuation**	Bashir I.,Taherzadeh S., Shin H.-C., Attenborough K.	Journal of the Acoustical Society of America , 137(1)	2015	20

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Growing demand on transportation, road, and railway networks has resulted in increased levels of annoyance from road traffic. Optimized use of green surfaces in combination with vegetation may be desirable as a method for reducing the *noise* impact of road traffic in urban and rural environments. *Sound* propagation over soft ground and through crops has been studied through *outdoor* measurements at short and medium ranges and through predictions. At lower frequencies, ground effect is dominant, and there is little or no attenuation due to crops. At higher frequencies above 3-4 kHz, the attenuation in crops is dominant. It was also found that the ground effects and

	Document title	Authors	Source	Year	Citations

the influence of crops can be treated independently and can be added to obtain the total effect. *Sound* attenuation by crops is the result of multiple scattering between the stems and leaves, loss of coherence, and viscous and *thermal* losses due to foliage. The major contribution is associated with viscous and *thermal* losses. A model for *sound* attenuation by vegetation is proposed. Insertion losses for a typical road traffic *noise* source have been calculated that result either by replacing hard ground with different types of *acoustically* soft ground or by growing crops along the road sides.

Conference Paper

514 **Data downloading on the coverage of visible light communication** — Chowdhury H., Chaudary B.,Saud M.S., Bashir N.,Katz M. — Proceedings of 21st European Wireless Conference European Wireless 2015, 7147666 — 2015 — 0

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In this work, we provide the methodology to evaluate the performance of visible light communication (VLC) hotspot networks in the context of data downloading on the move by using throughput-distance relationship model. At first, throughputdistance relationship model has been derived analytically by taking into account the optical design parameters such as field-ofview (FOV) of the receiver, semi-angle of transmitter, background *noise* related parameters such as ambient and *thermal noise*. Mobility related parameters such as velocity of the user is also taken into account in performance analysis. Secondly, analytical throughput-distance model is approximated to throughput distance model such as polynomial throughput-distance model. Simulations are also performed to evaluate the performance of such hotspot network in the context of data download on move scenario. Performance of VLC hotspot networks is also quantified in terms of received file size. Simulation results reveal that there is a large impact of background *noise* on the performance of VLC hotspots. As expected, in indoor and *outdoor* environments VLC hotspot performs better at night than at day time due to the prominent day light solar radiation presence in the *outdoor* environment.

Conference Paper

515 **A simplified model for temperature prediction in a ventilated wall module** — Rakotomahefa T.M.J., Zhang T.,Wang S. — Proceedings Healthy Buildings 2015 America Innovation in A Time of Energy Uncertainty and Climate Adaptation Hb 2015, 2015-July — 2015 — 0

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A ventilated wall is a smart building envelope component that can filter the intake air, reduce heat loss/gain and insulate *outdoor noises*. It is composed of a solar board, serpentine air cavities, and an air filter. This investigation proposed an efficient matrix method to compute heat transfer in a ventilated wall module. A *thermal* network was established by implementing energy balance to each network node, which results in a system of algebraic equations of temperatures. The *outdoor* temperature, indoor temperature and the solar irradiation intensity were specified as boundary parameters. The results reveal that the proposed model can provide reasonably accurate temperature profiles with a gap of 1.2 °C to 3°C between the prediction and the measurement. When compared to CFD modeling, the saved computing expense is quite significant but without sacrificing much accuracy.

Conference Paper

516 **Can wavefront coding infrared imaging system achieve decoded images approximating to in-focus infrared images?** — Feng B.,Zhang C.,Xu B., Shi Z. — Proceedings of SPIE the International Society for Optical Engineering , 9795, 97951B — 2015 — 6

Hide abstract Find It @ UCL View at Publisher Related documents

	Document title	Authors	Source	Year	Citations

Artefacts and *noise* degrade the decoded image of a wavefront coding infrared imaging system, which usually results in the decoded image being inferior to the in-focus infrared image of a conventional infrared imaging system. The previous letter showed that the decoded image fell behind the in-focus infrared image. For comparison, a bar target experiment at temperature of +20°C and two groups of *outdoor* experiments at temperatures of +28°C and +70°C are respectively conducted. Experimental results prove that a wavefront coding infrared imaging system can achieve the decoded image being approximating to its corresponding in-focus infrared image.

Article

517 **A questionnaire survey on sleep environment conditioned by different cooling modes in multistorey residential buildings of Singapore** — Yang B.,Olofsson T. — Indoor and Built Environment, 26(1) — 2015 — 26

Hide abstract Find It @ UCL View at Publisher Related documents

A good sleep environment is essential to maintain a person's health and daily working performance. How to create comfortable and healthy sleep environments with less energy use is worth exploring. Findings, based on one questionnaire survey on sleep environments conditioned by different cooling modes have been reported. The study investigated the use of different cooling devices in relation to the effect on sleep. Human responses to *thermal* environments and air quality created by different cooling modes were also studied. Totally 229 completed questionnaires were statistically analysed. The results show that most of the respondents prefer to use air conditioner 3-6 months in a year with relatively low temperature settings especially for respondents living below fifth floor. It is better to choose relatively high temperature settings to reduce air conditioning intensity especially for elders and *outdoor* workers, which can not only avoid cold *thermal* discomfort but also reduce electric energy use. For elders, *outdoor* workers and persons living in higher floor levels, there was an increase usage of electrical fans or natural ventilation. *Thermal* comfort can be maintained by raised air movement and the perceived air quality could improve obviously by introducing *outdoor* air, which would create a good sleep environment to ensure sleep quality.

Article • *Open access*

518 **Residents' experiences of privacy and comfort in multi-storey apartment dwellings in subtropical Brisbane** — Kennedy R.,Buys L., Miller E. — Sustainability Switzerland, 7(6) — 2015 — 74

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Dwellings in multi-storey apartment buildings (MSAB) are predicted to increase dramatically as a proportion of housing stock in subtropical cities over coming decades. The problem of designing comfortable and healthy high-density residential environments and minimising energy consumption must be addressed urgently in subtropical cities globally. This paper explores private residents' experiences of privacy and comfort and their perceptions of how well their apartment dwelling modulated the external environment in subtropical conditions through analysis of 636 survey responses and 24 interviews with residents of MSAB in inner urban neighbourhoods of Brisbane, Australia. The findings show that the availability of natural ventilation and *outdoor* private living spaces play important roles in resident perceptions of liveability in the subtropics where the climate is conducive to year round "*outdoor* living". Residents valued choice with regard to climate control methods in their apartments. They overwhelmingly preferred natural ventilation to manage *thermal* comfort, and turned to the air-conditioner for limited periods, particularly when external conditions were too noisy. These findings provide a unique evidence base for reducing the environmental impact of MSAB and increasing the acceptability of apartment living, through incorporating residential attributes positioned around climate-responsive architecture.

Article • *Open access*

519 **Modern housing retrofit: Assessment of upgrade packages to enerphit standard for 1940-1960 state houses in Auckland** — Leardini P.,Manfredini M. — Buildings, 5(1) — 2015 — 9

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New Zealand state housing includes a significant portion of problematic buildings constructed after the public housing scheme launched in 1936. Most of these houses are still uninsulated, thus, cold, draughty, mouldy, and progressively decaying; however, as they are fundamental elements of the country's culture, society, and

	Document title	Authors	Source	Year	Citations

environment, and are built with good quality materials and *sound* construction, they are suitable candidates for effective energy upgrades. This paper presents findings of a study on problems and opportunities of retrofitting the state houses built between 1940 and 1960 in the Auckland region. It advocates strategic national policies and initiatives for retrofitting, based on more challenging performance thresholds. The research defines and virtually implements an incremental intervention strategy including different retrofit packages for a typical 1950s stand-alone house. Indoor and *outdoor* environmental parameters were monitored over a year, and data used to establish a base case for *thermal* simulation. The upgrade packages were then modelled to assess their impact on the house's *thermal* performance, comparing heating requirements and comfort of various insulation and ventilation options. The paper reports on effective ways of preserving the integrity of such a house, while improving its *thermal* performance to the EnerPHit standard, and discusses the benefits of introducing this holistic approach into New Zealand retrofit practice.

Conference Paper • *Open access*

☐ 520 **Marble ageing characterization by acoustic waves**

Boudani M.E., Wilkie-Chancellier N., Martinez L., ... Verges-Belmin V., Serfaty S.

Physics Procedia, 70

2015

6

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In cultural heritage, statue marble characterization by *acoustic* waves is a well-known non-destructive method. Such investigations through the statues by time of flight method (TOF) point out *sound* speeds decrease with ageing. However for *outdoor* stored statues as the ones in the gardens of Chateau de Versailles, ageing affects mainly the surface of the Carrara marble. The present paper proposes an experimental study of the marble *acoustic* properties variations during accelerated laboratory ageing. The surface degradation of the marble is reproduced in laboratory for 29 mm thick marble samples by using heating/cooling *thermal* cycles on one face of a marble plate. *Acoustic* waves are generated by 1 MHz central frequency contact transducers excited by a voltage pulse placed on both sides of the plate. During the ageing and by using ad hoc transducers, the marble samples are characterized in transmission, along their volume by shear, compressional TOF measurements and along their surface by Rayleigh waves measurements. For Rayleigh waves, both TOF by transducers and laser vibrometry methods are used to detect the Rayleigh wave. The transmission measurements point out a deep decrease of the waves speeds in conjunction with a dramatic decrease of the maximum frequency transmitted. The marble acts as a low pass filter whose characteristic frequency cut decreases with ageing. This pattern occurs also for the Rayleigh wave surface measurements. The speed change in conjunction with the bandwidth translation is shown to be correlated to the material de-structuration during ageing. With a similar behavior but reversed in time, the same king of phenomena have been observed trough sol-gel materials during their structuration from liquid to solid state (Martinez, L. et all (2004). "Chirp-Z analysis for sol-gel transition monitoring". Ultrasonics, 42(1), 507-510.). A model is proposed to interpret the *acoustical* measurements.

Conference Review

☐ 521 **E3S Web of Conferences**

[No Authors Found]

E3s Web of Conferences , 4

2014

0

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The proceedings contain 390 papers. The topics discussed include: evaluation of integrated time-temperature effect in pyrolysis process of historically contaminated soils with cadmium (Cd) and lead (Pb); testing single and combinations of amendments for stabilization of metals in contrasting extremely contaminated soils; risk-based approach for *thermal* treatment of soils contaminated with heavy metals; effects of chemical applications to metal polluted soils on cadmium uptake by rice plant; characterization of natural adsorbent material for heavy metal removal in a petrochemical site contamination; economics and science in higher education: the need for a common language; future scenarios of urban mobility and their sustainability; VELUXlab: a sustainable prototype of active house for innovation; nature, economics, property and commons. subversive notes inspired by Elinor Ostrom; towards identifying contribution of wake turbulence to inflow *noise* from wind turbines; hybrid turbulence models for atmospheric flow: a proper comparison with RANS models; satisfaction perception of indoor environment of low-cost

	Document title	Authors	Source	Year	Citations

housing: a case study of flat Taman Desa Sentosa; criteria for affordable housing performance measurement: a review; *outdoor* environment of low-cost housing: a case study of flat Taman Desa Sentosa; simulations of the muon flux sensitivity to rock perturbation associated to hydrogeological processes; and muon telescope based on micromegas detectors: from design to data acquisition.

Book • *Open access*

522 **Eco-efficient Masonry Bricks and Blocks: Design, Properties and Durability** — Pacheco-Torgal F., Lourenco P.B., Labrincha J.A., Kumar S., Chindaprasirt P. — Eco Efficient Masonry Bricks and Blocks Design Properties and Durability — 2014 — 42

Hide abstract Find It @ UCL View at Publisher

Masonry walls constitute the interface between the building's interior and the *outdoor* environment. Masonry walls are traditionally composed of fired-clay bricks (solid or perforated) or blocks (concrete or earth-based), but in the past (and even in the present) they were often associated as needing an extra special *thermal* and *acoustical* insulation layer. However, over more recent years investigations on *thermal* and *acoustical* features has led to the development of new improved bricks and blocks that no longer need these insulation layers. Traditional masonry units (fired-clay bricks, concrete or earth-based blocks) that don't offer improved performance in terms of *thermal* and *acoustical* insulation are a symbol of a low-technology past, that are far removed from the demands of sustainable construction. This book provides an up-to-date state-of-the-art review on the eco-efficiency of masonry units, particular emphasis is placed on the design, properties, performance, durability and LCA of these materials. Since masonry units are also an excellent way to reuse bulk industrial waste the book will be important in the context of the Revised Waste Framework Directive 2008/98/EC which states that the minimum reuse and recycling targets for construction and demolition waste (CDW) should be at least 70% by 2020. On the 9th of March 2011 the European Union approved the Regulation (EU) 305/2011, known as the Construction Products Regulation (CPR) and it will be enforced after the 1st of July 2013. The future commercialization of construction materials in Europe makes their environmental assessment mandatory meaning that more information related to the environmental performance of building materials is much needed. Provides an authoritative guide to the eco-efficiency of masonry units. Examines the reuse of waste materials. Covers a range of materials including, clay, cement, earth and pumice.

Article

523 **PROMETHEUS: heterogeneous sensor database in support of research on human behavioral patterns in unrestricted environments** — Ntalampiras S., Arsic D., Hofmann M., Andersson M., Ganchev T. — Signal Image and Video Processing, 8(7) — 2014 — 6

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The multi-modal multi-sensor PROMETHEUS database was created in support of research and development activities [PROMETHEUS (FP7-ICT-214901): http://www.prometheus-FP7.eu] aiming at the creation of a framework for monitoring and interpretation of human behaviors in unrestricted indoor and *outdoor* environments. The distinctiveness of the PROMETHEUS database comes from the unique sensor sets, used in the various recording scenarios, but also from the database design, which covers a range of real-world applications, correlated to smart-home automation and indoors/outdoors surveillance of public areas. Numerous single-person and multi-person scenarios, but also scenarios with interactions between groups of people, motivated by these applications were implemented with the help of skilled actors and supernumerary personnel. In these scenarios, the actors and personnel were instructed to implement a range of typical and atypical behaviors, and simulations of emergency and crisis situations. In summary, the database contains more than 4 h of synchronized recordings from heterogeneous sensors (an infrared motion detection sensor, *thermal* imaging cameras, overview/surveillance video cameras, close-view video cameras, a 3D camera, a stereoscopic camera, a general-purpose camcoder, microphone arrays, and motion capture equipment) collected in common setups, simulating smart-home environment, airport, and ATM security environment. Selected scenes of the database were annotated for the needs of human detection and tracking. The entire audio part of the database was annotated for the needs of *sound* event detection, *sound* source enumeration, emotion recognition, etc.

Article

	Document title	Authors	Source	Year	Citations
524	**Description and quantification of uncertainty in outdoor sound propagation calculations**	Keith Wilson D.,Pettit C.L., Ostashev V.E., Vecherin S.N.	Journal of the Acoustical Society of America , 136(3)	2014	29

Hide abstract Find It @ UCL View at Publisher Related documents

The accuracy of *outdoor sound* propagation predictions is often limited by imperfect knowledge of the atmospheric and ground properties, and random environmental variations such as turbulence. This article describes the impact of such uncertainties, and how they can be efficiently addressed and quantified with stochastic sampling techniques such as Monte Carlo and Latin hypercube sampling (LHS). Extensions to these techniques, such as importance sampling based on simpler, more efficient propagation models, and adaptive importance sampling, are described. A relatively simple example problem involving the Lloyd's mirror effect for an elevated *sound* source in a homogeneous atmosphere is considered first, followed by a more complicated example involving near-ground *sound* propagation with refraction and scattering by turbulence. When uncertainties in the atmospheric and ground properties dominate, LHS with importance sampling is found to converge to an accurate estimate with the fewest samples. When random turbulent scattering dominates, the sampling method has little impact. A comprehensive computational approach is demonstrated that is both efficient and accurate, while simultaneously incorporating broadband sources, turbulent scattering, and uncertainty in the environmental properties.

Conference Review

525	**E3S Web of Conferences**	[No Authors Found]	E3s Web of Conferences , 3	2014	0

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The proceedings contain 390 papers. The topics discussed include: evaluation of integrated time-temperature effect in pyrolysis process of historically contaminated soils with cadmium (Cd) and lead (Pb); testing single and combinations of amendments for stabilization of metals in contrasting extremely contaminated soils; risk-based approach for *thermal* treatment of soils contaminated with heavy metals; effects of chemical applications to metal polluted soils on cadmium uptake by rice plant; characterization of natural adsorbent material for heavy metal removal in a petrochemical site contamination; economics and science in higher education: the need for a common language; future scenarios of urban mobility and their sustainability; VELUXlab: a sustainable prototype of active house for innovation; nature, economics, property and commons. subversive notes inspired by Elinor Ostrom; towards identifying contribution of wake turbulence to inflow *noise* from wind turbines; hybrid turbulence models for atmospheric flow: a proper comparison with RANS models; satisfaction perception of indoor environment of low-cost housing: a case study of flat Taman Desa Sentosa; criteria for affordable housing performance measurement: a review; *outdoor* environment of low-cost housing: a case study of flat Taman Desa Sentosa; simulations of the muon flux sensitivity to rock perturbation associated to hydrogeological processes; and muon telescope based on micromegas detectors: from design to data acquisition.

Book Chapter

526	**Imaging and Display of Human Size Scenes by Long Wavelength Digital Holography**	Locatelli M.,Pugliese E., Paturzo M.,... Meucci R., Ferraro P.	Multi Dimensional Imaging, 9781118449837	2014	0

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In this chapter it is shown how the imaging of human size object as well as live people through smoke and flame curtains is possible thanks to the infrared digital holography (IRDH). This technique exploits the advantages of the long infrared wavelength source, i.e. low sensitivity to seismic *noise* and large field of view capability, and those of the digital holographic technique, i.e. lensless acquisition and discharge of non-coherent radiation, allowing to see through flames, also in *outdoor* conditions. The acquired holograms can be numerically or optically reconstructed. In the first case, the image is shown on a computer display, while in the second case a visible laser source and some spatial light modulators are used to project the image on a screen or to obtain a floating in space object reproduction.

	Document title	Authors	Source	Year	Citations
	Conference Review				
527	**E3S Web of Conferences**	[No Authors Found]	E3s Web of Conferences, 2	2014	0

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The proceedings contain 390 papers. The topics discussed include: evaluation of integrated time-temperature effect in pyrolysis process of historically contaminated soils with cadmium (Cd) and lead (Pb); testing single and combinations of amendments for stabilization of metals in contrasting extremely contaminated soils; risk-based approach for *thermal* treatment of soils contaminated with heavy metals; effects of chemical applications to metal polluted soils on cadmium uptake by rice plant; characterization of natural adsorbent material for heavy metal removal in a petrochemical site contamination; economics and science in higher education: the need for a common language; future scenarios of urban mobility and their sustainability; VELUXlab: a sustainable prototype of active house for innovation; nature, economics, property and commons. subversive notes inspired by Elinor Ostrom; towards identifying contribution of wake turbulence to inflow *noise* from wind turbines; hybrid turbulence models for atmospheric flow: a proper comparison with RANS models; satisfaction perception of indoor environment of low-cost housing: a case study of flat Taman Desa Sentosa; criteria for affordable housing performance measurement: a review; *outdoor* environment of low-cost housing: a case study of flat Taman Desa Sentosa; simulations of the muon flux sensitivity to rock perturbation associated to hydrogeological processes; and muon telescope based on micromegas detectors: from design to data acquisition.

	Conference Paper				
528	**Sensors positioning in outdoor environment with signal strength**	Hung F.H., Chi H.R., Li B.Y.S., Tsang K.F.	IECON Proceedings Industrial Electronics Conference, 7049087	2014	0

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Log-distance path loss model have been using as a simple positioning method because of its simplicity which is mainly depends on Received Signal Strength (RSS) along with path loss exponent and random Gaussian *noise* variable with zero-mean. This model can be extensively used in urban and remote area because it is related to energy representation. On the other hand, fingerprint positioning is also an alternative solution in positioning due to it reliable performance. It is noticed that antenna gain and background *thermal noise* should be included into the model such that the accuracy could be improved. In this investigation, a sub-urban route was selected as testing area and carried the Particle Swarm Optimization (PSO) for an optimal coordinate. Experimental results show that the new scheme was implemented successfully in RSS positioning resulting in about averaged 100 meters and 84 meters in daytime and evening time respectively in same experiment scene. This new scheme provides a more reliable way in calculating a sensors position.

	Conference Review				
529	**3rd International Conference on Civil Engineering and Transportation, ICCET 2013**	[No Authors Found]	Applied Mechanics and Materials, 507	2014	0

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The proceedings contain 175 papers. The special focus in this conference is on Civil Engineering and Transportation. The topics include: Enclosure exploration of tianjin wudadao district; study on internal layout of UK's day care unit for older people; study on site design of UK's day care unit for older people; study on the construction mode and quality monitoring of security housing in henan province; on the abstract metaphorical design in the transformation of traditional houses in Guanzhong region; analysis on modern residence design; study of architectural space efficacy based on the modern theory of validity; the application of digital technology in the international design competition of large sports center; an inquiry of teaching methods based on the psychological needs of comfort in small architectural space; architectural space evolution of railway passenger station; Himalayan water tower; London city hall and the values it promotes; on application of virtual space in architecture internal space; a brief discussion about effects of Chinese traditional culture on contemporary Chinese architectural design; interpretation of concerning comfort of office building design in cold regions; modern residential design; new changes in current architectural art

Document title	Authors	Source	Year	Citations

teaching and corrective measures; strategies of three-dimensional greening design in building sectors; the *acoustical* design of conference room based on speech *acoustic*; the *acoustic* test report of three *outdoor* courtyard in Shenyang palace museum; the building energy conservation design responses of urban underground space; research on the light design for the enclosed target lane; research on indoor *thermal* environment of dai nationality wood dwellings; slits' effects on *sound* insulation for lightweight partitions; an investment-benefit assessing approach for intelligent buildings; comparing the core evaluation system for green building at home and broad; study on the green construction of current constraints and promotion measures in china; study on natural lighting modification of existing steel-structure plants in cold regions; a middleware of IoT-based smart home based on service; the development for internet of things smart home service operation support platform; a reproducible study on the recycling project of old industrial buildings; advances in damage of concrete due to freeze-thaw circles; experimental research on seismic performance of pre-damaged RC frame T-beam strengthened with sprayed BFRP; experimental research on compressive strength of adobe with cement; effect of curing time and $CaSO_4$ whisker content on properties of gypsum board; development and evaluation for microdamage of concrete under uniaxial compressive load; a mixed-mode fracture criterion for brittle materials; experimental study on *sound* absorption property of porous concrete pavement layer; chloride ion penetrability of HPFRCC after loading; study on optimal ratio of compound corrosion inhibitors for reinforcing steel; experimental study on the strength of basalt fiber confined concrete under impact loading; experimental study on frost resistance of concrete in seawater; research on prediction of carbon steel in SO_2 atmospheric corrosion using grey model; the contrast of concrete Brazilian and flattened Brazilian disc specimen under quasi static diametral compression tests; different replacement rate of recycled aggregate concrete research; various replacement rate experimental study on recycled concrete; mechanical properties of stone masonry mechanical properties; a measurement method of tensile strength for irregular small stones; corrosion resistance performance of ethylamines in fly ash blended cement concrete; dynamic response test of reinforced concrete slab under blast loading; hydration properties of steel slag under different curing temperatures; research on sand gradation to ultra strength grouting material properties; experimental study on flexural behavior of damaged reinforced concrete beams strengthened with CFRP; experimental study on flexural capacity of prestressed RC hollow slabs strengthened by CFRP; research on deformation index limits of RC shear walls based on material damage; experimental study for CFRP sheets reinforcing high-titaniferous blast furnace slag concrete beams; experimental research on fireproofing measure of slabs strengthened by carbon fiber sheet bonded with inorganic adhesive; influence of high-temperature curing on the properties of the concrete containing ground iron and steel slag; state-of-the-art report of recycled aggregate concrete and rapid repair concrete; research on the cement matrix of super toughness fiber concrete; laboratory investigation on the strength characteristics of cement-treated base; frost scaling resistance and improvement of concrete; dry-mixed antifreeze concretes design of mixing and performance of antifreeze; study the strength regularity of recycled masonry mortar produced with recycled ceramic tiles; experimental study on the mechanical performance of pumice powder cement soil; fabrication and properties of new building materials by reutilization refractory materials; study on curing agent of chemical modification on dredged soil; compensation effect of expansive agent on shrinkage of self-compacting concrete; experiment study on the compressive strength of combined aggregate concrete; evaluation of green degree of concrete based on the gray relation; study on the mechanical properties of the latex-mycelium composite; the research on the numerical modeling of foam material; effect of different grain grading of manufactured-sands on properties of dry-mixed mortar; study on the performance of glazed hollow bead aerated concrete and its influence factors; preparation and properties of lime-based curing material for soil; finite element analysis for the crack self-repairing behavior of the concrete beam embedded shape memory alloy; design and study on integrated photovoltaic steel-structure industrial plants in cold regions; a study on the energy-saving evaluation index system of rural residential buildings; a study on the energy-saving evaluation index system of rural residential buildings; optimization of an invention based on a force multiplier mechanism for wave power generation; research on solar water heating system design of the high-rise residential buildings based on sunlight simulation; study on application of the integrated solar hot-water system to high-rise residential buildings in china; a study on integration design of solar hot-water systems in high-rise residential buildings; analyses of the suitability of ground-source systems in typical architecture in northern region; energy-saving technical measures of German low-energy buildings; study on energy efficiency design of self-insulation system with autoclaved aerated concrete block in cold regions; building energy audit and energy-saving analysis for a five-star hotel in Beijing; analysis to urban traffic jam from the perspective of city planning; about architectural aesthetics in the three gorges reservoir area planning; a study of protective exploitation of zhu des former residences tourism environment; technical system and application strategy of digital urban planning management; extraction of ningxia region new rural living environment to repair again;

Document title	Authors	Source	Year	Citations

development study on cultural creative industrial parks of Suzhou central city; analysis of spatial evolution characteristics in songyuan main urban area based on space syntax; study on evaluation of rural human settlement in Changchun; discussion about the planning of settlements supporting education facilities in Jinzhou city; historical architectures renovation strategy under the guidance of the ecologic concept; on the protection and utilization of the famous historic and cultural cities, towns and villages in central plain economic region; study on low-carbon settlements spatial form design strategy - the case of northern china; study of fractal structure characteristics of urban system in northern Shaanxi; a strategy research on the development of Tianjin tuanbo new town west district; regenerative protection of urban historic conservation area amid cultural prosperity; thinking on redevelopment of old urban housing area in the viewing angle of peoples livelihood happiness; concept renewal about koguryo cultural heritage protection with "cultural routes" the study on ancient waterway of yuanshui river and its culture from the perspective of cultural routes; humanization design of landscape for commercial pedestrian street; discussion about the landscape planning in contemporary campuses from the traditional Chinese landscape architecture; study on the community environment design for the aged; thinking of village construction in central region under the context of labor migration; research on the experimental device of combined two-phase ABR and SBR; resent advances in phosphorus removal from waste water; design of the reclaimed water reuse project for recirculating cooling water in a *thermal* power plant; emerging technologies for phosphorus removal and recovery; kinetic and *thermodynamic* studies of adsorption of phosphate on steel-slag filter material; synthesis of spherical CuS photocatalyst and the research on photo-degradation activity of methylene blue; effect of hydraulic retention time on pollutant removal performance of biological contact oxidation process treating hospital wastewater; research on flocculation of protein-bearing wastewater by alga flocculant; evaluation of aquatic plants for water purification in the xixi wetland in Hangzhou, china; chlorine decay models and influencing factors; organics and its control in reclaimed water.

Conference Review

☐ 530 **2013 3rd International Conference on Mechanical Engineering, Materials and Energy, ICMEME 2013** | [No Authors Found] | Applied Mechanics and Materials, 483 | 2014 | 0

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗

The proceedings contain 135 papers. The special focus in this conference is on Mechanical Engineering, Materials and Energy. The topics include: Numerical simulation of high speed single-grain cutting using a coupled FE-SPH approach; research on the technology for DC CMT welding of AZ31B magnesium alloy; a three-dimensional numerical analysis on the effective permittivity of composites including ellipsoids; effect of the process parameters of laser cladding on microstructure of Ni-Based titanium carbide; research on the activity of calcium oxides under various calcination temperatures; the new technology of preparation activated carbon from lignite; inductively coupled plasma modified silica material technology research; a simple and practicable PCR-microchip with bubble-free stationary chamber; synthesis and evalution of enteric aqueous polymethacrylate coating material; the new technology of precision grinding high performance ceramic materials; DWT-SVM on near-infrared spectra for moisture and volatile determination of coal; a brief review of cardanol-based surfactants; preparation of magnetic nanoparticles from byproduct of titanium dioxide-copperas; research on and applications of AU/PT *thermocouples*; Effect of Near β heat treatment process on microstructure of TC18 alloy; extraction study of nitro humic acid from lignite by dry and wet process; effect of particle size of BNT filler on dielectric and mechanical properties of LCP based composite; research on the rationality of the combined application of bamboo and wood in *outdoor* furniture; a 3-dimentional numerical analysis of solar-panel-chimney for house ventilation; numerical simulation on the effect of mist on impingement cooling of double chamber model; simulation and experimental study of oscillator for permanent magnet vibration-to-electrical power generator; numerical simulation of liquefied natural gas in rib-tube; research on the V/H inspection of spiral bevel gear before heat treatment; energy spectra of the low-lying state in N-layer quantum dots; study on an intelligent teaching dynamic balance technology; a surface roughness model in radial-mode abrasive waterjet turning for high-tensile steels; parametric FEM analysis of disc brake; *thermal* stress and modal analysis of engine exhaust manifold; energy consumption modelling for air-cushion vehicles; research on energy efficiency evaluation algorithm based on database system; numerical simulation and comparison on the inner-flow field in hydraulic feedback pump; analysis of hydrodynamic model of wave energy converter of inverse pendulum; the modeling and locating technology of assembly\disassembly tools in virtual environment; simulation of aerosol sampling inlet for different outlet angle; numerical simulation of the effect of spoiler ribs on the flow field in household soymilk maker; precise ultrasonic motor driver design and analysis; calculation of *thermal* analysis

Document title	Authors	Source	Year	Citations

kinetics using least mean square method; the modified iteration method for wet cooling towers to calculate the cooled water outlet temperature; environmental analysis on insulation resistance reduction case; selecting L and C of the passive dynamic harmonic filter; the mechanical model of the dynamic characteristics of the bituminous concrete compaction mechanism; *noise* analysis of disc brake for automobile; research on power flow interface oriented regional wind power forecasting method; influence parameters analysis of forming quality and system dynamics simulation of vertical steel bar benting; stability analysis of nonlinear stiffness rotor-bearing system with pedestal looseness fault; effect of bolt slippage and joint eccentricity on the response of lattice structure with non-uniform settlement; the finite element analysis concerning the slip coat of workover rig; lightweight design of tanker of LPG truck based on material strength equivalent substitution; study on the mechanical properties and seismic performance of beam-column joints of wooden structure building; stress analysis of asymmetric spur face-gear pair based on finite element method; boosting the efficiency of an 800 MW-class power plant through utilization of low temperature heat of flue gases; identifying the unit area dynamic characteristic parameters of fixed joint based on equivalent single DOF system; correlation between BMD and radio density of human bones for strength evaluation; effect of height and width of road humps on deceleration performance and driving safety; stability and oscillation phenomena analysis in microbial continuous fermentation time-delay system; research of vibratory feeder activated by the bimorph vibrator; an air wipe based grease removing and saving device for anti-twist rope; indentation tests based on gelatin phantom; the research and application on light shape detection of car's headlamp; oil-gas-water three-phase flow monitoring based on image fusion of ECT and ERT; optimal control for vibration of restrained cantilever fluid-conveying pipe; study on the comparison of two metrics used in the fields of object tracking; economic analysis of large-scale biogas project based on system dynamics; a method of optimal module partition in green product configuration design under uncertainty; optimization calibration of engine ignition advance angle based on engine model; multivariant optimization algorithm for multimodal optimization; design of onboard device to diagnose track fault online; analysis on game behavior of local governments in reverse logistics of ELV; remaining life prediction based on HOLT algorithm for civil aero-engine; topology optimization of thin-shell frame with surface loads; research on the design and extraction of vehicle brand form gene; knowledge tracing modeling for product design knowledge management; the affection of fasteners load constraint in topology optimization; use of logistic approach in optimizing the rotary kiln run; optimization of distributed systems' fusion based on climb-ant algorithm; design of PSO based PI controller for wind turbine systems; control system design of ducted fan aerial vehicle; alarm system for ancient tombs based on zigbee; urban environment monitoring and control system design; average wage prediction based on logistic model; an evaluation system of pulsed GMAW process; blind image quality assessment using latent dirichlet allocation model; model and algorithm for *thermal* performance calculation of power station boiler based on process steady-state simulation theory; utilization of improved RBF network in modeling of wave energy absorption system; design of a high force voice coil motor and multi-objective dynamic optimum design of crane gearbox.

Conference Paper

☐ 531 **Smart management of LTE Outdoor site support cabinet-migrate towards simplicity** | Kaptein R., Luo S., Wei Q., Wang O. | INTELEC International Telecommunications Energy Conference Proceedings , 2014-January(January), 6972167 | 2014 | 1

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Mobile Broadband (MBB) is in demand of social connection in everything and everywhere. LTE is booming technology of MBB. Remote Radio Unit Architecture with multi-mode (GSM, UMTS, and LTE) is the trend of LTE network deployment. There are some new challenges for management of LTE *Outdoor* site support cabinet: • How to improve multi-mode baseband and transmission reliability? •How to reduce *acoustic noise* for zero complains of residential? •How to embed power system alarm management into mobile network operation supporting system? •How to upgrade the power system efficiency with higher and higher rectifier's efficiency (96%-98%)? •How to face smart grid evolution with solar energy access and smart storage requirement Smart management will be the solution for above

	Document title	Authors	Source	Year	Citations

challenges with following units: Advanced *thermal* management; *Acoustic* control: •Site management; •Rectifiers Hybrid application (96%-98%) •MIMO(Multi-Input and Multi-Output).

Article

532 **Outdoor space quality: A field study in an urban residential community in central China** — Lai D.,Zhou C.,Huang J., ... Long Z.,Chen Q. — Energy and Buildings , 68(PART B) — 2014 — 165

Hide abstract Find It @ UCL View at Publisher Related documents

The quality of *outdoor* spaces in a residential community affects the quality of life of its residents. This paper presents the findings of a study on *outdoor thermal* comfort and space usage at a residential community in Wuhan, central China, through the monitoring of microclimate conditions, interviews with residents, and recording of occupants' activities. The data were used to develop a *thermal* sensation vote (TSV) model and a space usage rate model. An alternative design strategy was proposed for considering *thermal* environment and usage rate. The results show that *thermal* comfort is the most important factor in the quality of *outdoor* spaces. Other significant factors include air quality, *acoustic* environment, functionality, and convenience. The study provided valuable information for the design of *outdoor* spaces in residential communities. © 2013 Elsevier B.V.

Conference Paper

533 **Use of adaptive actions and thermal comfort in a naturally ventilated office** — Honnekeri A.,Pigman M.C., Zhang H.,... Zhai Y., Dutton S. — Indoor Air 2014 13th International Conference on Indoor Air Quality and Climate — 2014 — 6

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A naturally ventilated office building in Alameda, CA with operable windows and ceiling fans was monitored from Oct 2011 to Oct 2012. Physical environmental parameters such as dry-bulb air temperature, humidity, CO_2 levels, *outdoor* wind speed, hourly window positions, heater settings and fan settings were recorded. Occupants were surveyed regularly over a period of one year about their current *thermal* comfort, *thermal* acceptability, air movement satisfaction, clothing and *noise* satisfaction. Occupants wore clothing with a clo value of between 0.5-0.6 during summer, and 0.7-0.8 during winter. The clo value of the occupants clothing was most closely correlated with the *outdoor* running mean temperature. Occupants start opening windows when the *outdoor* is at 16 °C. Window opening was strongly related with occupant's arrival and *outdoor* temperature. Fans use was best explained by indoor temperature. Fans were typically turned on during the summer at indoor temperatures above 26 °C. Occupants voted that the *thermal* environments in the building were acceptable 98% of the time during the year-long survey period.

Conference Paper

534 **Optimization of the building envelope in hot and humid climates** — Nehme B.A.,Annan G.R. — 2014 International Conference on Efficient Building Design Materials and HVAC Equipment Technologies Icebd Met 2014 — 2014 — 0

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This paper aims to promote building sustainable designs to reduce energy consumption by implementing optimized building envelope for high performance buildings. Several building envelope case studies for office, airport, retail and residential buildings, located in hot and humid climates, are analyzed in WUFI, IES VE and CadnaA software in order to assess, simulate and optimize building envelope performance. A detailed building *thermal* energy performance simulation, assessing the impact of solar radiation and external shading devices, using IES VE, has been conducted to evaluate the effectiveness of the proposed building envelope designs and weigh the impact of sustainability measures on enhanced indoor environmental quality and building loads reduction. Also, a condensation analysis for external wall assemblies is conducted via WUFI software to study moisture influence on the *thermal* performance of the construction and long-term system behavior. Additionally, *sound* control software such as CadnaA assists in

	Document title	Authors	Source	Year	Citations

validating *noise* mitigation techniques from *outdoor* equipment to promote indoor occupant comfort. The simulation results also suggest building envelope composition optimized for hygrothermal and *acoustic* performance, window to wall ratios and appropriate shading strategies that promote reductions in building annual cooling energy for the different case studies to a range of savings varying from 5 % to 35% with respect to ASHRAE standard 90.1-2007 for hot and humid climates.

Conference Paper

535 **Experimental study of cabin thermal comfort and air quality at different seasons** — Li B.,Pei J.,Wang C.,Liu J.,Xiao X. — Indoor Air 2014 13th International Conference on Indoor Air Quality and Climate — 2014 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

To study cabin *thermal* comfort and air quality, objective parameters measurements and subjective questionnaire in a simulative short flight were simultaneously conducted in a real aircraft cabin environment research platform. Objective parameters measured included *noise*, illumination, air and wall temperature, air velocity, CO_2 and particle concentration. In spring and summer, two teams of volunteers (passengers and crew members) wrote questionnaires on cabin *thermal* comfort and air quality. In this research, we found that comfortable target operative temperature was changed by different season, and related to *outdoor* temperature and cloth *thermal* resistance. PMV-PPD index could not predict *thermal* comfort in cabin exactly. It should be modified maybe because of high population density in cabin.

Conference Paper

536 **Energy efficient free cooled telecommunications shelter** — Komljanec T. — INTELEC International Telecommunications Energy Conference Proceedings , 2014-January(January), 6972136 — 2014 — 3

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A limitation with traditional air conditioning controls is that they operate HVAC equipment to a fixed temperature and are intended for human comfort. In order to reduce HVAC energy consumption in a telecom shelter, controls can be set to a higher temperature, but fixed operation may not be ideal for the longevity of battery systems or equipment. A fixed, year round set point will not take advantage of cool *outdoor* conditions when they are available. With energy costs trending upwards, together with ongoing maintenance costs for air conditioners, there is strong incentive to eliminate mechanical cooling altogether. The Canadian moderate temperate climate is ideally suited for free cooling of telecommunications shelters. Field trials show that intelligently devised controls can mediate summer days when *outdoor* temperatures might reach 35C, while utilizing cooler overnight temperatures to average-down interior temperatures. Cooler seasons will allow a variable speed fan to operate at lower speeds, *noise* levels and energy consumption rates. Effective and energy efficient free cooling strategies require a systemic approach to shelter design. Careful selection of electronic equipment that can operate reliably in a -5C to +45C environment is key. Heat producing equipment is located in the hot zone closest to the exhaust. As the interior temperature rises, so does the air flow to minimize the delta temperature between indoor and *outdoor* air. For year round efficiency, heater operation in winter is set to 12C, which achieves a balance with the battery capacity derating. Instead of traditional fluorescent lighting, LED fixtures are utilized as their output is insensitive to temperature fluctuation. Everything associated with the shelter (except the baseboard heater) operates from the -48 VDC power plant. Referenced to utility electrical input, the free cooled telecom shelters can operate at 90% efficiency.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
537	**Radiometric active indoor imaging in the W-band**	Canavero M.,Murk A., Matzler C.,Notel D.,Huck J.	Journal of Infrared Millimeter and Terahertz Waves , 35(2)	2014	1

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Millimeter wave passive imaging systems constitute a good compromise between resolution and penetration depth for a variety of imaging applications. In an *outdoor* scenario, the cold sky radiation, interacting with the reflectivity characteristics of the targets, constitutes the main source of contrast in the acquired images. In indoor applications such a source is not available, and higher *thermal* sensitivity is required. Alternatively, one has to provide an artificial illumination to the scene in order to increase its dynamic range. The implementation of an active source for a passive radiometer can, under certain conditions, increase the contrast of the images acquired and add extra information to the measurement. With such a setup, *outdoor* systems can be used for indoor observations (the absence of cold sky radiation is compensated with active illumination). The subject of our study is to better understand which kind of source and which setup can provide a diffuse illumination over the targets. This topic was investigated by conducting observations of various indoor scenes with two radiometers in the W-Band, using *noise* and continuous wave (CW) sources as illumination. In this paper we present the results achieved and our conclusions in order to provide an efficient illumination for indoor environment. © 2013 Springer Science+Business Media New York.

	Conference Paper				
538	**Child Automatic Speech Recognition for US English: Child Interaction with Living-Room-Electronic-Devices**	Gray S.S.,Willett D.,Lu J., ... Maergner P., Bodenstab N.	4th Workshop on Child Computer Interaction Wocci 2014	2014	40

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Adult-targeted automatic speech recognition (ASR) has made significant advancements in recent years and can produce speech-to-text output with very low word-error-rate, for multiple languages, and in various types of noisy environments, e.g. car *noise*, living-room, *outdoor-noise*, etc. But when it comes to child speech, little is available at the performance level of adult targeted ASR. It requires a considerable amount of data to build an ASR for naturally spoken, spontaneous, and continuous child speech. In this study, we show that using a minimal amount of data we adapt multiple components of a state-of-the-art adult centric large vocabulary continuous speech recognition (LVCSR) system to form a child specific LVCSR system. The resulting ASR system improves the accuracy for children speaking US English to living room electronic devices (LRED), e.g. a voice-operated TV or computer. Techniques we explore in this paper include vocal tract length normalization, *acoustic* model adaptation, language model adaptation with childspecific content lists and grammars, as well as a neural network based approach to automatically classify child data. The combined initiative towards child-specific ASR system for the LRED domain results in relative WER improvement of 27.2% compared to adult-targeted models.

	Conference Paper				
539	**Design and testing out of an insulating floor element, composed of recycled rubber and inert demolition waste**	Buhagiar V.,Bonnici J., Borg S.P.	30th International PLEA Conference Sustainable Habitat for Developing Societies Choosing the Way Forward Proceedings, 2	2014	1

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Even if initially dominated by cost concerns, energy-awareness is today maturing towards a vision aimed at curtailing energy consumption and reducing carbon emissions. In consideration of today's large amount of tyres being used and disposed of, the purpose of this research was to investigate options for utilising recycled rubber as a building material (in combination with inert demolition waste), namely as insulation in flooring elements such as tiles, such a mix could replace new underlay insulating materials typically having a higher embodied energy content. This paper evaluates the potential of recycling used tyres in specially fabricated floor tiles for different design mixes and evaluates the

	Document title	Authors	Source	Year	Citations

success/failure of such a building element as a floor finish. The study also looks into the best combination of materials to form a durable, non-abrasive robust tile, yet also acting as a *thermal* and moisture barrier. An added value of the proposed tile is its *acoustic* property, where the shredded rubber makes it also resilient to impact and airborne *noise*. Its light colour also proves ideal for solar-exposed flat roofs, where, it is also aesthetically pleasing for the *outdoor* evening lifestyle in a Mediterranean climate.

Conference Paper

☐ 540 **Adaptive comfort and control protocols for natural ventilation**

Philip S.,Shameson A., Brown N.,Loisos G., Ubbelohde S.

30th International PLEA Conference Sustainable Habitat for Developing Societies Choosing the Way Forward Proceedings, 1

2014

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Recently there has been a shift in international standards towards adaptive *thermal* comfort. ASHRAE had formally incorporated adaptive comfort into ASHRAE Standard 55-2010. In adaptive comfort, the range of indoor temperatures perceived as comfortable drifts upwards in warm weather and downwards in cooler weather. This is especially true where the occupants have control over their *thermal* environment (for example, with clothing or operable windows) and when occupants are aware of the variable *outdoor* conditions. This paper presents the ventilation control strategies used to provide adaptive comfort without air conditioning (compressive cooling) in a major retrofit for an office and classroom building on the University of Hawaii campus. The building retrofit is designed for cross-ventilation while taking care to *acoustically* isolate the spaces from each other and to dampen *sound* from outside. A supplementary fan system is designed and configured to complement the cross-ventilation strategy. The fan system is controlled to actively draw just enough air to provide adaptive comfort conditions when wind is calm. If the wind is sufficient, the controls sense the speed and direction and shut down the fans allowing cross ventilation to do the cooling. The fan controls sense the conditions inside, conditions outside the building, and weather conditions in the recent past.

Conference Paper

☐ 541 **Tracking loop model and hardware prototype verification of GNSS/INS deep integration**

Zhang T.,Zhang H.,Ban Y., Niu X.,Liu J.

Lecture Notes in Electrical Engineering , 303 LNEE(VOL. 1)

2014

3

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There is no systematic and complete theoretical model for signal tracking loop of the GNSS/INS deep integration. And the performance of the deeply-coupled system based on hardware prototype hasn't been fully verified. These limitations block the progress and application of the GNSS/INS deeply-coupled technology. This paper studies the GNSS/INS deeply-coupled technology based on the scalar deep integration for GPS L1 receiver. It establishes the transfer functions between the error sources (including *thermal noise*, oscillator phase *noise*, inertial measurement unit (IMU) error, the delay of Doppler aiding information) and the tracking loop error of the deep integration. And then the steady state tracking model is proposed and analyzed. A hardware/software integrated GNSS/INS scalar deep-coupled prototype is successfully developed, and real-time optimizations are made in terms of the system operation and aiding information delay. The performance of the designed deeply coupled prototype is fully evaluated based on a GPS/IMU hardware simulator and *outdoor* tests. The result shows that the INS aiding could improve the steady state's tracking performance by extending the integration time to 20 ms and by compressing the bandwidth to 3 Hz under normal dynamic conditions. The proposed error models, designed methods, and hardware prototype developed in this paper can be further applied to the key performance study of the GNSS/INS deeply coupled system, such as the sensitivity and anti-interference under dynamic conditions. © 2014 Springer-Verlag.

Conference Review

	Document title	Authors	Source	Year	Citations
542	**4th International Conference on Applied Mechanics, Materials and Manufacturing, ICA3M 2014, ICAMMM2014**	[No Authors Found]	Applied Mechanics and Materials, 670-671	2014	0

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The proceedings contain 336 papers. The special focus in this conference is on Nanomaterials Science and Technology, Metals and Alloys, Optical and Magnetic Materials, Semiconductors and Technology and Composite Materials and Applications. The topics include: The function of PEG in the synthesis of nanomaterials; the synthesis and characterization of oxide free tin nanoparticles; effects of different heat treatments on transformation of residual austenite in bearing steel; the effect of water vapor on *thermal* oxide grown on Inconel 690; study on band gap varying of diamond photonic crystals by fabricating to bring the error of dielectric volume fraction; a two-dimensional square-lattice photonic crystal with rotated square cylinders and cross thin plates exhibiting wide photonic bandgap; preparation and application of grapheme; *thermal* strength of adhesion bond; the *outdoor* aging properties of polypropylene/mica composite; dyeing of mulberry silk using natural safflower yellow pigment; synthesis and preparation of desulfurization concerned naphthenic acid type ionic liquid; polymerization of maleic anhydride induced by non-equilibrium atmospheric pressure argon plasma; research of coalbed methane fracturing crack initiation and propagation law; large-scale growth of nitrogen-doped via solvothermal synthesis; research on the effect of prehydration of Portland cement stored in normal conditions; mechanism research of drying shrinkage of cement pastes based on the contact angle; deformation characteristics of concrete under uniaxial compression; research on mechanism of concrete creep; mirror grinding process parameter selection and design; finite poroelasticity with surface effect; analysis for the large precise equipment assembly; a kind of small quadcopter and *acoustic* cloaking with realizable mass.

Conference Paper • *Open access*

543	**Forecasting of microclimate in the course of buildings design and reconstruction**	Datsuk T.,Pukhal V.,Ivlev U.	Advanced Materials Research, 1020	2014	3

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Quality of human living environment is determined by combination of factors which constitute indoor space *thermal*, light and *acoustic* conditions. At the stage of building design it is essential to undertake complex analysis allowing revealing quantitative values of the governing factors and forecasting their joint influence affecting indoor microclimate formation.

Conference Paper

544	**Reversible constructive system for environmentally sensitive and energy efficient schools in different climate conditions**	Rossi M.,Barbera E., Nasini M.	30th International PLEA Conference Sustainable Habitat for Developing Societies Choosing the Way Forward Proceedings, 3	2014	0

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This research aims to develop a dry assembly, easy to build, flexible and reversible constructive system for the building of energy efficient (temporary) schools located in different climate conditions. With the intent to build schools characterized by good spatial and technological quality, high level of indoor comfort and low energy consumption it was necessary to develop an extremely simple (but at the same time extremely versatile) constructive system, able to easily adapt to different design, environmental and energy strategies. The developed system uses cross laminated timber and other natural materials and consists of small and light modular elements that can be quickly assembled according to different configurations. Spatial-functional, technological-constructive and energy-environmental performance of the developed system have been evaluated, tested and optimized in an intense “try and test” phase in different climate zones: cold (Helsinki, Finland), temperate (Rome, Italy) and hot (Nairobi, Kenya). In particular, the spatial-functional quality was evaluated by the application of the system in several different design configurations of kindergartens optimized with respect to following parameters: type of teaching activity, age and needs of users, spaces functionality and flexibility, paths and relationships between different (indoor and *outdoor*)

Document title	Authors	Source	Year	Citations

spaces and multifunctionality of the common areas. In relation to technological-constructive performance particular emphasis has been given to: rapidity and simplicity of assembly of building elements, expandability and modifiability of the building in response to the changing needs, cost and simplicity of maintenance over time and use of natural and low environmental impact materials. To reduce the annual energy consumption (for heating, cooling and lighting) different active and passive environmental strategies and devices have been developed for the three different chosen climatic zones. The level of hydrothermal, *acoustic* and visual indoor comfort has been evaluated and quantified with the support of *thermodynamic* simulations and the annual energy demand was also calculated and optimized trough appropriate software. In conclusion, tests (design application and simulations) conducted in this research have shown that the developed reversible constructive system (applicable in temporary schools) combines good spatial-functional and technological-constructive qualities with a high level of indoor comfort and a low energy consumption. Moreover, the system resulted extremely efficient also in very different climatic conditions, thanks to its easy adaptability to different configurations and strategies.

Conference Paper

545 **Brain order disorder 2nd group report of f-EEG** | Lalonde F.,Gogtay N., Giedd J.,... Krapels K.A., Dai L. | Proceedings of SPIE the International Society for Optical Engineering , 9118, 91180J | 2014 | 6

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Since the Brain Order Disorder (BOD) group reported on a high density Electroencephalogram (EEG) to capture the neuronal information using EEG to wirelessly interface with a Smartphone [1,2], a larger BOD group has been assembled, including the Obama BRAIN program, CUA Brain Computer Interface Lab and the UCSD Swartz Computational Neuroscience Center. We can implement the pair-electrodes correlation functions in order to operate in a real time daily environment, which is of the computation complexity of $O(N^3)$ for $N=10^{2\sim3}$ known as functional f-EEG. The daily monitoring requires two areas of focus. Area #(1) to quantify the neuronal information flow under arbitrary daily stimuli-response sources. Approach to #1: (i) We have asserted that the sources contained in the EEG signals may be discovered by an unsupervised learning neural network called blind sources separation (BSS) of independent entropy components, based on the irreversible Boltzmann cellular *thermodynamics*($\delta S < 0$), where the entropy is a degree of uniformity. What is the entropy? Loosely speaking, sand on the beach is more uniform at a higher entropy value than the rocks composing a mountain - the internal binding energy tells the paleontologists the existence of information. To a politician, landside voting results has only the winning information but more entropy, while a non-uniform voting distribution record has more information. For the human's effortless brain at constant temperature, we can solve the minimum of Helmholtz free energy (H = E TS) by computing BSS, and then their pairwise-entropy source correlation function. (i) Although the entropy itself is not the information per se, but the concurrence of the entropy sources is the information flow as a functional-EEG, sketched in this 2nd BOD report. Area #(2) applying EEG bio-feedback will improve collective decision making (TBD). Approach to #2: We introduce a novel performance quality metrics, in terms of the throughput rate of faster (δt) & more accurate (δA) decision making, which applies to individual, as well as team brain dynamics. Following Nobel Laureate Daniel Kahnmen's novel "Thinking fast and slowa", through the brainwave biofeedback we can first identify an individual's "anchored cognitive bias sources". This is done in order to remove the biases by means of individually tailored pre-processing. Then the training effectiveness can be maximized by the collective product $\delta t^* \delta A$. For Area #1, we compute a spatiotemporally windowed EEG in vitro average using adaptive time-window sampling. The sampling rate depends on the type of neuronal responses, which is what we seek. The averaged traditional EEG measurements and are further improved by BSS decomposition into finer stimulus-response source mixing matrix [A] having finer & faster spatial grids with rapid temporal updates. Then, the functional EEG is the second order co-variance matrix defined as the electrode-pair fluctuation correlation function $C(\tilde{s}, \tilde{s}')$ of independent *thermodynamic* source components. (1) We define a 1-D Space filling curve as a spiral curve without origin. This pattern is historically known as the Peano-Hilbert arc length a. By taking the most significant bits of the Cartesian product $a \equiv O(x^* y^* z)$, it represents the arc length in the numerical size with values that map the 3-D neighborhood proximity into a 1-D neighborhood arc length representation. (2) 1-D Fourier coefficients spectrum have no spurious high frequency contents, which typically arise in lexicographical (zig-zag scanning) discontinuity [Hsu & Szu, a"Peano-Hilbert curve," SPIE 2014]. A simple Fourier spectrum histogram fits nicely with the Compressive Sensing CRDT Mathematics. (3) Stationary power spectral density

	Document title	Authors	Source	Year	Citations

is a reasonable approximation of EEG responses in striate layers in resonance feedback loops capable of producing a 100, 000 neuronal collective Impulse Response Function (IRF). The striate brain layer architecture represents an ensemble <IRF< e.g. at V1-V4 of Brodmann areas 17-19 of the Cortex, i.e. stationary Wiener-Kintchine-Einstein Theorem. Goal#1: functional-EEG: After taking the 1-D space-filling curve, we compute the ensemble averaged 1-D Power Spectral Density (PSD) and then make use of the inverse FFT to generate f-EEG. (ii) Goal#2 individual wellness baseline (IWB): We need novel change detection, so we derive the ubiquitous fat-tail distributions for healthy brains PSD in *outdoor* environments (Signal=310°C; *Noise*=27°C: SNR=310/300; 300°K=(1/40)eV). The departure from IWB might imply stress, fever, a sports injury, an unexpected fall, or numerous midnight excursions which may signal an onset of dementia in Home Alone Senior (HAS), discovered by telemedicine care-giver networks. Aging global villagers need mental healthcare devices that are affordable, harmless, administrable (AHA) and user-friendly, situated in a clothing article such as a baseball hat and able to interface with pervasive Smartphones in daily environment. ©2014 SPIE.

Conference Paper

546 **Comparison of simulation tools for optimization and evaluation of green building performance in China** — Lu F.,Lin B.,Peng B. — Proceedings of Bs 2013 13th Conference of the International Building Performance Simulation Association — 2013 — 1

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Green building is developing very rapidly worldwide, and evaluation1 optimization of green building performance through simulation has become an important concern. In this paper, the significance of the simulation is pomted out through the comparison of simulation indicators required in the domestic and international green building evaluation standards, which includes building energy consumption, indoor and *outdoor noise*, lighting and *thermal* environment. Based on the survey of a number of China's domestic cases which have acquired green building identification, the condition of the use of simulation software is analysed, and some main problems in the simulation process are discussed in detail. Besides, tends towards the chosen of simulation tools for different simulation indicators in China are analysed taking building projects in the district of Shanghai as example. Finally, the standardization of building performance simulation was discussed in the study, expected to provide better technical supports to promote the application of simulation tools in the development process of green building in China.Copyright © 2011 by IPAC'11/EPS-AG. Copyright © 2011 by IPAC'11/EPS-AG.

Conference Review

547 **2nd International Conference on Green Building, Materials and Civil Engineering, GBMCE 2013** — [No Authors Found] — Applied Mechanics and Materials, 368-370(1) — 2013 — 0

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The proceedings contain 401 papers. The special focus in this conference is on Green Building, Materials and Civil Engineering. The topics include: Analysis of Yi people ancient town spatial form; curriculum innovation integration of digital technology and architecture; discussion of cities construction shape in urbanization; discussion on architectural design of green theatres in China; preserving the historical memory of Honolulu's Chinatown; research for aesthetic and visual quality management in highway landscape; research on large eaves' shading effect based on parametric design platform; selection and application of suburban residential evaluation based on rough set; some issues of BIM application in landscape architecture; study on adaptive pattern of eco-city green space structure; the basic ideas of the construction of ecological city in Baotou city; the development of green architectural heritage cave civilization; the residence maximum height according to the perfect building form coefficient; the present problems of the construction of residential area for aged in china; the research of the development of Xi'an urban space since 2000; to build a green university library - architectural design of Hefei institute library; the green building wisdom in Chinese traditional inhabitation and its significance; the application practice of green building technologies in landscape design; first exploration of disaster prevention and mitigation planning of small towns; systems

Document title	Authors	Source	Year	Citations

engineering problems related to slope ecological protection engineering; the study on feedback respone of urban land use; inner Mongolia grassland nomadic living space of modular theory; on re-understanding of waste landscape; conceiving of Harbin's ecologic residential quarters' water environment; risk analysis of regional energy efficiency in China based on DEA; research on pollutant loading of biological aerated filter; an analysis of economic and environmental impact of sewage treatment in Beijing city; study on the ecological characteristics of the seaweed house in Jiaodong peninsula; the research of rainfall error correction based on system response curve; water quality evaluation of a pumped-storage reservoir in Ningxia, China; study on development of circular economy in resource-based city; study on Poyang lake tourism plan under the low-carbon ecology; research on the key technology of vegetation roofing energy saving; relationship between Chinese WeiHai urbanization and climate change; study on the renewable toxic gas filtering apparatus; study on spatial distribution of surface water resources in Lhasa based on RS and GIS; human resident *acoustical* environment and *acoustical* insulation of residential building; implementation of environmental management on a construction site; industrialized precast construction low carbon design control; life cycle assessment on CO2 reduction of street house reuse; study on scheme optimization of low carbon construction based on value engineering; the ecological strategies of residential buildings in qinba mountain villages; research on evaluation model of sludge disposal method in sewage plant; the generation and analysis of carbon steel pickling sludge; wastewater containing copper precipitation research; study on the relationship between indoor air quality and children's health in Beijing; simulation study of planning layout effect on *outdoor* microclimate of cold residence; numerical simulation on ventilated cavity flow with different turbulence models; the economized design of modern indoor house; the research of green design for apartments for the elderly; the residential interior design in green ecological field; discussion on Huizhou woodcarving motifs and redesign it in modern interior design; construction and analysis of personalized air-conditioning system; fresh air exchange method research of residential house based on natural ventilation; field measurement on the *thermal* environment of greenway in Guangzhou; adsorption of phosphorus by Ce-modified lithium silica fume; adsorption of phosphorus by lithium silica fume; characteristics and applications of luminous fibers; development of a new non-chloride snow-melting agent; effect of niobium on microstructure of X80 grade pipe line steels; internal geometric and electronic structures of natural bornite crystal; mix design of sup-13 mixed with TOR and crumb rubber using dry process; modeling of blast-induced craters by multi-material ALE method; preparation and characterization of biodegradable cotton mulching film; thin-film non-polarizing filters' simulation analysis used in tilted incidence; multiplicity flow solutions in a four-sided lid-driven cavity; analysis of temperature control mode of concrete construction in winter; building blocks production with sludge; design research of wall and ground building materials based on the industrial design; evaluation of protective effectiveness and microstructure of silicate concrete sealer; an efficient environmentally-friendly materials for concrete building protection; influence of aggregates on cracking sensitivity of concrete; maturity of prestressed self-consolidating concrete; shear strength of concrete and gypsum composite walls; study on the influence of limestone dust on performance of cement mortar; workability of fresh cement-asphalt mastic; bond-slip relationship for EB-FRP joints by including free end slip; bending capacity of T beam with prestressed concrete-filled steel tube; experimental study of the recycled aggregate concrete beam flexural performance; experiments and analysis of the structure of concrete roof insulation; orthogonal design choices recycled concrete load-bearing hollow block; recycled concrete at elevated high temperature duration; research on mixing ratio of waste mud brick; analysis of green building properties of Shu Wang's works; analysis of the environmental benefits of green building; application of virtual construction technology in green construction; application of green construction in a project in Huangshi; the green residential building evaluation structural system based on gray clustering; a study on building energy-saving effect increasing with minimal cost; assessment of photovoltaic technologies integrated in building sectors in China; main parameters analysis of the solar street lamp in campus; optimization of building envelope *thermal* design for passive solar house; research of residential area ecological green roof for reduce the heat efficiency in Taichung; research on energy saving renovation of existing building in Tangshan; the construction of multi-light steady solar simulator experiment platform; the design of tension control system in production of the energy-saving windows film; the envelope energy saving technology of civil building and application; the research on energy efficiency of residential building in Guangzhou; the strategies of passive energy-efficiency design in low energy building; an improved maximum power point tracking method for solar PV system; assessment of solar *thermal* technology utilized in china; a solar power computerized platform for energy-saving management of green building; height control measures of highway subgrade fill in Xinjiang plain terrain; numerical simulation of atmospheric flow field around bridge; numerical simulations of the drifting ice sheets collision with the bridge pier; research on maintenance level decision of civil aero-engine module; research review on the anti-seismic characteristics of truss bridge structure; study of reinforced concrete bridge column ductility based on VB; study on the use of geothermal

Document title	Authors	Source	Year	Citations

energy in large space stadium; the dynamic characteristics of rigid frame single-rib arch bridge; 3D aided design solution for integrated pipe network of subway station; review on the ultimate bearing capacity of the aqueduct's supporting structure; resonance characteristics of stay cables under parametric excitation; active earth pressure on retaining wall due to local uniform load; design and field measurement analysis of steel template in axis rotating structures; mechanics analysis and construction measures on tension-chord beam architecture; study about mechanic parameter of slope rock mass' structure plane; research on structural stability of prilling tower in fertilizer industry; a general discussion of soil-water interaction in expansive soil; application of long anchor in support optimization of soft rock tunnel; application of rouse equation in coastal engineering; effects of flexible functional layer on the load stress of cement concrete pavement structure; evaluation of subgrade compactness; research of vibration analysis and monitoring during the pile driving; research on the bearing mechanism of mudstone with soft-hard alternant strata; seismic dynamic model tests on tunnel lining passing through the fault; stability analysis of an earth slope based on Taylor series method; structural design and analysis of a new soil moisture probe; study on the countermeasure of karst collapse in siding lead-zinc mining area; ultimate bearing capacity based on unified strength theory; application of BIM in construction and analysis of obstacles; passenger route choice behavior with congestion consideration; study on building safety economic incentive mechanism; the analysis of housing consumption; construction information management based on BIM technology; *noise* evaluation of concrete pavement and research on mobile coal mining safety monitoring system.

Conference Paper

548 **Thermal feedback identification in a mobile environment** | Wilson G.,Brewster S., Halvey M.,Hughes S. | Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics , 7989 LNCS | 2013 | 29

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Audio and vibrotactile feedback are not always suitable or desirable, as *noise* and/or movement may mask them, and so *thermal* feedback may provide a salient alternative. In this paper, the identification of '*thermal* icons' (structured *thermal* feedback) was tested as a means of conveying information when users were sitting and walking in an *outdoor* location. Overall identification rate for *thermal* icons was 64.6%, but identification of individual parameters was promising, at 94% accuracy for direction of *thermal* change (warming/cooling) and 73.1% accuracy for subjective intensity (moderate/strong). Results showed that walking outdoors did not significantly worsen icon identification compared to sitting outdoors, but the environmental temperature had a strong influence. Recommendations are given on how better to design and adapt *thermal* feedback for use in *outdoor* mobile scenarios. © 2013 Springer-Verlag.

Conference Review

549 **2013 International Conference on Mechatronic Systems and Materials Application, ICMSMA 2013** | [No Authors Found] | Applied Mechanics and Materials, 389 | 2013 | 0

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The proceedings contain 196 papers. The special focus in this conference is on Mechatronic Systems and Materials Application. The topics include: Influence of internal *noise* on rhythmic calcium bursting; preparation and characterization of hydroxyapatite whiskers; theoretical study on fluorescence spectra of three coumarin derivatives; an overview of bioconcrete for structural repair; research of molten carbonate fuel cells modeling based on neural computing; Beijing rural energy consumption and the use of renewable energy sources; economic performance model of new energy vehicles based on linear programming; modeling industrial cost-benefit matrix allocation of carbon emission reduction; energy price policy decision-making model under energy conservation environment; small wind-solar hybrid generation equipment design; application of improved FAHP for nuclear reactor accident emergency decision; analysis of oligopoly on electric power market under asymmetric information; application of

Document title	Authors	Source	Year	Citations

nanotechnology in anti-counterfeiting packaging; research on the ballistic model and simulation of a ejection cylinder; study on uncooled hybrid focal plane detector arrays; the curved plate flaw's location and direction detecting based on the transducer array; the application of soft-switching choppers in maglev trains; research on the performance of 3-state power amplifiers for magnetic bearings; purification experiment of silica sand from gold ore tailings; state of the art of shape memory materials and their applications; computational investigation of mechanical behaviour of FSW joints; bulk-sheet forming process for products with thin wall and thick rim; evaluation of residual iron of carbon nanotubes treated by acid; study on manufacturing technology of high-strength composite board; application of composite materials in children's *outdoor* playground; study on grinding burn in high efficiency deep grinding; purification experiment research of quartz sand; separation of weak magnetic iron ore with micro-fine particle; loss angle testing of polymer melts at high shear rate; research on driveline resistance of micro-automobile; a potential precipitating apparatus and its performance measuring; the design of comprehensive CAT test rig for electro-hydraulic servo valve; a new monitoring system based on microcontroller; modeling of vertical GaN based resonant cavity light-emitting diode; implementation of particular models of innovative servomechanisms; design of a high power programmable intelligent switch; PID control design for a maglev train system; the research of air bags overturning device based on fuzzy control; design and simulation of power system for hybrid sweeper truck; a compensation approach of dead-zone nonlinearity in dual-motor driving servo system; research on microelectronic packaging position detection based on machine vision; precision temperature control system based on PID algorithm; weak signal detection system based on lock-in amplifier technology; SAGE-ML joint estimation and detection algorithm in MIMO-OFDM systems; the diagnostis method of control valve stiction; research on the dynamic characteristics of switching power supply; semi physical simulation of diesel generator set rotational speed control; design of high gain CMOS folded cascode operational amplifier; design and electro-*thermal* analysis of a new micro-hotplate for gas sensor; high-efficiency charge pump LED driver circuit design; the design and implementation of experimental apparatus of magnetic levitation; design of control system of gemstone processing machine based on CAN-bus; stability analysis and design of a mono-pulse photoelectric detecting system; stability analysis switched systems; design and experiment of intelligent grape bagging robot; a new binocular vision stereo matching algorithm; advanced techniques for NC programs preparation; the realization of moving target tracking; the brief analysis on a broadband VCO; study on quadratic interpolation of motion control; reconfigurable spatial sub-chain evolution; energy efficient virtual machine allocation in cloud; design of ground power information system for EMU; T-wave alternans detection analysis and software design; 2-dimension polynomial fitting for the edge detection; web based learning system modeling research; a design of 3D EMC scanning system and the feature analysis for transformer reliability assessment.

Article

☐ 550 **Comparison of solvent extraction and thermal desorption methods for determination of volatile polyfluorinated alkyl substances in the urban atmosphere** — Wu Y.,Chang V.W.-C. — Analytical Methods, 5(13) — 2013 — 10

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Polyfluorinated alkyl substances (PFASs) are among emerging persistent organic pollutants (POPs) that are widely present in a variety of consumer products. The typical approach for determination of volatile PFASs in ambient air is based upon the high-volume active sampling and solvent extraction (method A), which suffers from the poor portability, high *noise* level and significant loss of low molecular weight analytes. All these limitations have restricted its application in urban regions, especially in residential areas when conducting source apportionment analysis. In this study, method A was compared with another approach using low-volume air sampling and *thermal* desorption (method B). In general, method B showed a better repeatability and recovery, while method A demonstrated a lower detection limit. Application of both methods in the *outdoor* urban environment of Singapore indicated a higher time resolution, higher portability and lower *noise* level of method B. With these advantages method B appeared to be more suitable for the quantitative analysis of targeted PFASs in regions with concentrated point and variable diffusive sources. © The Royal Society of Chemistry 2013.

Conference Review

	Document title	Authors	Source	Year	Citations
551	**E3S Web of Conferences**	[No Authors Found]	E3s Web of Conferences , 1	2013	0

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The proceedings contain 390 papers. The topics discussed include: evaluation of integrated time-temperature effect in pyrolysis process of historically contaminated soils with cadmium (Cd) and lead (Pb); testing single and combinations of amendments for stabilization of metals in contrasting extremely contaminated soils; risk-based approach for *thermal* treatment of soils contaminated with heavy metals; effects of chemical applications to metal polluted soils on cadmium uptake by rice plant; characterization of natural adsorbent material for heavy metal removal in a petrochemical site contamination; economics and science in higher education: the need for a common language; future scenarios of urban mobility and their sustainability; VELUXlab: a sustainable prototype of active house for innovation; nature, economics, property and commons. subversive notes inspired by Elinor Ostrom; towards identifying contribution of wake turbulence to inflow *noise* from wind turbines; hybrid turbulence models for atmospheric flow: a proper comparison with RANS models; satisfaction perception of indoor environment of low-cost housing: a case study of flat Taman Desa Sentosa; criteria for affordable housing performance measurement: a review; *outdoor* environment of low-cost housing: a case study of flat Taman Desa Sentosa; simulations of the muon flux sensitivity to rock perturbation associated to hydrogeological processes; and muon telescope based on micromegas detectors: from design to data acquisition.

Article

	Document title	Authors	Source	Year	Citations
552	**Experimental analysis on arc step voltage of fuse with rectangular notch**	Xiao Y.,Zhuang J.,Wang C.,Chen B.,Yuan Z.	Gaodianya Jishu High Voltage Engineering , 39(4)	2013	4

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Step voltages of a fuse dramatically influence the current-limiting action and electrical insulation of the fuse. For quantitatively describing the fuse electric characteristic during initial arcing stage, we experimentally studied step voltages produced in short circuit interruption with fuses containing rectangular notch. An approximately linear relationship between arc-starting current density, which ranged within 10~45 kA/mm^2, and the initial arc column voltage gradient was found, when pre-arcing energy remained the same as the packing density of silica sand. Meanwhile the effect of silica sand on the step voltage was discussed. Two more sets of comparative tests, in which fuse was linked with different notch lengths and notch numbers in series, showed the *sound* fitness of the linear relationship. The results can be applied in further investigations of small scale or full scale experiments on multiple notches in series for high voltage fuses.

Conference Paper • *Open access*

	Document title	Authors	Source	Year	Citations
553	**User perspectives on outdoor noise in open-plan offices with operable windows**	Goins J.,Chun C.,Zhang H.	Architectural Science Review , 56(1)	2013	4

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Recent research suggests that buildings with operable windows in general, and mixed-mode (MM) buildings in particular, can provide improved *thermal* comfort and control opportunities for users. However, there have been concerns about *outdoor noise* sources such as traffic or construction *noise* when windows are opened. Concerns like these may hinder the installation of operable windows in buildings. This article examines 23,000 office building occupants perspectives on *noise* from both sealed and naturally ventilated/MM buildings. The results suggest that occupants near operable windows are more satisfied than those near sealed windows or those far from either window type. Among occupants dissatisfied with *noise*, complaints about indoor *noise* sources (such as people talking to each other) are about 10 times more prevalent than *outdoor noise* complaints. © 2013 Copyright Taylor and Francis Group, LLC.

Article

	Document title	Authors	Source	Year	Citations
554	**Assessment of indoor environmental quality in open-plan offices**	Hwang T.,Kim J.T.	Indoor and Built Environment , 22(1)	2013	39

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This study aims to analyze the factors of indoor environment on occupants' health and comfort by post-occupancy evaluation (POE) and to accumulate the fundamental data of occupants' behaviour in relation to indoor environment, in an effort to ascertain the optimal building energy management system (BEMS). For this purpose, indoor temperature, relative humidity, vertical temperature distribution, air velocity, predicted mean vote (PMV), radiant temperature, *outdoor* temperature, and humidity were measured for evaluating *thermal* environment for nearly 2 years. Indoor air contaminants, such as PM_{10}, CO_2, formaldehyde, and volatile organic compounds (VOCs) were measured in the building and at *outdoor* locations. Working plane illuminance, window luminance, and the equivalent continuous a-weighted *sound* pressure level (Leq) were also monitored. In addition, the occupants in the intelligent office building were surveyed regarding the indoor environmental quality via the intranet. Results showed that all indoor environmental factors satisfied the recommendation, except the indoor relative humidity and PMV. The physical factors of *thermal* environment had been stable since two months after it was built. However, it should be necessary to improve the *sound* privacy caused by low background *noise*, seasonal air-conditioning for appropriate humidity, air freshness, and Video Display Terminal (VDT) glare in order to reduce the level of occupants' annoyance. © 2012 The Author(s).

Article

	Document title	Authors	Source	Year	Citations
555	**Case study health hazards of automotive repair mechanics: Thermal and lighting comfort, particulate matter and noise**	Loupa G.	Journal of Occupational and Environmental Hygiene , 10(10)	2013	13

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An indoor environmental quality survey was conducted in a small private automotive repair shop during May 2009 (hot season) and February 2010 (cold season). It was established that the detached building, which is naturally ventilated and lit, had all the advantages of the temperate local climate. It provided a satisfactory microclimatic working environment, concerning the *thermal* and the lighting comfort, without excessive energy consumption for air-conditioning or lighting. Indoor number concentrations of particulate matter (PM) were monitored during both seasons. Their size distributions were strongly affected by the indoor activities and the air exchange rate of the building. During working hours, the average indoor/*outdoor* (I/O) number concentration ratio was 31 for $PM_{0.3\text{-}1}$in the hot season and 69 for the cold season. However I/O $PM_{1\text{-}10}$number concentration ratios were similar, 33 and 32 respectively, between the two seasons. The estimated indoor mass concentration of $PM_{10}$for the two seasons was on average 0.68 mg $m^{-3}$and 1.19 mg m^{-3}, i.e., 22 and 36 times higher than outdoors, during the hot and the cold seasons, respectively. This is indicative that indoor air pollution may adversely affect mechanics' health. *Noise* levels were highly variable and the average $L_{EX,\ 8h}$of 69.3 dB(A) was below the European Union exposure limit value 87db (A). *Noise* originated from the use of manual hammers, the revving up of engines, and the closing of car doors or hoods. Octave band analysis indicated that the prevailing *noise* frequencies were in the area of the maximum ear sensitivity. Copyright © 2013 JOEH, LLC.

Review

	Document title	Authors	Source	Year	Citations
556	**Towards sustainability index for healthy buildings - Via intrinsic thermodynamics, green accounting and harmony**	Kim J.T.,Todorovic M.S.	Energy and Buildings , 62	2013	36

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As, fossil energy resources are closer to their exhaustion, global warming in raise, and more catastrophic weather extremes are occurring worldwide, there are more and more warnings that the risks to the Earth humanity survival are also in growth. The Earth/humanity as a whole is becoming more complex system than ever. Trying to find a way to make a turn from the current one-way irreversibility to sustainability, it is necessary to find universal schemes, quantities, indicators and criteria relevant for the Earth and humanity's resilience and sustainability. Presented is a

Document title	Authors	Source	Year	Citations

review of the multi-criteria sustainability analysis methods (intrinsic *thermodynamic* based on energy, exergy, sustainability index, analytic hierarchy process, etc.). It has been shown that crucial sensitivity of all methods is related to the selected sets of energy criteria (economical, social and environmental) and to the mathematical algorithms for the determination of the weighted factor and sub-indicators agglomeration. Also, shown is that concerning buildings sustainability is missing indicator relevant to the indoor comfort and people health, accompanied with a comprehensive, physically *sound* assessment methodology for building's synergetic environmental performance - twofold metrics with the reference to *outdoor* and indoor built environment concerning people health and comfort. A scheme of a structure and spatio-temporal vision of the Global sustainable development is presented - showing that the approach to the global sustainable development can be reliable only if it is based on a system of real human and ethical values applicable to every social, cultural and economic situation. Finally paper summarizes needs for future multidisciplinary, inter- and cross-disciplinary study on healthy sustainable buildings, interwoven with harmony and holistic people health and well-being research. That research is to be physically *sound* - intrinsic *thermodynamic*, harmonious green, and relevant metrics is to be free of the short-term politically shaped and quasi-economically distorted approaches, because the world is to get what is painfully missing - a standard metrics of the sustainability index, universally acceptable. © 2013 Elsevier B.V. All rights reserved.

Article

557 **Relationship between indoor environmental quality and building envelopes covered by plants: A review of the literature** — Fadeyi M.O. — International Journal of Environment and Sustainable Development, 12(4) — 2013 — 0

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It is not clear whether the benefits of improving *outdoor* environmental conditions inherent in urban plants apply to building indoor environment when plants are integrated on building envelopes. This paper reviews published research to date to clarify whether building envelopes covered by plants have effects on indoor environmental quality, and building occupants' health and comfort. The review exercise revealed that building envelopes covered by plants can (1) improve indoor *thermal* and *acoustic* conditions (2) compromise indoor light level. Evidences addressing the impact on indoor air quality (chemical, physical, and biological pollutants) is lacking in the literature. There are no evidences in the literature addressing the impact of improved IEQ conditions, as a result of plants integrated on building envelopes, on building occupants' health and comfort. This study is relevant to creating energy efficient and sustainable buildings. Copyright © 2013 Inderscience Enterprises Ltd.

Conference Paper

558 **User perspectives on outdoor noise in buildings with operable windows** — Goins J.,Chun C.,Zhang H. — Proceedings of 7th Windsor Conference the Changing Context of Comfort in an Unpredictable World — 2012 — 2

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Recent research suggests that buildings with operable windows in general, and mixed-mode (MM) buildings in particular can provide improved *thermal* comfort and control opportunities for users. Yet, there have been concerns about *outdoor noise* sources like traffic or construction *noise* when windows are opened. Concerns like these may hinder the installation of operable windows in buildings. This paper examines 23,000 office building occupants' perspectives on *noise* from both sealed and naturally ventilated/MM buildings. Results suggest that occupants near operable windows are more satisfied than those near sealed windows or those far from either window type. Among occupants dissatisfied with *noise*, complaints about indoor *noise* sources -like people talking- are about 10 times more prevalent than *outdoor noise* complaints.

Conference Paper

	Document title	Authors	Source	Year	Citations
559	**Numerical predictions for sustainable development of cities: Acoustic propagation in presence of urban vegetation**	Guillaume G.,Gauvreau B.,Fortin N.	41st International Congress and Exposition on Noise Control Engineering 2012 Inter Noise 2012, 7	2012	2

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Outdoor sound predictions in urban media are both a societal stake and a scientific issue, including sustainable development of cities. Thus SPL forecasts in such built-up areas will soon have to take into account particular boundary conditions, e.g. *acoustic* impedance of facades and/or roof surfaces in expectation of buildings greening or other use of vegetation on urban surfaces. In this context and in the framework of the "VegDUD" project, the Transmission Line Matrix (TLM) method has been used to model *acoustic* propagation in such complex media. This research project from the french research agency (ANR) involves different laboratories (Ifsttar, Météo-France, INRA, CSTB, ECN, etc.), various disciplinaries (*acoustics*, meterolorology, *thermics*, hydrologics, etc.), diverse approaches (physical, chemical and human sciences) and different meanings (experimental results and model predictions). Regarding *acoustics*, it aims at quantifying the best vegetation effort with respect to soundscape indicators, e.g. SPL, EDT, etc. This paper presents TLM predictions in (and out of) a canyon street for several configurations of roofs and/or facades vegetalization: ratio and location of vegetation covers, receivers positions, impedance characteristics of surface cover, etc.

	Article				
560	**Data mining approach to polymer selection for making SAW sensor array based electronic nose**	Jha S.K.,Yadava R.D.S.	Sensors and Transducers, 147(12)	2012	10

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In this paper, a simple application of principal component analysis and hierarchical clustering methods of classical data mining has been demonstrated for making selection of polymer coatings for surface *acoustic* wave (SAW) sensor array. The database consists of *thermodynamic* solvation parameters of the prospective polymers and the target vapors. The linear-solvation-energy relationship (LSER) has been used to calculate partition coefficients from the solvation parameters. The partition coefficient data for various vapor-polymer combinations is then arranged as a data matrix taking the polymers for instances (rows) and the vapors for variables (columns). Selection of polymer subset for optimal discrimination of target vapors from background interferents has been made by analyzing the principal component score and load plots. A simulation study for detection of trinitrotoluene (TNT) and dinitrotoluene (DNT) explosive vapors camouflaged in the background of 29 interferent organic vapors originating from varied sources such as soil, vegetation, body odor, and indoor and *outdoor* industrial/commercial environments has been carried out. The paper demonstrates that the application of data mining methods for the selection of polymeric sensor coatings could prove prudent in terms of reducing system complexity, cost and development time. © 2012 IFSA.

	Article				
561	**Implementation, characterization, and evaluation of an inexpensive low-power low-noise infrasound sensor based on a micromachined differential pressure transducer and a mechanical filter**	Marcillo O.,Johnson J.B.,Hart D.	Journal of Atmospheric and Oceanic Technology, 29(9)	2012	88

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The implementation, characterization, and evaluation of a low-cost infrasound sensor developed at the Infrasound Laboratory at the New Mexico Institute of Mining and Technology (Infra-NMT) are described. This sensor is based on a commercial micromachined piezoresistive differential pressure transducer that uses a mechanical high-pass filter to reject low-frequency outband energy. The sensor features a low-*noise*, 2.02-mPa rms (0.5-2 Hz), 5.47-mPa rms (0.1-20 Hz), or 5.62-mPa rms (0.05-20 Hz), flat response between 0.01 and at least 40 Hz; inband sensitivity of 45.13 ± 0.23 μV Pa^{-1}; and a nominal linear range from 2124.5 to 1124.5 Pa. Intended for *outdoor* applications, the influence of *thermal*

	Document title	Authors	Source	Year	Citations

changes in the sensor's response has been minimized by using a *thermal* compensated pressure transducer powered by an ultralow drift (<5 ppm °C $^{-1}$) and *noise* (<4μV from peak to peak) voltage reference. The sensor consumes a minimum of 24 mW and operates with voltages above 8 V while drawing 3 mA of current. The Infra-NMT specifications described above were independently verified using the infrasound test chamber at the Sandia National Laboratories' (SNL's) Facility for Acceptance, Calibration, and Testing (FACT), and the following procedures are for comparison calibration against traceable reference stands in voltage and pressure. Because of the intended broad frequency response of this sensor, the testing chamber was configured in a doublereference sensor scheme. A well-characterized microbarometer (with a flat-amplitude response between 0.01 and 8 Hz) and a microphone (with a flat-amplitude response above 8 Hz) were used simultaneously in this double-reference test configuration. © 2012 American Meteorological Society.

Article

☐ 562 **Innovative strategies for adaptive buildings in large cities** Pizzi E.,Iannaccone G., Ruttico P. International Journal for Housing Science and Its Applications, 36(2) 2012 5

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Changes characterizing our society include an ageing population, urban migration towards big cities, our lifestyle and work. These changes often make traditional building approaches obsolete. The existing building stock cannot totally satisfy the changed needs and new projects ask for careful valuations and new operating tools. In order to face the change, a feasible solution could be obtained by introducing the requisite of flexibility within both the design process and the construction technology of a building. By adopting a computational design methodology and an industrialized construction technology is possible to realize adaptive buildings, or buildings that can modify their characteristics according to the changing boundary conditions. The paper summarizes the management of adaptive/flexible buildings through the use of customized design tools, paying special attention to structural, energetic and technological issues. In order to design a building as an active, adjustable evolving system, is possible to develop a parametric program analysis visualization that allows to deeply understand complex architectural programs in four dimensions. The constantly evolving social mix is visualized with diagrams that automatically solve adjacency requirements and suggest planimetric and sectional relationships, enabling the user to explore different options and configurations. The algorithm can be implemented with arbitrary design choices and the programmatic use of collective spaces in order to reach the best feasible solution from the architectural and social standpoint. This raises questions as to how form emerges and how it continually differentiates, transforms and performs in relation to its specific environment. Several case studies show how it's possible to increase structural efficiency allowing a great flexibility in the interior transformation which could be obtained during the service life cycle of the building. Using internal lightweight walls, flexibility and transformation can be easily obtained without using too much energy for the necessary works. Prefabricated façade elements are designed and conceived with the same level of flexibility as the interior walls, by using innovative materials such as textiles. *Outdoor* spaces could be integrated in the interior spaces (e.g. balcony could change into loggia). A proper evaluation of changes in natural lighting, fire safety, *noise* reduction, solar gains, *thermal* insulation is needed in order to always guarantee high levels of comfort. Copyright©2012 IAHS.

Conference Paper • *Open access*

☐ 563 **A topological form study on passive design for energy-saving space of green cave dwellings** Wei Q.,Xu Y. Applied Mechanics and Materials, 174-177 2012 0

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This study solved the problems of cave dwelling physical environment on the Loess Plateau, such as poor ventilation, insufficient lighting, and moisture. Based on the principle that improved its microclimate by passive form design and the method of topological form transformation, the study adopted corresponding energy-saving measures to the control of the *outdoor* factors that could influence the indoor physical environment, such as solar radiation, air temperature, air flow and humidity. The result provides passive design system and form design strategies to improve cave dwelling *thermal* environment, light environment, *acoustic* environment from the aspects of shape, orientation, space layout, indoor partitions. © (2012) Trans Tech Publications.

	Document title	Authors	Source	Year	Citations
	Conference Paper				
564	**A Sonic boom propagation model including mean flow atmospheric effects**	Salamone J.,Sparrowa V.W.	Aip Conference Proceedings , 1474(1)	2012	4

Hide abstract Find It @ UCL View at Publisher Related documents

This paper presents a time domain formulation of nonlinear lossy propagation in onedimension that also includes the effects of non-collinear mean flow in the *acoustic* medium. The model equation utilized is an augmented Burgers equation that includes the effects of nonlinearity, geometric spreading, atmospheric stratification, and also absorption and dispersion due to *thermoviscous* and molecular relaxation effects. All elements of the propagation are implemented in the time domain and the effects of non-collinear mean flow are accounted for in each term of the model equation. Previous authors have presented methods limited to showing the effects ofwind on ray tracing and/or using an effective speed of *sound* in their model equation. The present work includes the effects of mean flow for all terms included in the augmented Burgers equation with all of the calculations performed in the time-domain. The capability to include the effects of mean flow in the *acoustic* medium allows one to make predictions more representative of real-world atmospheric conditions. Examples are presented for nonlinear propagation of N-waves and shaped sonic booms. [Work supported by Gulfstream Aerospace Corporation]. © 2012 American Institute of Physics.

	Article				
565	**Acoustic data of cross linked polyethylene (XLPE) and cured liquid silicone rubber (LSR) by means of ultrasonic and low frequency DMTA**	Buchholz U.,Jaunich M., Stark W.,Habel W., Petersson B.A.T.	IEEE Transactions on Dielectrics and Electrical Insulation , 19(2), 6180250	2012	12

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Partial discharges may cause damage to electrical insulation of high voltage equipment. They initiate elastic waves in the insulating material, e.g. in the stress cone of an *outdoor* termination. Localisation of the origin of such elastic waves can help to predict serious damaging processes in the electrical insulation. In order to measure and evaluate the wave propagation effects in typical multilayered elastomeric structures, knowledge of the material properties is required. The propagating velocity and the attenuation of longitudinal waves are important parameters. Values for these quantities found in the literature were not appropriate. Therefore, for cross-linked polyethylene (XLPE) and cured liquid silicone rubber (LSR), the longitudinal wave velocity and the attenuation were evaluated in the temperature interval from -20°C to 50°C and in the frequency range from 200 kHz to 600 kHz using a two-sample ultrasound technique. The loss factor was determined from these measured quantities. Additionally, low frequency Dynamic Mechanical *Thermal* Analysis (DMTA) was applied to investigate LSR and XLPE in a temperature interval between -100 and 50°C and to check qualitatively the ultrasound data. © 2012 IEEE.

	Article				
566	**A numerical approach to sound levels in near-surface refractive shadows**	Cheinet S.	Journal of the Acoustical Society of America , 131(3)	2012	21

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The present study formulates a consistent method to simulate the *outdoor*, near-surface *sound* propagation through realistic refractive conditions. The correlated atmospheric stratification and turbulence properties are derived from standard meteorological quantities through flux-profile similarity relationships. The propagation of a monochromatic *sound* field is simulated in presence of the turbulence and stratification effects and an impedance ground. The propagation model uses a numerical solution of a second-order moment parabolic equation, which is introduced and evaluated. The so-formed coupled atmospheric-*acoustic* model is used to systematically investigate the *sound* levels in near-surface refractive shadows. In an illustrative propagation scenario, the shadow zone *sound* levels are predicted to show significant variations with the meteorological conditions. Specifically, the *sound* levels decrease with the adverse wind, as a consequence of enhanced mean upward refraction. Conversely, they increase with the

	Document title	Authors	Source	Year	Citations

absolute value of the surface heat flux, as a consequence of enhanced turbulence scattering. Implications for the assessment of the *sound* levels in shadow zones are discussed. © 2012 *Acoustical* Society of America.

Book Chapter

567 **Air quality and human health** — Meijer A. — Sustainable Urban Environments an Ecosystem Approach — 2012 — 0

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This chapter deals with both the indoor and *outdoor* air quality of buildings and their effect on the health of the occupants. This is an important topic for sustainability of urban areas and buildings because of the long lifetime of buildings and the large amount of time that people spend inside buildings and in urban areas. Sources and parameters affecting the indoor and *outdoor* air quality are discussed. Other indoor environmental aspects such as *thermal* comfort, *noise* and lighting are also addressed briefly. Then, the effects of these parameters and aspects on the health and comfort of the occupants are described. Finally, guidelines are given for designing buildings with high indoor air quality and for solving existing problems with poor air quality and health complaints.

Conference Paper

568 **International standards related to the influence of the indoor environment on people's health, comfort and productivity** — Olesen B.W. — 7th International Cold Climate HVAC Conference — 2012 — 2

Hide abstract | Find It @ UCL | View at Publisher | Related documents

People spend in industrialized countries more than 90 % of their lives in an artificial indoor environment (home, transportation, work). This makes the indoor environment much more important for people health and comfort than the *outdoor* environment. In typical office buildings the cost of people is a factor 100 higher than energy costs, which make the performance of people at their work significantly more important than energy costs. The task is to optimize indoor environmental conditions for health, comfort and performance while conserving energy, since more than one third of current global energy consumption is used to maintain indoor environments. The paper presents an update on today's requirement for a healthy and comfortable environment. The paper will mainly be dealing with the indoor *thermal* environment and air quality. Several international standards and guidelines from ISO (International Organization for Standardization), CEN (European Committee for Standardization) and ASHRAE (American Society of Heating, Refrigerating and Air Conditioning Engineers) are specifying requirements related to comfort and health. Several of these standards have been developed mainly by experts from Europe, North America and Japan, thus guaranteeing a worldwide basis. Are there, however, special considerations related to other parts of the world (lifestyle, *outdoor* climate, and economy), which are not dealt with in these standards and which will require revision? Critical issues such as adaptation, effect of increased air velocity, personalized systems, adapted/non-adapted occupants, ventilation effectiveness, use of air cleaning etc. are still being discussed, but in general these standards can be used worldwide. The productivity of people is however not taken into account in these documents. Studies are showing that comfortable room temperatures, increased ventilation above normal recommendation, reduction of indoor pollution sources and more effective ventilation increases the performance of people. It has been decided by ISO TC163 and TC205 to start a work item to develop an international standard for the indoor environment based on a revision of EN1525, which specifies indoor environmental input parameters for design and assessment of energy performance of buildings addressing indoor air quality, *thermal* environment, lighting and *acoustics*. The paper presents several issues to be discussed in the planned revision. The presented issues are dealing with the *thermal* environment (when to use adaptive model and criteria for personalized systems), indoor air quality (ventilation effectiveness, air cleaning, adapted/non-adapted occupants, and personalized ventilation), *acoustic* (introduction of categories), lighting (introduction of categories, daylight factor) and finally occupant behaviour.

Conference Paper

569 **Application of acoustical indicators and index for Urban quality evaluation** — Magrini A., Ricciardi P., Soares P.F. — 18th International Congress on Sound and Vibration 2011 Icsv 2011, 2 — 2011 — 3

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

The evaluation of the urban quality is very complex, since it is influenced by numerous factors. The environmental comfort is one of these fundamental elements: it can allow to quantify the influence of the *outdoor* environment on the human activities and the men's health, on the basis of *thermo*-hygrometrical, *acoustical*, lighting comfort and Air Quality. The correlations among *outdoor* comfort, urban landscape and architectural features offer wide perspectives, as they could represent very useful means to give a qualitative and quantitative judgment on the existing and reachable urban quality level. In fact these correlations could represent an important tool for choosing the major actions for the renovation of the urban environment. In order to obtain the urban quality global judgment, all the playing factors must be correctly weighted and associated. To quantify the influence of the environmental aspects, a first attempt to apply a weighting and correlating methodology has been performed. Therefore the approach based on indicators and indexes, previously developed, has been applied in the field of the *acoustical outdoor* comfort evaluation. The *Acoustical* Quality Index AQI obtained by calculations will be considered in further investigations with other indexes (pollution, climatic conditions, urban elements, etc.) by means of a proper weighting system, to calculate a global urban index. The potentialities of the methodology have been experimented in this study elaborating *noise* levels distribution obtained by simulations to calculate the AQI of a small town. Copyright © (2011) by the International Institute of *Acoustics* & Vibration.

Conference Paper

570 **How DFV process can improve split AC installation process** Lopes C.A. 51st Annual Conference of SAVE International 2011 2011 1

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The Split Air Conditioning product has become very popular in Brazil in recent years. The system consists of two units physically separated - the indoor unit contains the evaporator and the *outdoor* unit contains the condenser. Copper tubes connect both units which transfers the refrigerant fluid from the indoor unit to the *outdoor* unit. The main benefit of this system over a traditional single unit system is the lower *sound* level combined with the improved *thermal* comfort. The customer concern about purchasing this product is the installation process. It requires a specialist to install it and the cost can reach 30% to 40% of the product's price. Consumers are dissatisfied with the installation cost and the need to depend on a technical specialist. Most customers wish they could install it by themselves if the product would allow it. A Value Engineering Workshop was held in February, 2010 with a multifunctional team of fourteen people in Joinville, Brazil, to discuss the installation main issues and generate solutions to improve this complex process using the Value Methodology. This Function Analysis approach generated 64 ideas, where at least 27 were considered easy to implement with high saving or increased value. Seven Business Cases were developed by the team requiring $147K in investment with an annual risk weighted estimated $21.5M of savings for the company and consumers. This paper will demonstrate the success of using the Value Engineering approach to improving the design of a product for simpler installation.

Conference Paper

571 **Methodology for measurement of objective and subjective appreciation of multicriteria comfort in residential buildings** Boulet S. 7th International Symposium on Heating Ventilating and Air Conditioning Proceedings of Ishvac 2011, 2 2011 0

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The notion of environmental high quality takes into account both the indoor ambient environment of users, with health and comfort preoccupations, and the impact on the *outdoor* environment. It aims to respect sustainable development principles but it also must base on human factors. Nowadays, the various approaches are essentially *thermal*, *acoustical*, visual or olfactory but the impact of one on the perception of the other one is important and must be taken into account. We want to break with this practice, developing an evaluation of the comfort with use of multi-criteria analysis as a decision-making tool, through the definition of a coherent family of criteria representing different components of the comfort. The objective of the presented study is the identification of parameters affecting

	Document title	Authors	Source	Year	Citations

comfort in moderate environment. We will analyse results through the multicriteria decision analysis, a confrontation between measurements and inquiries will be realized in case of validation.

Article

572 **Controlling Cabin and Envelope Air Flows and Pressure Differentials to Prevent Envelope Condensation, Enable Cabin Humidification, Improve Fire Safety, and Decrease Fuel Use**

Walkinshaw D.S., Preston K.F.

SAE International Journal of Aerospace, 4(2)

2011

6

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The uncontrolled flow of cabin air into the aircraft envelope caused by indoor-*outdoor* temperature gradients and associated stack pressure differentials causes cabin air to circulate between the cabin and the envelope, producing condensation on the cold fuselage behind the insulation with a number of adverse consequences. These include an inability to practically maintain cabin humidity at normal levels, a reduction in ventilation effectiveness, microbial growth, metal corrosion and structural failures, insulation performance degradation, *thermal* discomfort, increased engine *noise* transmission, additional fuel consumption and electrical failures and fires. This paper explains the origin of these stack pressures and the air flows they create between the cabin and the envelope, and indicates means to control the flows and associated cabin-envelope pressure differentials and thereby minimize condensation and its associated problems, improve *thermal* comfort and fire safety, and coincidentally decrease fuel use. © 2011 SAE International.

Article

573 **Design and reliability of acoustic wedge transducer assemblies for outdoor touch panels**

Son K.T.,Lee C.C.

IEEE Transactions on Components Packaging and Manufacturing Technology, 1(8), 5940997

2011

3

Hide abstract Find It @ UCL View at Publisher Related documents

Acoustic wedge transducer assemblies (AWTA) for generating and detecting surface *acoustic* waves (SAW) on non-piezoelectric substrates were designed, fabricated and evaluated for *outdoor* touch panel applications. An AWTA consists of a piezoelectric *acoustic* transducer plate bonded to an *acoustic* prism that is bonded to a glass panel. The *acoustic* prism converts a longitudinal *acoustic* wave into a SAW on the glass. This conversion is based on the *acoustic* analogy of optics Snell's law. Thus, the prism material must have low longitudinal *acoustic* velocity and low *acoustic* attenuation. For nearly all SAW-based touch panels on the market, the *acoustic* prisms are made of Lucite. An issue with Lucite is high coefficient of *thermal* expansion (CTE), much higher than that of glass panels and piezoelectric ceramics. In *outdoor* environments with large temperature variations, this large CTE mismatch causes the Lucite prism to break off the glass panel and the piezoelectric transducer to break apart from the Lucite prism. After extensive search, we identified liquid crystal polymer (LCP) and bismuth (Bi) as potential *acoustic* prism materials having low CTE. Wedge transducer assemblies with Lucite, LCP, and Bi prisms, were fabricated and evaluated. Following Telcordia GR-468-Core recommendation, the wedge assemblies went through *thermal* cycling between -45°C and 85°C for 1000 cycles. The wedge assemblies with Lucite prism broke before 50 cycles. Wedge assemblies with LCP and Bi prisms survived 1000 cycles. In terms of *acoustic* performance, wedge assembly with LCP prisms is as good as wedge assembly with Lucite prism. Accordingly, LCP is the best prism material for SAW-based touch panels in *outdoor* applications. © 2011 IEEE.

Article

	Document title	Authors	Source	Year	Citations
574	**Impact of portable HEPA filtration units on fungal exposure in a post transplant unit during | Einfluss dezentraler HEPA filtrationsgeräte auf die schimmelpilz-exposition in einer transplantations-nachsorgeeinheit während des abbruchs eines benachbarten bettenhauses**	Engelhart S.,Simon A., Hirschberg S.,Exner M.	Hygiene Medizin, 36(5)	2011	0

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Background: Building construction, renovation, or demolition works are commonly associated with a significant release of dust and fungal spores, including an increased risk of invasive aspergillosis in immunocompromized patients. Method: During the demolition of an old patient care building, we conducted an infection control risk assessment and initiated measures to reduce dust and fungal emissions and to minimize exposure of patientsat-risk. The nearby post-transplant-unit was equipped with portable HEPA-filtration units (MedicCleanAir® Pro). During the demolition period, we monitored air counts of *thermotolerant* fungi in filtered and unfiltered air using the Merck MAS-100, and conducted an active surveillance for invasive aspergillosis in patients-at-risk. Results: *Outdoor* fungal emissions were only moderately elevated during demolition, indicating the effectiveness of emission *thermoreducing* measures. Moreover, by means of the HEPA filtration units, indoor fungal exposure could be reduced by about one order of magnitude and did not exceed the mean indoor reference level. No cases of invasive aspergillosis occurred. Conclusion: We conclude that a *sound* infection control risk assessment and appropriate infection control measures can successfully minimize the excess risk of invasive aspergillosis in patients-at-risk. Portable HEPA-filtration units can be useful for limited periods of increased exposure, but *noise* level, indoor climate, and patient compliance restrict the use of these devices.

Article

575	**A simple outdoor criterion for assessment of low frequency noise emission**	Broner N.	Acoustics Australia, 39(1)	2011	24

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A criterion for examining low frequency *noise* (LFN) emission is presented. The modulation effect is clearly observable and can be seen to begin once the screens start operating. The modulation period is approximately 60 seconds. Transportation *noise* sources such as aircraft and diesel trains also are sources of LFN. Based on empirical and laboratory studies, it can be shown that the primary effect due to LFN appears to be annoyance and that this affect is greater than would be expected based on the A-weighted level alone. One obvious reason for this is that the A-weighting network significantly decreases the contribution of low frequency energy in a *sound* due to the reduced loudness sensitivity of a person's hearing at low frequencies. It has been known for many decades that gas turbines, boilers, forced draft fans and other sources can produce low frequency *noise* which can cause feelings of annoyance due to nausea, headache and uneasiness and vibration induced rattle.

Article

576	**Literature survey on how different factors influence human comfort in indoor environments**	Frontczak M.,Wargocki P.	Building and Environment , 46(4)	2011	1,129

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The present paper shows the results of a literature survey aimed at exploring how the indoor environment in buildings affects human comfort. The survey was made to gather data that can be useful when new concepts of controlling the indoor environment are developed. The following indoor environmental conditions influencing comfort in the built environment were surveyed: *thermal*, visual and *acoustic*, as well as air quality. The literature was surveyed to determine which of these conditions were ranked by building users as being the most important determinants of comfort. The survey also examined the extent to which other factors unrelated to the indoor environment, such as individual characteristics of building occupants, building-related factors and *outdoor* climate including seasonal changes, influence whether the indoor environment is evaluated as comfortable or not. The results suggest that when developing systems for controlling the indoor environment, the type of building and *outdoor* climate, including season, should be taken into account. Providing occupants with the possibility to control the indoor environment

	Document title	Authors	Source	Year	Citations

improves *thermal* and visual comfort as well as satisfaction with the air quality. *Thermal* comfort is ranked by building occupants to be of greater importance compared with visual and *acoustic* comfort and good air quality. It also seems to influence to a higher degree the overall satisfaction with indoor environmental quality compared with the impact of other indoor environmental conditions. © 2010 Elsevier Ltd.

Article

577 **Dynamic spectrum access operational parameters with wireless microphones** — Erpek T.,McHenry M.,Stirling A. — IEEE Communications Magazine, 49(3), 5723798 — 2011 — 10

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This article provides a comprehensive analysis of dynamic spectrum access operational parameters in a typical hidden node scenario with protected wireless microphones in the TV white space. We consider all relevant effects and use an analysis framework that properly combines probabilistic technical factors to provide specific policy recommendations including the exclusion zone distances and the sensingbased DSA threshold detection levels. First, man-made *noise* measurements were taken in different locations and the amount of interference from man-made *noise* in potential wireless microphone channels was analyzed. Data collection results show that man-made *noise* levels can be up to 30 dB above the *thermal noise* floor. Furthermore, indoor-to-*outdoor* path loss measurements were conducted to determine the required exclusion distance for DSA-enabled TV band devices to ensure reliable wireless microphone operation in a typical application. The results show that the required exclusion zone can be safely and conservatively set at around 130 m when the results from man-made *noise* measurements and wireless microphone propagation measurements are used. Additionally, we developed a simulation to determine the required DSA sensing threshold levels for impairment-free wireless microphone operation. An indoor-to-*outdoor* path loss model was created based on the above path loss measurement results. This statistical path loss model was used to determine the received signal level at TV band devices and the interference level at the wireless microphone receiver. Our results show that the sensing threshold can be set at around -110 dBm (in a 110 kHz channel) for impairment- free wireless microphone operation when man-made *noise* and representative propagation models are used. © 2006 IEEE.

Article • *Open access*

578 **The most problematic variable in the course of human-biometeorological comfort assessment - The mean radiant temperature** — Kantor N.,Unger J. — Central European Journal of Geosciences, 3(1) — 2011 — 335

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This paper gives a review on the topic of the mean radiant temperature T_{mrt}, the most important parameter influencing *outdoor thermal* comfort during sunny conditions. T_{mrt} summarizes all short wave and long wave radiation fluxes reaching the human body, which can be very complex (variable in spatial and also in temporal manner) in urban settings. *Thermal* comfort researchers and urban planners need easy and *sound* methodological approaches to assess T_{mrt} . After the basics of the T_{mrt} calculation some of the methods suitable for obtaining T_{mrt} also in urban environments will be presented.. Two of the discussed methods are based on instruments which measure the radiation fluxes integral (globe *thermometer*, pyranometer-pyrgeometer combination), and three of the methods are based on modelling the radiation environment with PC software (RayMan, ENVI-met and SOLWEIG). © 2011 Versita Warsaw and Springer-Verlag Wien.

Article

579 **Temperature Aging, Compression Recovery, Creep, and Weathering of a Foam Silicone Sealant for Bridge Expansion Joints** — Malla R.B.,Shrestha M.R.,Shaw M.T.,Brijmohan S.B. — Journal of Materials in Civil Engineering, 23(3) — 2011 — 20

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Silicone foam was investigated as a sealant for small movement expansion joints in bridge decks. This paper presents results from the laboratory assessment of a model foam sealant subjected to *thermal* aging (exposure to high and low temperatures and temperature cycling), fatigue conditions, and *outdoor* weathering. Parallel tests were performed on a commercial solid-silicone bridge-joint sealant. Test results showed that the solid sealant recovered

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faster than the foam sealant after being subjected to prolonged compression at elevated temperature. When subjected to a constant tensile force, both the foam and solid sealants exhibited high initial creep rates (rate of elongation), but appeared to reach an equilibrium level of elongation at longer times. The foam sealant creeps at a slower rate and takes more time to get to the equilibrium elongation, whereas the solid sealant creeps much faster initially and reaches equilibrium faster. *Thermal* aging was found to have significant effects on the sealant modulus (increases with temperature); however, the effects on ultimate stress and strain were not apparent for both types of sealants. Temperature cycling between 24 and -29 C was observed to diminish the ultimate stress and strain of both the sealants by roughly 25%; however, no significant changes in modulus were found. Results from the tests on a limited number of *outdoor*-weathered sealant specimens showed that weathering appeared to produce an increase in sealant tensile and shear moduli (i.e., hardening effects because of weathering) and a decrease in ultimate strain. The weathered samples show tension loading-unloading behavior similar to the unaged samples. The tensile stress relaxation rate of the *outdoor*-weathered sealants could not be distinguished from those laboratory-cured (unweathered) counterparts. © 2011 American Society of Civil Engineers.

Article

☐ 580 **Performance evaluation of traditional bath buildings via empirically tested simulation models** — Orehounig K.,Mahdavi A. — Journal of Building Performance Simulation, 4(1) — 2011 — 8

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In a recent research effort, we studied the performance of a certain class of traditional buildings in Egypt, Turkey, Morocco, Syria, and Algeria, namely public baths ('hammams'). Over a period of 1 year, indoor and *outdoor* climate data as well as visual and *acoustical* performance information was collected for a selected number of these buildings. To further explore indoor environmental conditions, the monitored data were used to calibrate digital performance simulation models of a number of the objects studied. Specifically, *thermal* performance simulation models were generated for three hammams in Egypt, Turkey, and Algeria. Likewise, *acoustical* performance simulation models were generated for three baths in Algeria. These calibrated *thermal* and *acoustical* simulation models were shown to reliably reproduce measurements. Thus, they could be parametrically applied to evaluate alternative options for the renovation, restoration, reuse, and adaptation of these buildings. © 2011 International Building Performance Simulation Association (IBPSA).

Article

☐ 581 **A critique of European Standard EN 15251: Strengths, weaknesses and lessons for future standards** — Nicol J.F.,Wilson M. — Building Research and Information, 39(2) — 2011 — 54

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European Standard EN 15251 has broken new ground. Firstly, it assembles existing information about optimizing *thermal*, air quality, *acoustic* and visual comfort so that designers can make energy calculations which embody all the best information about aspects of the environment that impinge on energy use. Secondly, it follows other standards in recognizing the different expectations occupants have of the *thermal* environment in mechanically and naturally ventilated buildings. Potentially this makes EN 15251 an important standard in the design and the evaluation of new and existing buildings. Therefore, it is essential to be clear about its strengths and weaknesses and to indicate ways in which future updates might improve on it. This paper considers the way in which the Standard is written, its underlying philosophy and purpose, and the ways in which it defines unacceptable *thermal* conditions and how they might be misunderstood. Questions are raised regarding the use of *outdoor* temperature to define indoor comfort as the basis for providing a comfortable indoor *thermal* environment. Suggestions are made for improving the Standard. © 2011 Taylor & Francis.

Conference Paper

	Document title	Authors	Source	Year	Citations
582	**Controlling cabin and envelope air flows and pressure differentials to prevent envelope condensation, enable cabin humidification, improve fire safety, and decrease fuel use**	Walkinshaw D.S., Preston K.F.	SAE Technical Papers	2011	1

Hide abstract Find It @ UCL View at Publisher Related documents

The uncontrolled flow of cabin air into the aircraft envelope caused by indoor-*outdoor* temperature gradients and associated stack pressure differentials causes cabin air to circulate between the cabin and the envelope, producing condensation on the cold fuselage behind the insulation with a number of adverse consequences. These include an inability to practically maintain cabin humidity at normal levels, a reduction in ventilation effectiveness, microbial growth, metal corrosion and structural failures, insulation performance degradation, *thermal* discomfort, increased engine *noise* transmission, additional fuel consumption and electrical failures and fires. This paper explains the origin of these stack pressures and the air flows they create between the cabin and the envelope, and indicates means to control the flows and associated cabin-envelope pressure differentials and thereby minimize condensation and its associated problems, improve *thermal* comfort and fire safety, and coincidentally decrease fuel use. Copyright © 2011 SAE International.

Article

583	**[The environment of the Intensive Care Center and the work of the nursing staff]. | Ambiente do Centro de Terapia Intensiva e o trabalho da equipe de enfermagem.**	Chavaglia S.R., Borges C.M., do Amaral E.M., Iwamoto H.H.,Ohl R.I.	Revista Gaucha De Enfermagem EENFUFRGS , 32(4)	2011	3

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This is a descriptive exploratory study with a quantitative approach. It aims to characterize the environment of the Intensive Care Center (ICC) in regard to its physical area, material resources and equipments, and to identify factors concerned to the work of nurses. It investigates environmental factors that contribute to an aesthetically harmonious, functional and humanized space and that favor the performance of nursing work. The units that make up the ICC meet the recommendations of the National Health Surveillance Agency (ANVISA) in the evaluated items. he following favorable work conditions were highlighted: *thermal* conditions, color of the ceiling, walls and floors, luminosity. The following unfavorable work conditions were highlighted: *outdoor* spaces, privacy and individuality of clients and *noise*. The conclusion is that the facilities of the units meet the minimum recommendations of ANVISA. Both favorable and unfavorable environmental work conditions were identified. The creation of better environmental conditions allows a better staff performance, influencing positively quality, safety, and job satisfaction.

Conference Paper

584	**Smart insulation for thermal control in buildings**	Knotts W.,Miller D.,Mo C., Schaefer L.A.,Clark W.W.	ASME 2011 Conference on Smart Materials Adaptive Structures and Intelligent Systems Smasis 2011, 1	2011	5

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Buildings are a significant source of energy consumption in the US and worldwide, and conditioning building interiors occupies a major portion of that expenditure. At the building level, a particular scenario can occur where it becomes more advantageous for a structure's walls to be "open" to the outside, as opposed to remaining "closed", in an insulating state. For instance, a cool night may follow a hot summer day after the sun sets, but traditional insulation captures heat built up inside a home. A clear, sunny day may also heat a building's exterior during a cold winter day, and heat transfer to the interior would be more desirable than insulation. This paper presents a study of a concept, termed "smart insulation", that could be used in such scenarios to take advantage of beneficial *thermal* gradients in order to save heating and cooling costs. Enabling heat transfer without moving air can also help control humidity within interior spaces, as well as limit *noise* either from moving air or through open windows. Unlike traditional insulation that maintains a specific insulating value during its useful life, smart insulation changes between an insulating and a conducting state depending on the *thermal* gradient. Some design concepts have been fabricated

	Document title	Authors	Source	Year	Citations

and were tested in an insulated chamber that provides a "hot side" and a "cool side". In the test chamber, temperature measurements were taken and the insulating capacity of each design specimen was calculated. Results indicate that the designs have potential to operate between two states (with the best case ranging from 11% to 61% of a benchmark insulation case) and effectively provide sustainable heating or cooling by capitalizing on ambient *outdoor* conditions. © 2011 by ASME.

Conference Paper

585 **Acoustical indicators and index for urban quality evaluation** — Magrini A., Alfano F.R.D. — 20th International Congress on Acoustics 2010 Ica 2010 Incorporating Proceedings of the 2010 Annual Conference of the Australian Acoustical Society, 3 — 2010 — 4

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In the urban health evaluation, the environmental comfort (*thermo*-hygrometrical, *acoustical* and lighting comfort and IAQ) represents a fundamental aspect to quantify the influence of the climate and of the human activities on the men's health. The correlations among *outdoor* comfort, urban landscape and architectural features offer wide perspectives, as they could represent very useful means to give a qualitative and quantitative judgment on the existing real estate or to choose the major actions for the restoration of the urban environment. To obtain urban health global judgment, all the playing factors must be correctly weighted and associated. An analysis based on homogeneous quantities is needed, avoiding multiplying judgment scales and using, for example, one scale for each involved parameter comparable with the others. To quantify and correlate the various environmental elements, a first attempt was performed aimed at a global assessment on the basis of a mutual judgment system. In this paper a proposal of schematisation based on indicators and index has been developed. With the definition of them, relating to the *acoustical outdoor* comfort evaluation, it is possible to obtain a global quality environment evaluation considering also other aspects like, for example, *thermo*-hygrometrical and atmospheric pollution effects. First analyses to validate the method on the basis of experimental and simulated *sound* pressure levels have been developed. Moreover, *noise* level data, traffic and population density were considered as basic parameters on which it is possible to develop single indicators and a global *acoustic* quality index. In the meantime a subjective investigation was performed, to correlate the values of the *acoustical* index to the individual sensation of the people living in the zones of the investigation.

Conference Paper

586 **Adaptive comfort and thermal expectations-a subjective evaluation in hot humid climate** — Rajasekar E., Ramachandraiah A. — Proceedings of Conference Adapting to Change New Thinking on Comfort Windsor 2010 — 2010 — 45

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Subjective and experimental evaluation of *thermal* comfort characteristics have been carried out in naturally ventilated apartment buildings located in a hot humid climatic region of India. Comfort and acceptability limits of environmental parameters under these conditions have been evaluated on the basis of adaptive comfort theory. The residents are found to accept a wider range of environmental conditions as being comfortable, compared to the limits suggested by traditional *thermal* comfort standards. Factors like age, *thermal* expectation, economic status and past experience with *thermal* comfort play a significant role in determining the comfort perception. A comparative study of the *thermal* performance of these buildings with respect to Fanger's extended adaptation theory has been carried out. Constraints for adaptation like *outdoor noise* levels and their influence on the *thermal* comfort perception have also been examined in this study.

	Document title	Authors	Source	Year	Citations

Conference Paper

587 **Compensated fiber-optic frequency distribution equipment** — MacDonald J.,Conway G. — 42nd Annual Precise Time and Time Interval Ptti Systems and Applications Meeting 2010 — 2010 — 6

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One-way frequency distribution of atomic clock standards has been widely studied, and several architectures have emerged that employ the use of a return path signal to compensate for the effects of temperature and phase *noise*. These architectures have generally relied upon temperature-controlled delay lines to compensate for fiber *thermal* effects, and polarization scrambling techniques to minimize the impact of polarization mode dispersion (PMD). These compensation techniques can limit the performance of the system by introducing spurious *noise* products onto the optical carrier. Uncompensated one-way fiber optic distribution systems are currently deployed with a demonstrated stability of $2{\cdot}10^{-13}$ (1 sec). Without temperature compensation, this equipment is suitable for intra-facility distribution of hydrogen maser references. However, operation over moderate length inter-facility links (IFL) will degrade the stability significantly, due in large part to the temperature effects of the *outdoor* or underground fiber cable. The stability can be improved through the use of temperature-compensated fiber cable, but this type of cable is typically an order of magnitude more costly than standard fiber. Furthermore, many locations already have installed dark fiber whose use would be more cost effective. We are developing a compensated one-way frequency distribution system that is compact, easy to install, and requires neither polarization scramblers nor large, *thermally* controlled delay lines. The system employs forward and reverse path microwave modulated optical carriers on a single fiber. Electronic compensation is used to correct for path length phase variations, and a novel technique to moderate the PMD without the need for polarization scramblers. © 2010 by Precise Time and Time Interval (PTTI) - Time Service Department.

Article • *Open access*

588 **Probability of error of linearly modulated signals with Gaussian cochannel interference in maximally correlated Rayleigh fading channels** — Rugini L.,Banelli P. — Eurasip Journal on Wireless Communications and Networking, 2010, 193183 — 2010 — 5

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We evaluate the probability of error of linearly modulated signals, such as phase-shift keying (PSK) and quadrature amplitude modulation (QAM), in the presence of Gaussian cochannel interference (CCI) and Rayleigh fading channels. Specifically, we assume that the fading channel of the CCI is maximally correlated with the fading channel of the signal of interest (SOI). In practical applications, the maximal correlation of the CCI channel with the SOI channel occurs when the CCI is generated at the transmitter, such as the multiuser interference in downlink systems, or when a transparent repeater relays some *thermal noise* together with the SOI. We analytically evaluate the error probability by using a series expansion of generalized hypergeometric functions. A convenient truncation criterion is also discussed. The proposed theoretical approach favorably compares with alternative approaches, such as numerical integration and Monte Carlo estimation. Among the various applications of the proposed analysis, we illustrate the effect of nonlinear amplifiers in orthogonal frequency-division multiplexing (OFDM) systems, the downlink reception of code-division multiple-access (CDMA) signals, and the *outdoor*-to-indoor relaying of Global Positioning System (GPS) signals. Copyright © 2010 Luca Rugini and Paolo Banelli.

Conference Paper

589 **DSA operational parameters with wireless microphones** — Erpek T.,McHenry M.A.,Stirling A. — 2010 IEEE Symposium on New Frontiers in Dynamic Spectrum Dyspan 2010, 5457881 — 2010 — 6

	Document title	Authors	Source	Year	Citations

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This paper provides a comprehensive analysis of Dynamic Spectrum Access (DSA) operational parameters in a typical "Hidden Node" scenario with wireless microphones in the TV white space situation. We consider all relevant effects and use an analysis framework that properly combines probabilistic technical factors to provide specific policy recommendations including the exclusion zone distances and the sensing-based DSA threshold detection levels. First, man-made *noise* measurements were taken in different locations and the amount of interference from man-made *noise* in potential wireless microphone channels was analyzed. Data collection results show that man-made *noise* levels can be up to 30 dB above the *thermal noise* floor. Furthermore, indoor-to-*outdoor* path loss measurements were conducted to determine the required exclusion distance for DSA devices to ensure reliable wireless microphone operation in a typical application (a church). The results show that the required DSA radio exclusion zone can be safely and conservatively set at around 130 m when the results from man-made *noise* measurements and wireless microphone propagation measurements are used. Additionally, we developed a simulation to determine the required DSA sensing threshold levels for impairment-free wireless microphone operation. An indoor-to-*outdoor* path loss model was created based on the above path loss measurement results. This statistical path loss model was used to determine the received signal level at DSA devices and at the interference level at wireless microphone receiver. Our results show that the sensing threshold can be set at around -110 dBm (in a 110 kHz channel) for impairment free wireless microphone operation when man-made *noise* and representative propagation models are used. ©2010 IEEE.

Conference Paper

☐ 590 **Near the ground laser communication system: Fried parameter estimation from the WFS measurements** — Sergeyev A., Roggemann M., Demars C. — IEEE Aerospace Conference Proceedings, 5446897 — 2010 — 18

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Near the ground laser communication systems must operate in the presence of strong atmospheric turbulence. The effects of atmospheric turbulence on the laser beam which are relevant to optical communications are a broadening of the laser footprint, random jitter of the laser beam, and high spatial frequency intensity fluctuations referred to as scintillation. The overall goal of our program is to improve the performance and extend the range of optical communications systems by exploring the use of adaptive optics and channel coding. Knowledge of the turbulence conditions and the ability to describe its properties are the key aspects to make these improvements effective. The multiphase approach described here is directed to statistically describe atmospheric turbulence based on results derived from experimentally collected data. To model the performance of a real world system, we have developed an *outdoor* path partially over water, spanning 3 km turbulence measurement and monitoring system. The system described in this paper has two transmitters and two receivers. The transmitter side is equipped with a laser and a bank of 14 horizontally, in-line mounted LEDs. The laser transmitter and LED operate at 808 nm with laser power of 2mW. The receiver side consists of two channels. The WFS channel captures the signal from the laser transmitter and is specifically designated for the wavefront sensing. The wavefront sensor channel consists of relay optics, a Hartmann WFS, and a CCD camera. The PSF channel is designed to capture the signal from LED bar and consists of relay optics and a CCD camera. The data collected from the PSF channel is used for measurements of anisoplanetic effects. In this paper we emphasize the data collection and consecutive analysis from the WFS channel only. Statistics of the Fried parameter r_0 are derived from 6TB of data collected over 50 days under various day and night atmospheric conditions. Significant fluctuations of r_0 are found with the values ranging from 2mm and up to 15cm. ©2010 IEEE.

Article

☐ 591 **Pedestrian tracking by fusion of thermal-visible surveillance videos** — Leykin A., Hammoud R. — Machine Vision and Applications, 21(4) — 2010 — 63

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In this paper we introduce a system to track pedestrians using a combined input from RGB and *thermal* cameras. Two major contributions are presented here. First is the novel probabilistic model of the scene background where each pixel is represented as a multi-modal distribution with the changing number of modalities for both color and *thermal*

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input. We demonstrate how to eliminate the influence of shadows with this type of fusion. Second, based on our background model we introduce a pedestrian tracker designed as a particle filter. We further develop a number of informed reversible transformations to sample the model probability space in order to maximize our model posterior probability. The novelty of our tracking approach also comes from a way we formulate observation likelihoods to account for 3D locations of the bodies with respect to the camera and occlusions by other tracked human bodies as well as static objects. The results of tracking on color and *thermal* sequences demonstrate that our algorithm is robust to illumination *noise* and performs well in the *outdoor* environments. © Springer-Verlag 2008.

Article

☐ 592 **Adaptive use of natural ventilation for thermal comfort in Indian apartments** Indraganti M. Building and Environment, 45(6) 2010 135

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Thermal comfort research in India is in its nascent stage. Indian codes specify uniform comfort temperatures between 23 and 26 °C for all types of buildings. About 73% of energy in Indian residences is consumed for ventilation and lighting controls. Therefore, a *thermal* comfort field survey was conducted in apartment buildings in Hyderabad, which included information on the use of building controls. The present analysis is based on this database. Due to the poor availability of adaptive opportunities, 60% of the occupants were uncomfortable in summer. The comfort range obtained (26.0-32.5 °C) was way above the standard. The occupants adapted through clothing, metabolism and the use of various controls like windows, balcony and external doors and curtains. The subjects operated the controls, as the indoor temperature moved away from the comfort band. At comfort temperature, maximum use of openings was found, which correlated robustly with indoor/*outdoor* temperature and *thermal* sensation. Use of controls was critically impeded by lack of privacy and safety and non-availability of controls. Several design and non-*thermal* factors, such as operation and maintenance of controls, mosquitoes, *noise*, and occupant's attitude, age and tenure impacted the occupant's adaptive behaviour and *thermal* comfort significantly. The building's 'restrained adaptive opportunity' seriously hampered the occupant's *thermal* satisfaction and adversely affected the sensation vote. © 2009 Elsevier Ltd. All rights reserved.

Article

☐ 593 **Scoping indoor airborne fungi in an excellent indoor air quality office building in Hong Kong** Mui K.W.,Chan W.Y., Wong L.T.,Hui P.S. Building Services Engineering Research and Technology, 31(2) 2010 4

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This study aims to investigate the differences in indoor airborne fungal exposure between a selective subset of air-conditioned offices within a building classified with excellent indoor air quality (IAQ) in Hong Kong and typical offices in Hong Kong and other countries. Assessments of the indoor and *outdoor* fungal exposure levels and compositions, and indoor-to-*outdoor* (I/O) fungi count ratios were conducted in 82 offices of a building with excellent IAQ. The indoor airborne fungi were detected in 97.6% air samples with airborne fungi count (AFC) ranged from 2 to 92 CFU m^{-3}, and with a mean I/O ratio of 0.5. It was reported that the indoor AFC was significantly lower than other typical offices in Hong Kong and other countries. Aspergillus, Cladosporium, Penicillium and yeast were dominant indoors but an *outdoor* dominant fungi Alternaria was not detected in the offices. This study reveals that better control of *thermal* conditions and proper maintenance of the air-conditioning system would lower the AFC. In addition, a significantly higher relative abundance (RA) of Aspergillus and lower RA of Cladosporium were also reported. It presents a useful reference for IAQ assessment regarding the quantification of airborne fungal exposure in air-conditioned offices in a building with excellent IAQ. Practical application: An assessment of exposure levels to common indoor airborne fungi, such as Aspergillus, Cladosporium, Penicillium and yeast in Hong Kong air-conditioned offices reveals that an excellent office environment with good *thermal* and IAQ control would associate with a significantly lower airborne fungi count. A significant different pattern of indoor airborne fungi were observed that a significantly higher relative abundance of Aspergillus and lower relative abundance of Cladosporium were reported. © 2010 The Chartered Institution of Building Services Engineers.

	Document title	Authors	Source	Year	Citations
594	Conference Paper **Outdoor synthetic aperture acoustic ground target measurements**	Bishop S.,Ngaya T.-A., Vignola J.,... Soumekh M., Rosen E.	Proceedings of SPIE the International Society for Optical Engineering , 7664, 76641Y	2010	3

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A novel *outdoor* synthetic aperture *acoustic* (SAA) system consists of a microphone and loudspeaker traveling along a 6.3-meter rail system. This is an extension from a prior indoor laboratory measurement system in which selected targets were insonified while suspended in air. Here, the loudspeaker and microphone are aimed perpendicular to their direction of travel along the rail. The area next to the rail is insonified and the microphone records the reflected *acoustic* signal, while the travel of the transceiver along the rail creates a synthetic aperture allowing imaging of the scene. Ground surfaces consisted of weathered asphalt and short grass. Several surface-laid objects were arranged on the ground for SAA imaging. These included rocks, concrete masonry blocks, grout covered foam blocks; foliage obscured objects and several spherical canonical targets such as a bowling ball, and plastic and metal spheres. The measured data are processed and ground targets are further analyzed for characteristics and features amenable for discrimination. This paper includes a description of the measurement system, target descriptions, synthetic aperture processing approach and preliminary findings with respect to ground surface and target characteristics. © 2010 Copyright SPIE - The International Society for Optical Engineering.

	Document title	Authors	Source	Year	Citations
595	Conference Paper **Fusion of acoustic and optical sensor data for automatic fight detection in urban environments**	Andersson M., Ntalampiras S.,Ganchev T., ... Ahlberg J.,Fakotakis N.	13th Conference on Information Fusion Fusion 2010, 5712105	2010	34

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We propose a two-stage method for detection of abnormal behaviours, such as aggression and fights in urban environment, which is applicable to operator support in surveillance applications. The proposed method is based on fusion of evidence from audio and optical sensors. In the first stage, a number of modality-specific detectors perform recognition of low-level events. Their outputs act as input to the second stage, which performs fusion and disambiguation of the first-stage detections. Experimental evaluation on scenes from the *outdoor* part of the PROMETHEUS database demonstrated the practical viability of the proposed approach. We report a fight detection rate of 81% when both audio and optical information are used. Reduced performance is observed when evidence from audio data is excluded from the fusion process. Finally, in the case when only evidence from one camera is used for detecting the fights, the recognition performance is poor.

	Document title	Authors	Source	Year	Citations
596	Article **In situ calibration of hot-film probes using a collocated sonic anemometer: Implementation of a neural network**	Kit E.,Cherkassky A., Sant T.,Fernando H.J.S.	Journal of Atmospheric and Oceanic Technology , 27(1)	2010	31

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Although the integral quantities of atmospheric turbulence are conveniently measured using sonic anemometers, obtaining relevant finescale variables such as the kinetic energy dissipation using conventional hot-film/wire techniques remains a challenge because of two main difficulties. The first difficulty is the mean wind variability, which causes violation of the requirement that mean winds have a specific alignment with the hot-film/wire probe. To circumvent this problem, a combination of collocated sonic and hot-film anemometers, with the former measuring mean winds and aligning the latter in the appropriate wind direction via an automated platform, is successfully designed and implemented. The second difficulty is the necessity of frequent and onerous calibrations akin to hot-film anemometry that lead to logistical difficulties during *outdoor* (field) measurements. This is addressed by employing sonic measurements to calibrate the hot films in the same combination, with the output (velocity) to input (voltage) transfer function for the hot film derived using a neural network (NN) model. The NN is trained using low-pass-filtered

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hot-film and sonic data taken in situ. This new hot-film calibration procedure is compared with the standard calibration method based on an external calibrator. It is inferred that the sonic-based NN method offers great potential as an alternative to laborious standard calibration techniques, particularly in the laboratory and in stable atmospheric boundary layer settings. The NN approximation technique is found to be superior to the conventionally used polynomial fitting methods when used in conjunction with unevenly spaced calibration velocity data generated by sonic anemometers. © 2010 American Meteorological Society.

Conference Paper

597 **Outdoor elements providing urban comfort: The role of fountains in the soundscape** — Semidor C.,Venot-Gbedji F. — PLEA 2009 Architecture Energy and the Occupant S Perspective Proceedings of the 26th International Conference on Passive and Low Energy Architecture — 2009 — 5

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Urban planners and other town designers need information to improve the environmental quality of cities. It is well known that water constitutes an important element for mitigating *outdoor thermal* comfort, especially in hot climates. The use of water takes different forms: the evapotranspiration effect from vegetation, river currents or the water jets from a fountain. It also concerns the urban *acoustic* comfort. As the matter of fact knowing how to provide pleasant and quiet outside areas in order to ensure pleasant and quiet inside areas is one of the challenges for a better urban environment. So, it is necessary to know what type of urban design enables such areas to exist and how to characterise their *sound* quality. This paper deals with the data analysis from different case studies (Barcelona, Bristol and Genoa) depending on the type, the place, and the surroundings of the fountains. In each case, some soundwalks were carried out in order to highlight the quality of the urban soundscape.

Conference Paper

598 **Optimization of the acoustic performances of lightweight sandwich roof elements** — Dijckmans A.,Vermeir G., Niggebrugge J.W. — 38th International Congress and Exposition on Noise Control Engineering 2009 Inter Noise 2009, 3 — 2009 — 5

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Lightweight roof elements are commonly used in buildings because of the ease and speed of installing and the potential for very high and reliable *thermal* insulation. The *sound* insulation of the sandwich panels however may be too low. In fact, the combination of low mass and high stiffness typically results in a lower *sound* transmission loss over a large frequency band. But regularly the presence of higher *outdoor* levels is reflected in stricter requirements, so making improvements on the *sound* insulation of these roof elements necessary. The *sound* transmission loss of different lightweight sandwich panels with a core of expanded polystyrene has been measured in the lab. Also the structure-borne *sound* sensitivity has been measured with a mini-tapping machine. These measurements are compared with an analytical model based on the transfer matrix method and a numerical, wave based model. Primarily the influence of geometrical size of the panel and of the composition of the interlayer is discussed in view of optimizing the results.

Conference Paper

	Document title	Authors	Source	Year	Citations
599	**Field studies in the united states and united kingdom of the effectiveness of mechanical ventilation systems on indoor air quality in hospitality venues where smoking is allowed in relation to outdoor air conditions**	Sterling E.,Glassco M.	ASHRAE Transactions, 115 PART 2	2009	0

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Mechanical ventilation has been used since the middle 19th century as an effective method of providing acceptable conditions of indoor air in buildings. Mechanical ventilation systems have traditionally been the method of choice by which to ventilate hospitality venues where smoking occurs indoors. Mechanical ventilation systems work by introducing air into and moving air through buildings to achieve indoor air quality that is comparable to *outdoor* air quality. The Federation of European Heating and Air-Conditioning Associations (REHVA) in Guideline 4 "Ventilation and smoking: Reducing the exposure to ETS in buildings" provides guidance on best practices to achieve effective ventilation for hospitality venues and other buildings where smoking occurs (REHVA 2004). REHVA recognizes this topic as extremely important in respect to indoor air quality and energy consumption. The recently approved European Union (EN) EN/DIN Standard 15251 "Indoor environmental input parameters for design and assessment of energy performance of buildings addressing indoor air quality, *thermal* environment, lighting and *acoustics* "recommends specific ventilation rates for spaces in buildings where smoking is allowed (CEN- EN/DIN Standard 15251). Questions have been recently raised about the effect of mechanical ventilation systems on indoor air quality in hospitality venues where smoking is allowed. We have reviewed the literature available about effective mechanical ventilation and tobacco smoke and have found that very few studies purporting to establish the effect of ventilation on tobacco smoke in hospitality venues have actually measured and quantified ventilation parameters. In fact, many of these studies have not measured and quantified indoor air quality or tobacco smoke. Further although there are a few studies that have measured and reported ventilation, indoor air quality and environmental tobacco smoke (ETS) parameters, most of the hospitality venues included in these studies do not have effective ventilation systems. In addition, some of the studies appear to have misclassified air filtration (cleaning) and air conditioning systems as ventilation systems. There is a dearth of data with which to determine the relationship between effective ventilation and tobacco smoke in hospitality venues where smoking occurs. These field studies of hospitality venues in the United States and the United Kingdom was undertaken in order to obtain field data on this relationship. The objectives of these field studies were to evaluate the performance of mechanical ventilation systems in various types and configurations of hospitality venues in both the United States and the United Kingdom to determine the effectiveness of these systems in providing acceptable indoor air quality in spaces where smoking is allowed. These field studies of six hospitality venues on two continents included ventilation and indoor air quality measurements of three casinos, two pubs and one restaurant. This paper presents measurements of ventilation and indoor air quality including environmental tobacco smoke parameters sampled in several locations within each of the six venues to assess the effectiveness of mechanical ventilation. The indoor air quality measurements included carbon dioxide, carbon monoxide, total volatile organic compounds, temperature, relative humidity and respirable suspended particles. Measurements were also taken in the *outdoor* air as a means of comparing *outdoor* air quality with the indoor air quality. The data from the field studies shows that mechanical ventilation is an effective means of providing indoor air quality conditions that are comparable to the *outdoor* air quality in hospitality venues where smoking is permitted. ©2009 ASHRAE.

Book Chapter

	Document title	Authors	Source	Year	Citations
600	**Numerical methods to model infrasonic propagation through realistic specifications of the atmosphere**	Norris D.,Gibson R.,Bongiovanni K.	Infrasound Monitoring for Atmospheric Studies	2009	45

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To model infrasonic propagation, two areas must be adequately addressed. First, the environment must encompass a domain spanning from the ground to the lower reaches of the *thermosphere*, and its properties must be resolved at a scale comparable to the *acoustic* wavelengths of interest. Second, the relevant fundamental physics that influence the wave propagation must be captured in the numerical models that are applied. To address these challenges, a large amount of work has been completed by researchers in fields spanning from traditional *acoustics* to upper atmospheric physics. In this chapter, this body of work is reviewed with an emphasis on its applicability to the modeling and interpretation of infrasonic observations. Global climatological models have largely been replaced in

Document title	Authors	Source	Year	Citations

current infrasound modeling practice by atmospheric specifications that combine output from numerical weather prediction models for the lower atmosphere with empirical models for the upper atmosphere. Recently developed specifications incorporate higher-resolution, regional or mesoscale weather analysis products in order to improve fidelity below altitudes of 50 km. Specifications that utilize a terrain-following coordinate system, including high-resolution topography, enable the incorporation of additional physical effects. These specifications are unable to resolve all fine-scale stochastic phenomena, e.g., atmospheric irregularities smaller than the model resolution, fine-scale structures at altitudes above 50 km, and gravity wave fluctuations. In particular, gravity waves are of interest because their spatial scales are of the same order as infrasonic wavelengths. Because atmospheric fine-scale structure is inherently turbulent, spectral formulations are used to capture the energy distribution across space-time scales relevant to infrasound and to generate representative realizations of variable fields. The fundamental physical processes that affect infrasound, much like those of higher frequency *acoustics*, include refraction, diffraction, scattering, absorption, dispersion, and terrain. In addition, nonlinear effects can become significant near the source and at high altitudes. Several numerical approaches have been used for modeling infrasonic propagation often based on analogous models for underwater and *outdoor sound* propagation. They can be loosely classified under the headings of geometric (ray tracing, Gaussian beam, tau-p), continuous wave (parabolic equation or PE), full-wave (normal mode, finite-difference time domain, time-domain PE), and nonlinear (e.g., NPE). A good introductory source to the underlying computational aspects is (Jensen F, Kuperman W, Porter M, Schmidt H (1994) Computational ocean *acoustics*. AIP Press, New York). (Although this reference is primarily intended for ocean *acoustics*, many of the models carry over to the atmosphere and the presentation is superb.) A general overview of the propagation models is presented here with their corresponding strengths and weaknesses highlighted in the context of infrasonic studies. Also included are examples of modeling studies using high-resolution atmospheric specifications. © 2009 Springer Science+Business Media B.V.

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